

20

Mechanical Circulatory Support

FRANCIS D. PAGANI AND JAMES K. KIRKLIN

SECTION I: ADULT MECHANICAL CIRCULATORY SUPPORT	788		
Definition	788	Warfarin	807
Historical Note	788	Antiplatelet Therapy	807
Biological Barriers	789	Anticoagulation for Continuous Flow Pumps	807
General Concepts	789	Anticoagulation for the Total Artificial Heart	807
Coagulation System	789	Blood Pressure Management with Continuous Flow Pumps	807
Engineering Concepts in Pump Design	790	Ventricular Suction	807
General Concepts	790	Pump Thrombus in Continuous Flow Pumps	808
Pulsatile (Volume Displacement) Versus Continuous Flow (Rotary) Pumps	791	Pump Stoppage of Continuous Flow Pumps	808
Continuous Flow Rotary Pump Design: Axial Versus Centrifugal Pumps	791	Results	808
Bearing Design/Impeller Suspension	792	Survival	808
Mechanical Bearing	792	Major Adverse Events	812
Noncontact Bearing Designs	792	Neurologic Events	812
Hydrodynamic Bearing	793	Infection	814
Magnet Bearing	794	Gastrointestinal Bleeding	814
Hydrodynamic Performance of Continuous Flow Pumps	794	Right Heart Failure	815
Interaction of the Continuous Flow Rotary Pump and Native Heart	794	Patient Reported Outcomes	815
Parallel and Series Circulation	796	Indications and Patient Selection for Durable Mechanical Circulatory Support	815
Flow Estimation	796	Advanced Age	816
Limitations in Flow Control with Continuous Flow Rotary Pumps	798	Obesity	816
Durable Circulatory Support Devices	798	Psychosocial Risk	816
Pulsatile Ventricular Assist Devices	798	Right Heart Failure	816
Continuous Flow Ventricular Assist Devices	798	Timing of Therapy	816
HeartMate 3	798	Risk Scores	816
Total Artificial Heart	798	MCS Support Strategy: Short-Term and Long-Term Applications	816
SynCardia Total Artificial Heart	799	Special Situations and Controversies	817
Device Considerations	799	Adult Congenital Heart Disease, Including Fontan Circulatory Failure	817
Techniques of Operation	799	Myocardial Recovery	817
Intracorporeal Continuous Flow Left Ventricular Assist Device	799	SECTION II: PEDIATRIC MECHANICAL CIRCULATORY SUPPORT	817
Concomitant Surgical Procedures	805	Historical Note	817
Management of Acute Right Heart Failure	805	Clinical Features	818
Total Artificial Heart	805	Cardiomyopathies	818
Special Features of Postoperative Care	806	Myocarditis	818
Anticoagulation	806	Congenital Heart Disease	818
Heparin	806	Neonates and Infants	818
		Modes of Support	818
		Temporary Ventricular Assist Device Support	818

Paracorporeal Pulsatile Support	818	Stroke	826
Implantable Continuous Flow Devices	820	Infection	827
Technique of Operation	820	Renal Dysfunction	827
Cannulation Strategies for Temporary Support	820	Liver Dysfunction	827
Paracorporeal Pulsatile Ventricular Assist Devices (Berlin Heart EXCOR)	821	Indications, Timing, and Decision-Making for Device Implantation	827
Special Features of Postoperative Care	824	Congenital Heart Disease in Two-Ventricle Situations	827
Circulatory Management	824	Pulmonary Hypertension	827
Anticoagulation	824	Infants with Two Ventricles Considered for Support as Bridge-to-Transplant	827
Results	825	Infants with Functionally Single Ventricle Considered for Support as Bridge-to-Transplant	827
Survival	825	Ventricular Assist Device Therapy in Failing Fontan	828
General Results and Risk Factors	825	Special Situations and Controversies	828
Implantable Continuous Flow Devices	825	Atrial Versus Ventricular Cannulation for VAD Inflow	828
Paracorporeal Devices	825	Indications for Biventricular Support, Addition of Membrane Oxygenator, and Renal Replacement Therapy	828
Single Ventricle Before or After Stage I Palliation	825		
Single Ventricle After Stage II Palliation	826		
Failing Fontan Circuit	826		

SECTION I: ADULT MECHANICAL CIRCULATORY SUPPORT

Definition

Mechanical circulatory support (MCS) is a means of imparting energy for the forward flow of blood in the body using manmade devices. Its intent is to remove some or all work of cardiac output from either the left and/or right ventricle. MCS devices (MCSDs) can be used to provide *temporary* ventricular assistance, with the assumption that ventricular function will recover, or a suitable heart donor is identified within days to weeks. Durable mechanical devices can be used for prolonged (months to years) circulatory support with the intent to bridge-to-transplant, bridge to recovery (sufficient ventricular recovery to permit device explant), or provide permanent or lifetime (destination) therapy. The current distinction that separates temporary MCSDs from durable MCSDs is the ability to permit safe patient discharge from the hospital.

Mechanical pumping mechanisms can be placed internally (*implantable*) or external to the body (*paracorporeal* or *extracorporeal*). Typically, short-term devices are extracorporeal or paracorporeal pumps while durable devices are implantable (intracorporeal) systems. Power sources for devices can be *electric* or *pneumatic*, located outside the body or completely within it, with electric power conducted transcutaneously (totally implantable pump systems) or through a percutaneous cable. Pump flow characteristics can be *pulsatile* or *continuous* flow.

This chapter does not include use of biological means for MCS nor passive (non-energy imparting) devices used primarily to address ventricular remodeling. This chapter also does not include short-term *temporary* devices and focuses on long-term durable MCSDs.

Historical Note

Early descriptions of mechanical support of the human circulation date back at least to the early 19th century in the writings of Le Gallois,¹ Carrel and Lindberg,² as well as Demikhov,³ who reported experimental application of MCS systems in animal models in the 1930s. However, major interest in mechanical support of human circulation would await the dawn of open-heart surgery in the 1950s.

With successful application of cardiopulmonary bypass (CPB) for a cardiac surgical operation by Gibbon in 1953 and the subsequent first successful series of cardiac operations using CPB by Kirklin and colleagues at the Mayo Clinic,⁴ the stage was set for rapid proliferation of the technology of temporary MCS (in this case, the heart-lung machine) for repair of cardiac malformations. Failure to successfully wean some patients from CPB stimulated surgeons to seek additional methods of MCS while awaiting myocardial recovery. Roller-pump technology⁵ was complicated by trauma to blood elements and difficulty in modulating pump speed in response to fluctuations in atrial filling pressures. The first application of a true ventricular assist device (VAD) was attributed to Michael DeBakey, who in 1966 reported the successful application of a pneumatically driven diaphragm pump for 10 days in a 37-year-old woman unable to be weaned from CPB following aortic and mitral valve replacements. Cooley subsequently reported the first successful bridge to cardiac transplantation in a 47-year-old man for 64 hours while awaiting heart transplantation with a pneumatically driven artificial heart, the Liotta Heart,⁶ developed by the DeBakey-Baylor-Rice research team.

MCS research focused on pneumatic, electric, and even nuclear-powered designs through the 1980s.⁷⁻⁹ The experimental work of Kolff, Olsen, Jarvik, and others¹⁰⁻¹³ paved the way for the first permanent total artificial heart (TAH) implant in Dr. Barney

Clark by DeVries and his team in 1982.^{14,15} Five patients received permanent TAHs under a US Food and Drug Administration (FDA) protocol, with a maximum reported survival of 620 days. The close scrutiny of this initial trial of the Jarvik-7 TAH stimulated intense and sometimes acrimonious debate among ethicists, economists, and healthcare experts about the application of expensive and human-intensive technologies in end-of-life situations.¹⁶⁻¹⁸

Just as open-heart surgery using CPB paved the way for early application of MCS, cardiac transplantation provided the stimulus for proliferation of ventricular assist systems as a bridging therapy to transplantation. With nearly 30% of patients dying while awaiting cardiac transplantation in the early 1980s, a clear need developed for effective and durable MCSs that could safely support patients until suitable donor hearts could be identified. The improving outcomes following cardiac transplantation and the scarcity of available organs provided the impetus for a major collaborative effort among the heart transplantation community, the National Institutes of Health (NIH), and scientists and clinicians dedicated to the development of durable MCS systems.¹⁹ In 1984, Oyer, Portner, and colleagues²⁰ reported the first successful cardiac transplant following bridging with a Novacor (WorldHeart Corp., Oakland, Calif.) left VAD (LVAD). Hill and colleagues subsequently published successful transplantation following support with a Pierce-Donachy pneumatic LVAD.²¹ About the same time (1985), Copeland and colleagues performed the first planned TAH implant as a bridge to transplantation. Continuing advancements in MCS over the ensuing decade led to the first FDA-approved implantable device as a bridge to transplantation in 1994.

Despite the early focus on MCS as a bridging therapy to transplantation, the clear intent of the scientific and engineering community was the development of devices capable of long-term safe circulatory support. The landmark feasibility study of long-term MCS was the Randomized Evaluation of Mechanical Assistance for Treatment of Congestive Heart Failure (REMATCH) trial. As reported by Rose and colleagues in 2001,²² the HeartMate VE VAD provided significant survival benefit at 1 and 2 years compared with medical therapy for patients with very advanced heart failure (HF) who were not eligible for cardiac transplantation. This NIH-sponsored multi-institutional trial provided the impetus for FDA approval of this device for destination therapy in 2002. This set the stage for multiple clinical trials in the application of long-term MCS.

In recent years, device technology has increasingly focused on smaller, simpler, and more durable continuous flow (CF; rotary) pumps that lack the pulsatile characteristics of earlier pumps. Following the earlier experimental work of Saxton and Andrews,²³ Wampler,²⁴ DeBakey,¹¹ and others,²⁵⁻²⁸ recent clinical applications have focused on CF pumps with centrifugal design.²⁹

Biological Barriers

General Concepts

A *biomaterial* is a natural or artificial material that remains in contact with one or more internal components of the human body for the purpose of replacing organ function or treating an abnormal condition. *Biocompatibility* refers to the effect of a specific biomaterial on exposed host tissues, whereas *hemocompatibility* refers to the specific effects of a biomaterial or circulatory support system on blood components, coagulation cascade, and

the tendency for thrombus formation. Successful blood pump design requires special knowledge of microlevel interactions between blood elements and the contact surface as well as macrolevel considerations that include choice of prosthetic valves (if required), inflow and outflow port design, and blood flow pathways within the pump.³⁰ The ideal biocompatible surface for blood is functioning endothelium, but the creation of a functioning endothelial layer on a bioprosthetic surface remains elusive. Minimizing thrombogenicity requires avoidance of highly thrombogenic biomaterials, specific features of pump design, and pharmacologic inhibition of the coagulation cascade. The requirement for blood-exposed pump components that minimize thrombogenicity has limited compatible materials for blood-exposed surfaces to titanium, polymers (primarily polyurethanes), silicone, graphite, and pyrolytic carbon.³¹

A fundamental concept for the understanding of blood-pump surface interaction is the process of protein adsorption to biomaterial surfaces.³² Following exposure of pump surfaces to circulating blood in vivo, a protein layer develops that covers the biomaterial surface. The make-up of this protein layer is determined by the protein composition of the patient's blood, the chemical composition of the biomaterial surface (more specifically, surface charge and hydrophobicity), and surface topography (rough vs. smooth surface, porous vs. nonporous).³⁰ Concentration of proteins in blood, net protein charge relative to the biomaterial surface, distribution of charges on the protein surface, and ability of the protein to undergo conformational changes all contribute to the propensity for a given protein to adsorb to the pump surface.^{33,34} Protein interactions with the biomaterial vary over time and are therefore dynamic. The change in composition of proteins that adsorb to the pump surface over time is termed the *Vroman effect*.³⁵ The specific details of these protein-surface interactions contribute directly to the likelihood of pump thrombogenicity, because these proteins are biologically active and can initiate platelet adhesion and activation and trigger coagulation cascades.

Both smooth and rough surface designs have been used successfully in pump design. The textured titanium surface of the HeartMate series of LVADs stimulates the formation of a thin, stable coagulum that, although counterintuitive, has proven effective in minimizing development of pump thrombus.^{22,29,30}

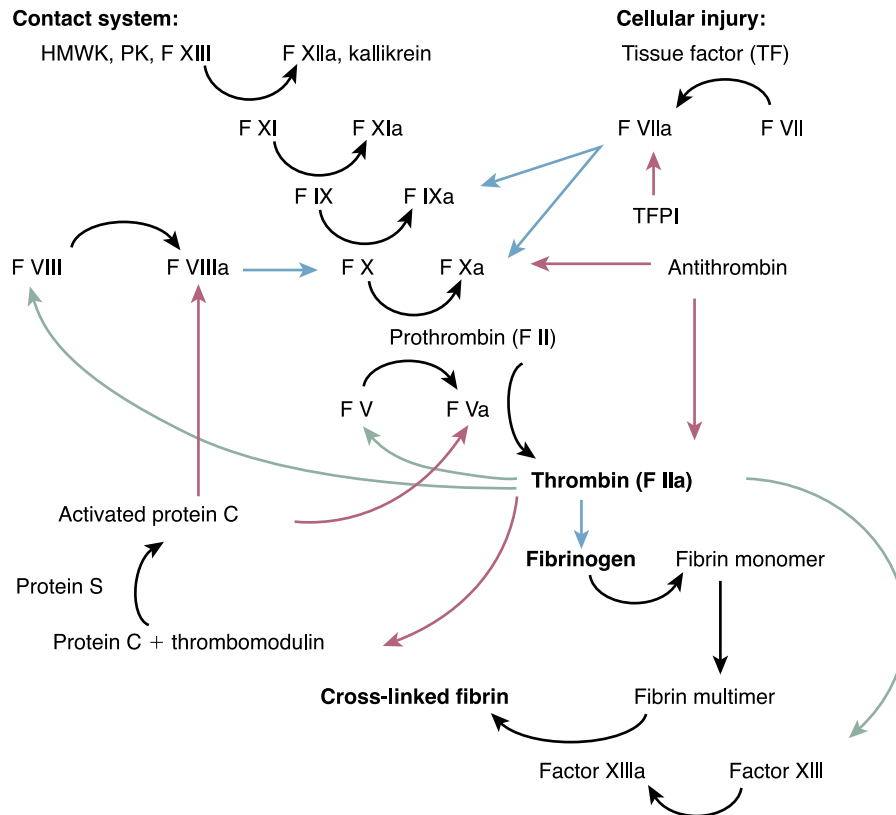
Application of computer simulations called *computational fluid dynamics* (CFD) analyses has greatly facilitated the ability to predict the effects of shear stresses in the pump flow pathway and areas of relative stasis on platelet activation and thrombus formation.

Coagulation System

Contact between the pump surfaces and specific plasma proteins, including factor XII (Hageman factor), prekallikrein, and factor XI, can initiate the coagulation cascade via the *intrinsic clotting system* (also called the *contact system*) particularly in areas of relative blood stagnation (Fig. 20.1).

The additional critical component of thrombus formation is *platelet adhesion, aggregation, and activation*. Normal endothelium is antithrombogenic, in part related to active biochemical reactions involving nitric oxide and prostacyclins. In areas of low shear rates, fibronectin functions as the adhesive system for platelets, whereas von Willebrand factor (VWF; factor VIII) is the active adhesive protein for platelets at high shear rates,³⁶ particularly in combination with fibrinogen.

Adhesion of platelets to vascular subendothelium is facilitated by VWF, which forms a bridge between collagen fibrils in



• **Figure 20.1** The coagulation cascade. Intrinsic clotting system (also called *contact system*) is shown on left; extrinsic clotting system (cellular injury) is shown on right. Red arrows indicate inhibitory influences on coagulation cascades. *F*, Factor; *HMWK*, high-molecular-weight kininogen; *PK*, prekallikrein. (From Holman WL, Teitel ER, Itescu S. *Biologic barriers to mechanical circulatory support*. In: Frazier OH, Kirklin JK, eds. *ISHLT Monograph Series: Mechanical Circulatory Support*. Philadelphia: Elsevier; 2006:9.)

the vessel wall and platelet receptors. Following contact activation, platelets undergo changes in shape in which they display spreading pseudopods and release the contents of their α granules (which contain fibrinogen, fibronectin, thrombospondin, VWF, β -thromboglobulin, platelet factor IV, and platelet-derived growth factor), which stimulate thrombin generation. Platelet aggregation is stimulated by the release of adenosine diphosphate (ADP). Under laminar flow conditions, the released granule contents cannot accumulate, but when flow is nonlaminar, platelets are more likely to aggregate and accumulate. As newly recruited platelets release their granular contents, thrombin formation is facilitated by the generation of fibrinogen/fibrin bridges. Increasing thrombin generation accelerates the formation of thromboxane A_2 and the release of ADP, which further promotes conversion of fibrinogen to fibrin and platelet activation and aggregation. Damaged red cells from shear stress-induced hemolysis also release ADP, further perpetuating platelet activation.

Preserved vascular endothelial cell function is critical for prevention of intravascular thrombus. Synthesis and release of prostacyclins inhibit platelet activation. Thrombomodulin, an endothelial cell product, neutralizes the procoagulant properties of thrombin and activates protein C, a potent anticoagulant that destroys factors Va and VIIIa. Antithrombin III binds to the endothelial cell plasma membrane and inactivates thrombin.

Engineering Concepts in Pump Design

General Concepts

Specific technologic barriers challenging successful MCS include development of corrosion-resistant materials with minimal toxicity and a high level of structural integrity, management of specific blood-contacting surfaces to minimize thrombogenicity and damage to blood elements, blood pump design, and methods to store energy.

Energy to generate flow from circulatory assist pumps requires conversion of either electrical or pneumatic energy (compressed gas) into kinetic energy (energy of motion). Pulsatile or volume-displacement pumps are usually driven by electric motors that either transfer power directly to a pusher plate mechanism or compress gas or liquid in order to transfer energy to the blood sack or pusher plate. CF (rotary) pumps, currently used in clinical practice, use electric motors to transmit kinetic energy to the blood.

The principles of Starling's law also apply to circulatory pumps, in that the pump must respond to higher inflow into the pump by increasing output. As in the natural heart, this balance is maintained in pulsatile pumps by variations in stroke volume or pump rate and to some degree with CF pumps by changing the pressure difference across the pump (referred to as ΔP) by changes in preload and afterload.

The major cause of hemolysis in blood pumps is rapid acceleration or deceleration of red cells through the pump, which can induce red cell membrane fracture. In general, pump-induced hemolysis is considered acceptable if the plasma free hemoglobin is maintained at less than 19 mg/dL. The rate of pressure increases and flow-channel velocities are maintained at levels designed to avoid high shear stresses within the blood pathways within an MCS.

Proper application of fluid dynamics is critical to minimize thrombus formation. Because blood stasis, particularly flow cessation promotes clot formation, stationary vortex flow must be avoided because the central stagnant portion of the vortex can become a nidus for thrombus formation.

Power sources and alarms must provide reliability and durability backed up by software programs designed to activate appropriate alarm systems when deviations from normal function occur. Approximately 1.6 watts of power are needed to pump 6 L/min at 120 mmHg. Power in excess of 1.6 watts is both wasted and converted to heat that must safely dissipate within the body.

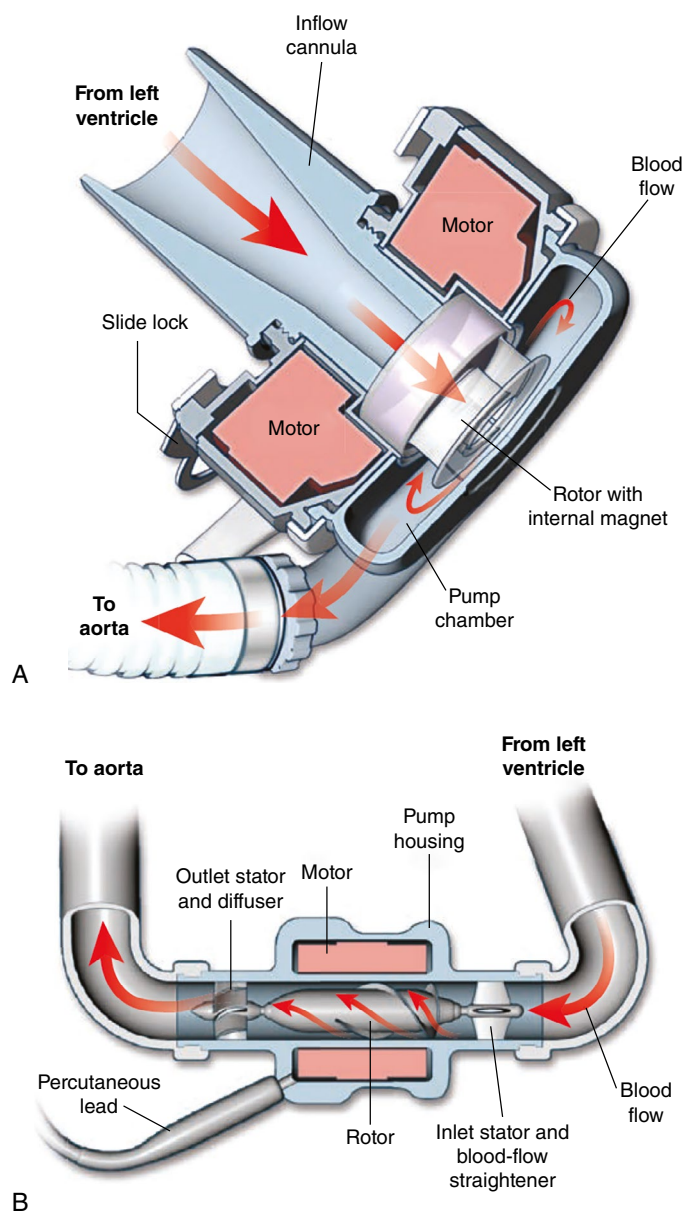
Pulsatile (Volume Displacement) Versus Continuous Flow (Rotary) Pumps

MCS designs have evolved considerably over time. Early technology utilized pulsatile (volume displacement) designs that were replaced by more modern CF (rotary) pumps, which include axial flow and centrifugal pump designs. Despite the controversies surrounding the issues of maintaining pulsatile or nonpulsatile circulation, the field of MCS has evolved from the use of volume displacement pulsatile pumps to CF rotary pumps. The evolution to CF rotary pumps has been associated with major improvements in device reliability resulting in major improvements in patient survival and reduction in adverse events.³⁷ Rotary pumps designs included both axial and centrifugal pumps, with rotary pumps with centrifugal flow design and with total magnetic levitation of the internal rotor demonstrating greater safety compared to other rotary pump designs.^{29,38} Currently, the use of pulsatile pumps and early generation of rotary pumps with axial flow design has been completely abandoned in adult patients, and these pumps are of historical interest, having been completely replaced by contemporary rotary pumps with centrifugal flow and total magnetic levitation of the internal rotor.²⁹

A CF rotary pump consists of blood inlet and outlet ports and a single internal rotating element (i.e., rotor or impeller) that is suspended within a pump housing that propels blood forward by spinning the impeller at high speeds, imparting significant kinetic energy to the blood that overcomes outflow resistance to the pump. The spinning of the impeller is accomplished by sequentially actuating an electrical current and creating a magnetic field that is coupled to the internal magnets within the impeller.^{39,40} CF rotary pumps in clinical use today are powered by a brushless DC motor and currently require an external power source (most often provided by rechargeable batteries or an AC power cord) that is transmitted to the internal pump through a percutaneous cable or driveline. In the future, wireless energy transfer systems should eliminate the need for the percutaneous lead to power implantable pumps.

Continuous Flow Rotary Pump Design: Axial Versus Centrifugal Pumps

There are essentially two types of CF rotary pumps in clinical use today: a centrifugal flow pump and axial flow pump (Fig. 20.2).



• **Figure 20.2** (A) Diagram of a centrifugal blood pump where blood enters the pump inlet cannula along the axis of rotation or centerline of the impeller and is driven outward, tangentially to the outlet of the pump. (B) Diagram of an axial flow pump. Blood enters at the inlet end of the rotor and is driven along the axis of rotation or centerline of the rotor to the outlet end of the pump. (From Mehra MR, Naka Y, Uriel N, et al. A fully magnetically levitated circulatory pump for advanced heart failure. *N Eng J Med*. 2017;376:440-450.)

The primary difference between centrifugal flow and axial flow pumps lies in the design of their rotating element. In a centrifugal pump, the outlet path is positioned 90 degrees relative to the axis of rotation or centerline of the impeller and the rotating element acts as a spinning disk with blades that capture fluid and propel the blood from impeller blades to the outflow cannula along a tangential course. In an axial flow pump, the inlet and outlet blood paths are positioned parallel relative to the axis of rotation or centerline of the impeller and the rotating element operates like a propeller in a pipe that pushes fluid forward. In both cases, blood exits opposite to the direction of thrust generated by the pump motor.

The rate at which a pump adds energy to a fluid is⁴¹:

$$\dot{W} = \frac{dV}{dt} \times \Delta P = Q \times \Delta P$$

where $Q = \frac{dV}{dt}$ is the flow and ΔP is difference in pressure between inlet and outlet orifices of the pump. The efficiency of a pump is defined as the ratio of the useful power output to the required power input:

$$\eta_{\text{pump}} = \frac{Q \times \Delta P}{\dot{W}_{\text{input}}}$$

Centrifugal devices generally have greater hydraulic efficiency at energy transfer and provide CF at rotational speeds that are much slower, approximately 2000 to 6000 rpm compared with 8000 to 15,000 rpm for pumps with axial flow designs.^{39–41} This generally means that a lower shear stress is exerted on the blood elements and that there is a lower risk of hemolysis.⁴² The hydraulic efficiency, defined earlier as the ratio of power imparted to the fluid divided by the power input to the impeller, is related to the ability of a pump to transport fluid with minimal power loss over the blood flow path. Hydraulic efficiency is an important, but not sole, determinant of overall LVAD system efficiency. LVAD systems require power supplies and controllers that have unique methods of operating the pump. It is the sum of the efficiency of the motors and controllers and the hydraulic efficiency that contribute to the overall system efficiency.

Bearing Design/Impeller Suspension

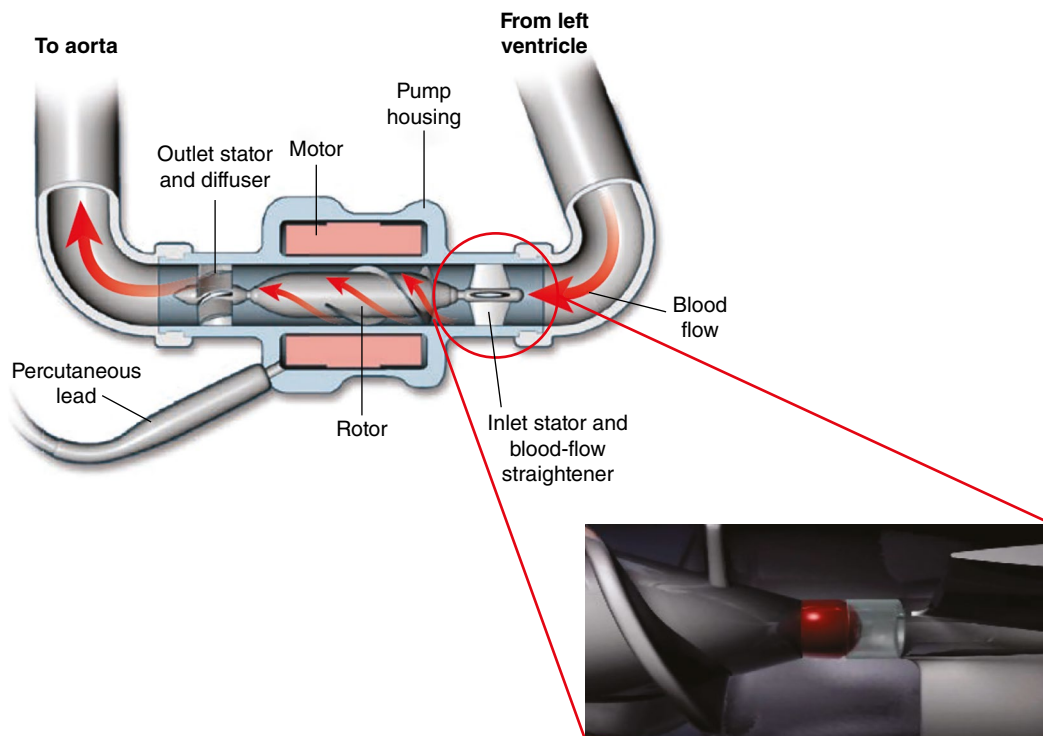
CF rotary pump designs can be further distinguished by the mechanism of impeller suspension or levitation and include use of: (1) mechanical bearings (Fig. 20.3); (2) hydrodynamic bearings (fluid forces); (3) hydrodynamic bearings working in synergy with magnetic suspension (Fig. 20.4); or (4) variations of active and/or passive magnetic suspension (Fig. 20.5).⁴³

Mechanical Bearing

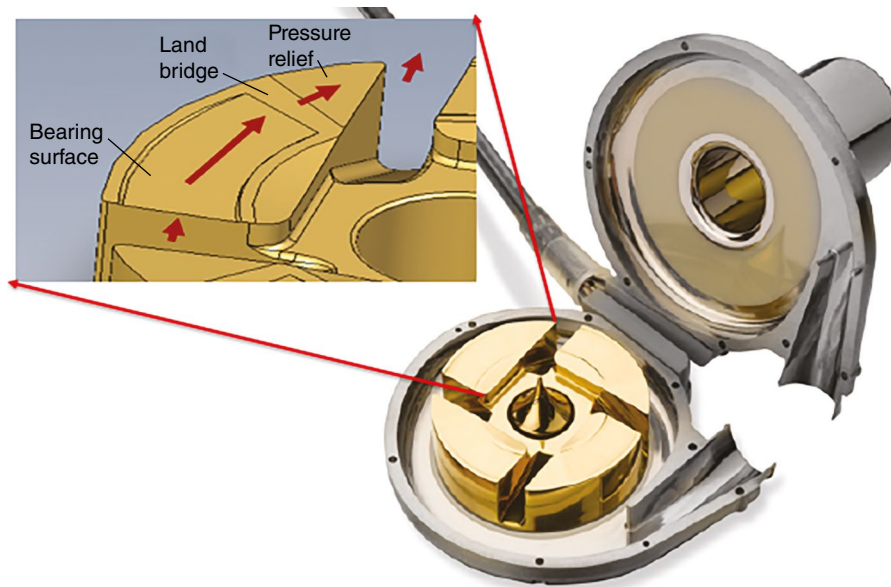
A mechanical pivot design or contact bearing utilizes mechanical bearings on spherical surfaces rotating in sockets (see Fig. 20.3). These bearings are made of ceramic components with low friction coefficient (i.e., ceramic or ruby) and have long durability attributed to, in part, boundary lubrication from blood immersion.^{43,44} Although simplistic in design, a major limitation of this mechanism of impeller support is that the point of contact of the mechanical bearings is a point of friction and heat generation that represent potential sites for fibrin deposition with thrombus formation.⁴⁵ Washing of these points by blood or fluid is necessary to provide adequate dissipation of heat. In addition, the concentration of hydrodynamic loads on these bearings, especially at stress concentration points, makes these contact points theoretically susceptible to wear and fatigue.

Noncontact Bearing Designs

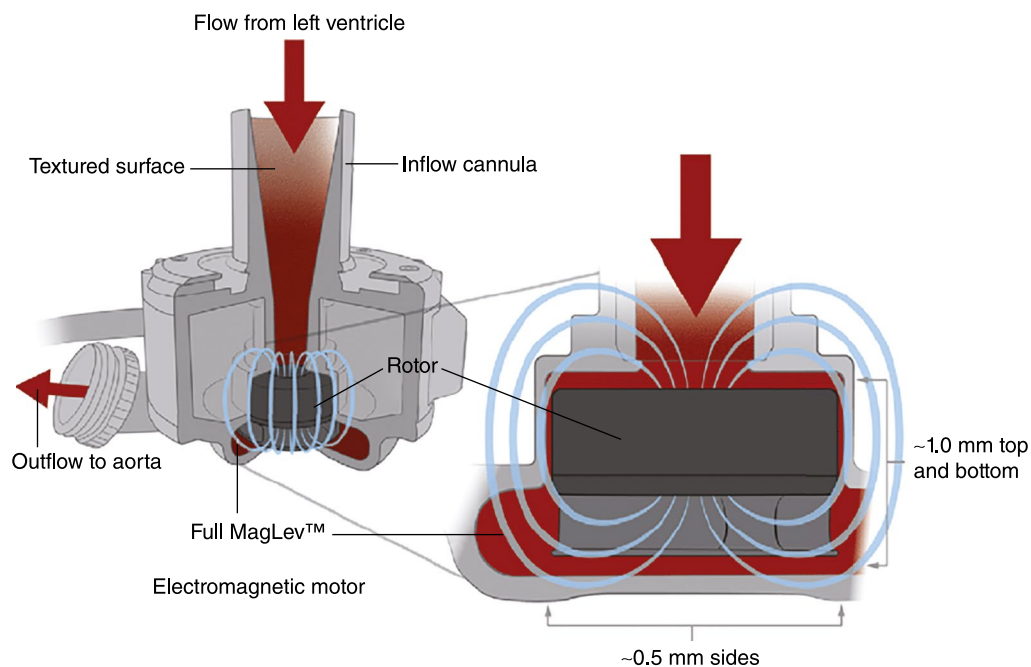
Noncontact bearing designs incorporate either hydrodynamic or magnetic forces or a combination of both to achieve suspension



• **Figure 20.3** Diagram of a continuous flow rotary pump (axial pump) using a mechanical bearing or pivot design to suspend the internal rotor. Insert at lower right-hand corner of the figure demonstrates a “ball and socket” mechanical bearing design. The surfaces of the bearing are interspersed with a thin film of fluid to reduce friction and wear of the bearing. (Reproduced with permission of Abbott. © 2025 Abbott. All rights reserved.)



• **Figure 20.4** Picture of a continuous flow rotary pump (HVAD; Medtronic Inc., Minneapolis, MN) utilizing a combination of magnetic and hydrodynamic suspension of the internal rotor. The insert in the upper left-hand corner of the picture details the hydrodynamic surface of the impeller that utilizes fluid forces to oppose magnetic forces generated from opposing magnets within the center post and impeller. (With permission from Medtronic Inc., Minneapolis, MN.)



• **Figure 20.5** Diagram of a continuous flow rotary pump with total magnetic levitation of the internal impeller. (Reproduced with permission of Abbott Laboratories. © 2025 Abbott. All rights reserved.)

of the internal impeller without the use of mechanical contact supports.^{39,40,46} Without the need for mechanical contacts, these suspension systems do not have points of friction and theoretically have an infinite or limitless lifespan. These levitation systems must control for six degrees of freedom of the impeller: three rotational movements (rotation around “x”, “y” and “z” axes) and three translational movements (displacement without rotation). Rotation around the axis of

rotation or centerline of the pump is achieved by electromagnetic coupling of the motor stator and internal magnet of the impeller.

Hydrodynamic Bearing

Hydrodynamic bearings utilize pressure generated from a fluid film that acts to separate the rotating surfaces of the impeller

from the stationary base of the pump (see Fig. 20.4).^{39,40} However, in order to generate a hydrodynamic force, the surface of the impeller must be moving and when the impeller is stationary (e.g., at pump startup or stop, impeller and stationary surfaces, such as the pump housing, will come into contact that could result in damage to the surfaces of the impeller or pump housing, resulting in potential niduses for thrombus formation). Contact can also occur if rotating impeller speeds are not sufficient to overcome outflow pressures resulting in impeller contact with the housing or, in the absence of contact, cause a significant decrease in the distance between impeller and pump housing resulting in increased shear and friction of blood elements passing between impeller and pump housing. Importantly, hydrodynamic forces dissipate quickly as a function of distance, thus this limits the distances between the stationary base (pump housing) and rotating impeller. As a result, the load-bearing fluid film is prone to higher shear stresses, that theoretically can result in more damage to blood elements with hemolysis. Hydrodynamic suspension systems do not utilize position sensors resulting in a less complicated electronic design that enhances the ability to miniaturize pump sizes.

Magnet Bearing

Magnetic forces may be passive without the consumption of power (permanent magnet) or active (induction of magnetic field with electricity) in design (see Fig. 20.5).^{39,40} Passive magnets utilize rare earth magnets such as neodymium boron-iron magnets within the rotor and along the housing to achieve suspension of the impeller. Permanent magnet bearings allow for a larger gap between the static motor armature and the rotor, while eliminating the need for mechanical bearings, lubrication, sealing or purging fluid. However, the magnetic forces are dependent on the instantaneous position of the rotor (i.e., magnetic forces are greater when magnets—impeller and pump housing—are at closer distances). Magnetic bearings are often used in conjunction with hydrodynamic or electromagnetic bearings. Control of any axis can be active or passive. In active mode, the repelling force is adjusted by changing the supplied current to the electromagnets based on the instantaneous position or other feedback. In a passive mode, the supplied current to the electromagnets is constant. The electromagnetic force adjusts automatically based on the position itself (i.e., as the impeller approaches the magnet in the pump housing, the repelling force increases). However, electromagnetic systems are complex, requiring position sensors, electromagnets and extra conductors, connector pins, and electronics to execute the dedicated position control algorithm. If, for example, there is an electrical contact failure in a connector pin, or momentary instability encountered in the control algorithm, pump failure can occur. To make the system fail-safe, electromagnetic bearing elements frequently utilize hydrodynamic bearings as a fail-safe, which require no control. Distances between impeller and pump housing maintained by magnetic fields are a function of the strength of the magnetic field and hence, a function of the size of the magnet. Thus, overall pump size and degree of miniaturization are a function of the size of the magnet needed to maintain the desired gap between impeller and pump housing. Overall efficiencies of the pump are also determined, in part, by magnet strength. The major advantages of magnetic bearings are that distances between impeller and pump housing are generally larger and the degree of shear stresses on blood elements is less compared to mechanical contact bearings or hydrodynamic suspension systems.

Hydrodynamic Performance of Continuous Flow Pumps

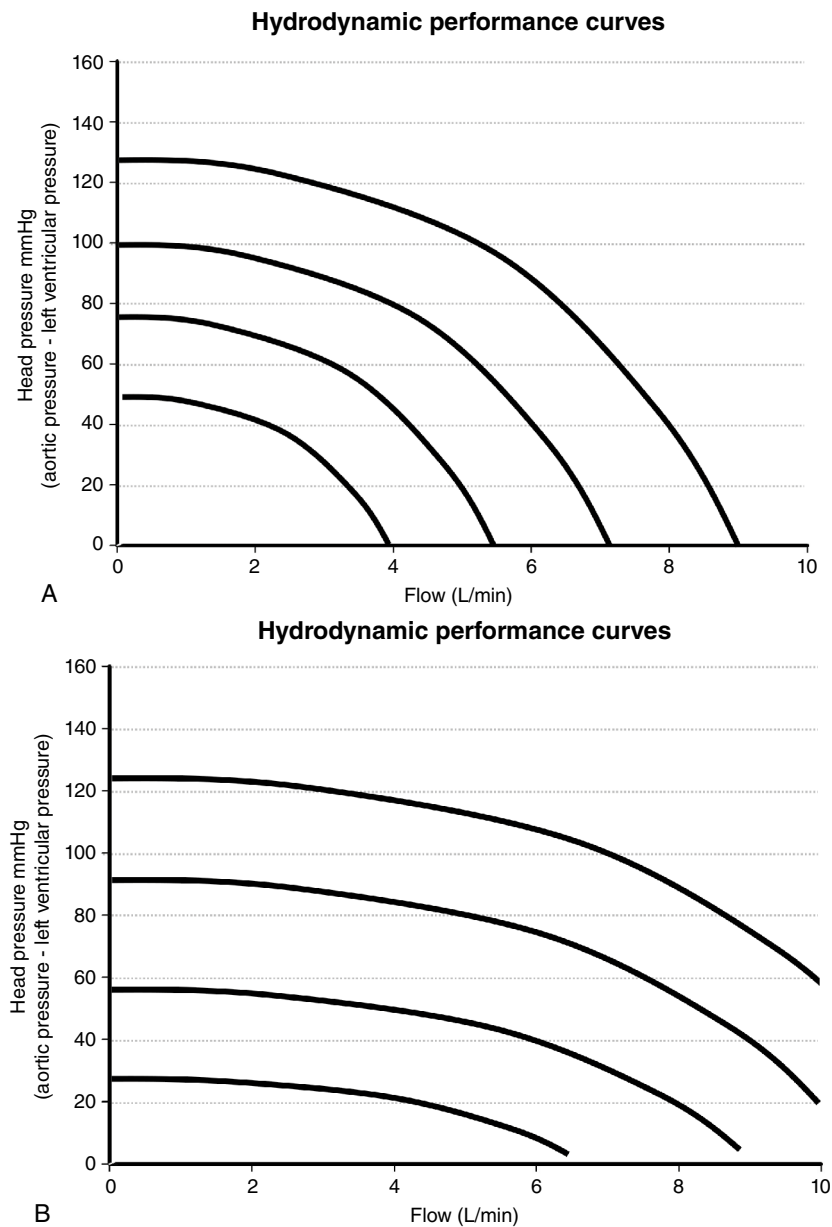
Blood flow through CF pumps, both axial and centrifugal, is directly proportional to pump speed and inversely proportional to the pressure difference across the inlet (e.g., left ventricular [LV] pressure) and outlet (e.g., aortic pressure) orifices of the pump.^{39,40} This is referred to as the “ ΔP ” or “head pressure” and represents the work performed by the pump on the blood. A combination of torque and velocity allows the impeller to transfer energy to the blood and generate flow and pressure to overcome head pressure.

The relationship between ΔP and flow can be displayed in a series of curves reflecting blood flow over varying pressure gradients at differing pump speeds (i.e., the pressure-flow relationship or “head curve”) (Figs. 20.6 and 20.7). This relationship between ΔP and flow is unique to each continuous rotary pump, analogous to an individual’s fingerprint. Generally speaking, most axial flow rotary pumps in clinical use to date have been designed with steeper pressure-flow relationships compared to centrifugal flow rotary pumps, but the generalizations do not hold for every pump design. A steep relationship between ΔP and flow is often represented by an almost linear relationship. A shallow or “flat” head curve represents greater sensitivity between the pressure-flow relationship such that small changes in pressure elicit greater changes in flow with any given change in pressure.

For a CF rotary pump with a flat head curve, the same ΔP will elicit a larger change in flow during systole relative to diastole compared to a CF rotary pump with a steep head curve. There are several important advantages of a flat head curve compared to a steep head curve. The more responsive pressure-flow relationship in CF rotary pumps with a flat head curve results in a greater degree of flow variability across the cardiac cycle (less flow in diastole and more flow in systole). In theory, the benefits of a rotary pump with a flat head curve versus that with a steep head curve are a greater aortic pulse pressure. Further, as ΔP increases in settings of lower preload, flow in a rotary pump with a flat head curve will decrease substantially more than a rotary pump with a steep head curve. Thus, there is theoretically less propensity to create ventricular collapse or a “suction event” resulting from the greater drop in pump flow as LV diastolic pressures decrease. Ventricular collapse or “suction” events elicited by a CF rotary pump may cause ventricular arrhythmias or increase the propensity for thrombus formation on bearings as a result of low flow or on the endocardium owing to trauma or increase the risk of damage to blood elements through generation of high negative pressures.

Interaction of the Continuous Flow Rotary Pump and Native Heart

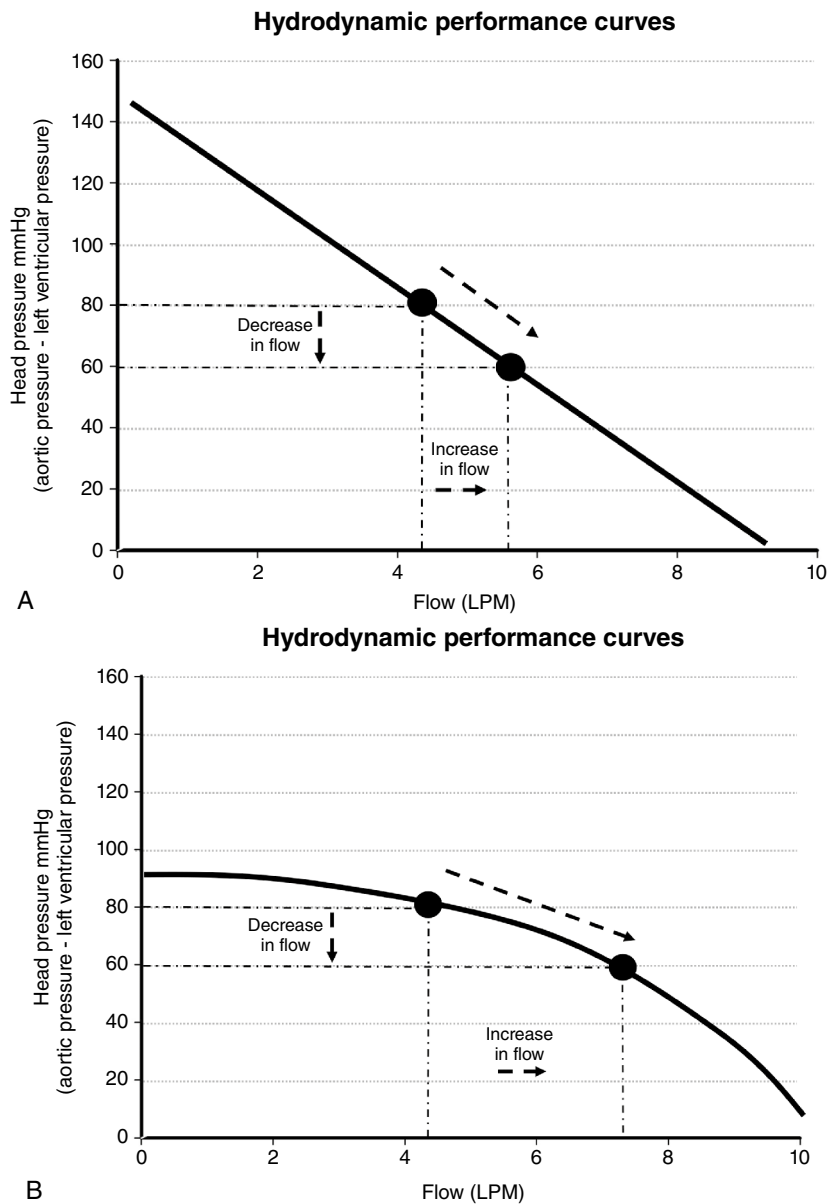
CF rotary pumps do not generate pulsation. However, CF rotary pumps operate in conjunction with the native ventricles that influence both hemodynamics and mechanical interactions with changes in ventricular function and systemic pulsatility. Flow through a CF rotary pump is continuous throughout the cardiac cycle but has superimposed phasic changes in pump flow caused by the interaction of the heart with the CF rotary pump.^{39,40,43} Depending on hemodynamic conditions, contractile state of the left ventricle, pump speed and hydrodynamic properties of the pump, flow through a CF rotary pump may be more or less pulsatile. The hydrodynamic properties of the steepness of the head curve have important effects on the flow pattern of the CF rotary pump and its



• **Figure 20.6** (A) Diagram of a pressure-flow relationship or “head curve” for an axial flow rotary blood pump. The relationship between pressure and flow is nearly linear and represents a blood pump with a “steep” pressure-flow relationship. Each line on the graph represents differing pump speeds. (B) Diagram of a pressure-flow relationship or “head curve” for a centrifugal flow rotary blood pump. The relationship between pressure and flow is flat and represents a blood pump with greater sensitivity to changes in pressure. Each line on the graph represents differing pump speeds.

interaction with the native cardiac contraction. During the cardiac cycle, pump flow is greater during native cardiac systole because the native LV contraction raises intracardiac pressure, thereby lowering the ΔP (e.g., the difference between aortic and LV pressures) the pump must overcome to generate forward flow. During diastole, the difference between LV pressure and aortic pressure is increased (i.e., ΔP is greater) and thus pump flow is less relative to systole (Fig. 20.8). These phasic changes in blood flow with a CF rotary pump impart a pulse to the native circulation. The magnitude of pulse pressure typically is diminished compared with the pulse pressure that is generated with a native heart contraction or a pulsatile flow pump. The magnitude of a pulse generated with a CF rotary pump is dependent on the relative differences in pressure changes during the cardiac cycle and contributions of the native heart to the

total cardiac output. Thus, conditions that create greater LV pressure generation (i.e., greater preload [Frank-Starling mechanism]; inotrope therapy; native LV recovery) will create a greater pulse pressure. Under normal circumstances (i.e., pump working in conjunction with the native heart contraction), the aortic flow pattern with a CF rotary pump is more accurately described as being continuous, rather than using the description of nonpulsatile flow. In circumstances in which there is absence of native heart contraction (e.g., ventricular fibrillation), the flow through a CF rotary pump is nonpulsatile. Importantly, CF rotary blood pumps do not contain unidirectional valves. In circumstances where afterload exceeds pressure generated by the pump and/or if pump speed is inadequate, backflow through the pump may occur during portions of the cardiac cycle or throughout the entire cardiac cycle.



• **Figure 20.7** (A) With a “steep” pressure-flow relationship, any given change in pressure is associated with a relatively small change in flow. (B) In a “flat” pressure-flow relationship, a small change in pressure is associated with a relatively larger change in flow. Pumps with a “flat” pressure-flow relationship are more sensitive to changes in preload and afterload conditions and, in general, impart a greater degree of pulsatility to the systemic circulation.

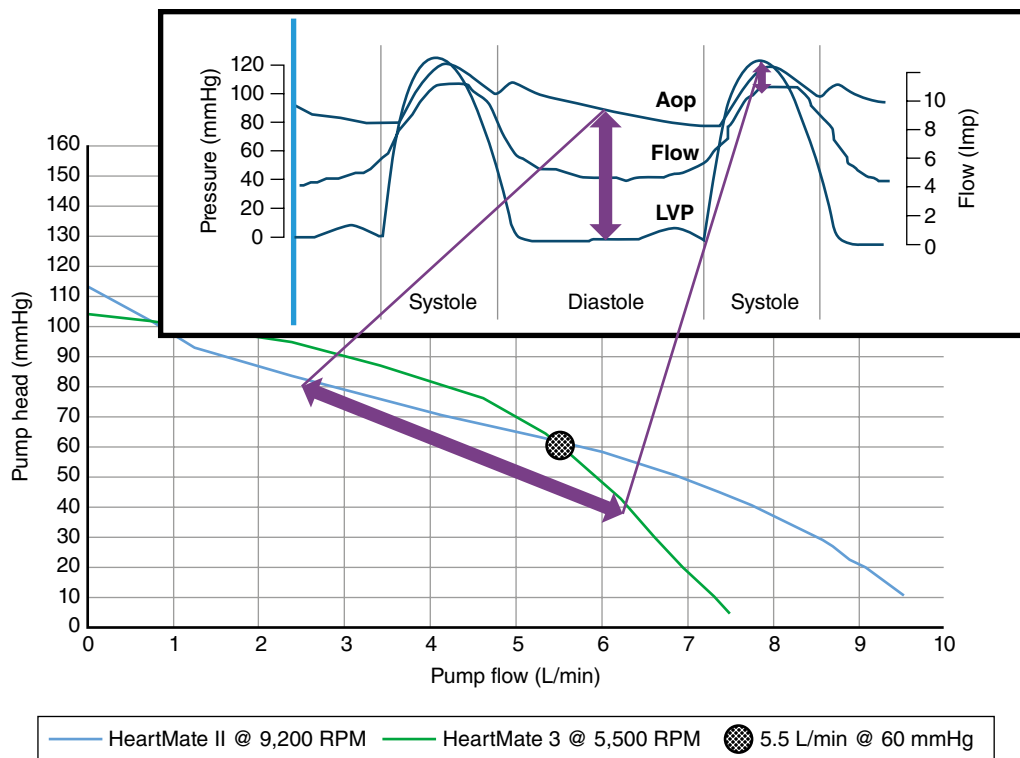
Parallel and Series Circulation

Depending on pump design and hemodynamic conditions, CF rotary pumps can either provide full support (pump responsible for all cardiac output) or partial support (pump supplies a portion of the total cardiac output). In circumstances of full pump support the circulation is described as in “series” (Fig. 20.9). Conversely, in circumstances of partial pump support the circulation is describe as in “parallel.” The presence of series or parallel circulations can vary during support. For example, a series circulation may exist early following pump implantation, but as native LV function recovers, a parallel circulation may be created. Changes in a patient’s preload caused by changes in volume status or activity could impart moment to moment cycling between series and

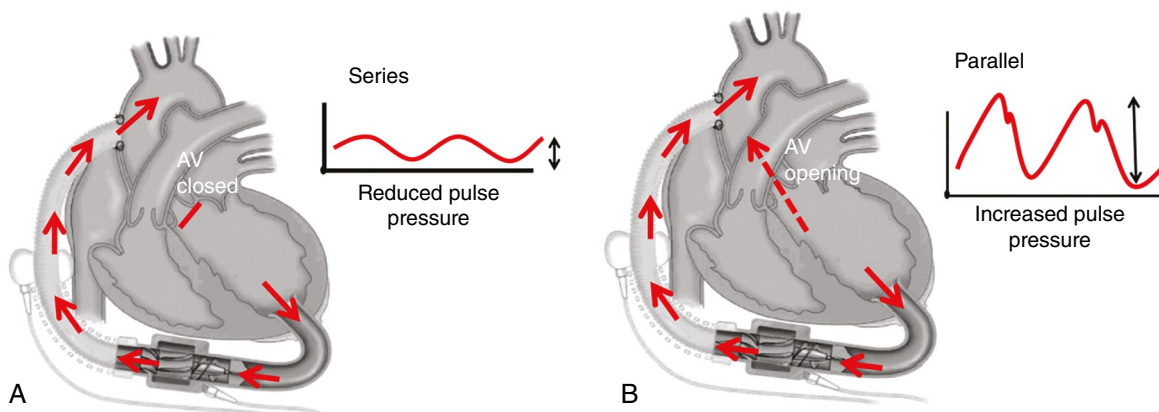
parallel circulations. With a series circulation, the aortic valve is continuously closed during the cardiac cycle and has been associated with development of physiologic important aortic insufficiency during pump support. Thus, there may be relative benefits of a parallel circulation compared to a series circulation.

Flow Estimation

Flow through a rotary blood pump is proportional to motor current or pump power thus permitting motor current to be used as a sensorless index of pump flow and virtual index of LV pressure during the cardiac cycle. However, the relationship between motor current and pump flow is not linear throughout the entire range of flows and the relationship between motor current and flow differs



• **Figure 20.8** Flow through a CF rotary pump varies with the cardiac cycle owing to the changes in head pressure during diastole and systole. During diastole, ΔP is greater and flow through the pump is less relative to systole where the difference in pressure between the left ventricle and aorta is less and ΔP is smaller. This flow variation across the cardiac cycle imparts a pulse to the systemic circulation. The size of the pulse imparted to the systemic circulation is dependent of the degree of flow variation during the cardiac cycle and dependent on contributions of the native heart to total cardiac output. The diagram shows two head curves, one for the HeartMate II pump (axial) and HeartMate 3 pump (centrifugal). (Reproduced with permission of Abbott. © 2025 Abbott. All rights reserved.)



• **Figure 20.9** (A) represents a circulation in “series” with the rotary pump. All cardiac output is delivered by the rotary pump and there is a small contribution of native LV ejection to flow variation. In this circumstance, the systemic pulse pressure is significantly reduced compared to the native circulation. (B) represents a circulation in “parallel” with the rotary blood pump. In this circumstance, both rotary pump and native heart contractions, with substantial changes in flow variation through the cardiac cycle, contribute to total cardiac output and a greater degree of pulsatility. AV, aortic valve.

between axial and centrifugal pumps. In addition, the point of power measurement may differ between different pump designs. Power output for centrifugal pumps tends to be more linear in the operating range of the pump compared to axial flow pumps, particularly at low flow rates. Thus estimation of pump flow for centrifugal pumps tend to be more accurate as compared to axial flow pumps.^{39,40,43} In addition, viscosity has important effects on pump flow and power, with increasing hematocrit increasing motor current at any given flow. Incorporation of the hematocrit into the flow estimation thus increases the accuracy of the flow estimate.

Limitations in Flow Control with Continuous Flow Rotary Pumps

LVAD support with CF rotary pumps is associated with adverse physiologic sequelae that, in large part, represent the limitations of the current design of clinical LVADs operating in a “fixed speed” mode algorithm that cause diminished arterial pulse pressure.⁴⁷⁻⁴⁹ Further, this operating algorithm requires frequent provider interaction and assessment with clinical, echocardiographic, or right heart catheterization data to determine appropriate pump speeds necessary for clinical optimization. Changes in pump flow in CF rotary pumps can occur without changes in pump speed (“fixed speed” mode; reliance on the Starling mechanism to alter pump output) or occur as a consequence of changes in pump speed based upon an algorithm-based adjustment from a signal input (physiologic flow adjustments). The absence of autonomous speed control in LVADs eliminates the potential for creating pulsatility matching that of the normal physiologic state or significantly hinders the ability of current LVADs to respond to widely varying changes in preload and afterload conditions. Although current clinical LVADs possess hydrodynamic properties (Starling mechanism) that permit changes in pump output to altering preload and afterload conditions, this response, under some circumstances, is not appropriate as the hydrodynamic properties of CF rotary pumps are less sensitive to changes in preload (i.e., small changes in pump output relative to large changes in preload) and more sensitive to changes in afterload (i.e., large changes in pump flow relative to small changes in afterload) as compared to the native heart.⁵⁰ Thus, more appropriate changes in pump output are limited without alteration of pump speed. This “uninformed” mode of speed operation limits CF rotary pumps from autonomously and appropriately responding to widely varying changes in loading conditions and precludes: (1) optimizing preload in the setting of increased physiologic demand; (2) maintenance of adequate pump flows in times of high afterload (i.e., hypertension); (3) maintenance of adequate LV volumes in times of low preload (i.e., prevention of “suction events”); and (4) appropriate flow balancing in applications with biventricular assist devices. Incorporation of physiologic pump speed adjustments based upon physiologic input (e.g., left atrial or LV pressure) to increase the safety of CF rotary pumps or to induce greater pulsatility to the systemic circulation has proved challenging. Although CF rotary pumps in use today do not have speed algorithms to adjust flow to physiologic demand, systems do have speed adjustment algorithms to respond to “suction events” that utilize power assessments and the relationship between pump power and flow.

Assessment of LV volume status during axial or centrifugal flow pump support is facilitated by echocardiography, in which ventricular filling, presence or absence of aortic valve opening, and any

tendency toward LV chamber collapse around the inflow cannula (suction event) can be assessed. Ideally, pump speed should be adjusted to permit intermittent aortic valve opening, which minimizes the risk of a suction event and may promote more effective washout of the sinuses of Valsalva, decreasing the likelihood of thrombus formation along the aortic valve. Further, intermittent aortic valve opening has been demonstrated to reduce the risk of late development of late aortic regurgitation that is due to pathologic changes of the aortic valve from residing in a continuously closed position (i.e., blood flow being completed directed through the mechanical pump). Optimal operating conditions to maximize exercise reserve and promotion of reverse remodeling remain controversial.

Durable Circulatory Support Devices

Pulsatile Ventricular Assist Devices

Durable pulsatile VAD designs have all been abandoned for use in the adult population and are of historical interest only. Previous implantable, durable pulsatile VADs designed for intracorporeal use included the HeartMate VE (Abbott Labs, Chicago, IL) and its updated version, the HeartMate XVE (Abbott Labs, Chicago, IL) device, the Novacor (WorldHeart, Ottawa Canada) device, and IVAD was unique in that the device had the capability to support both the right ventricle and left ventricle or either ventricle, separately. Durable VADs with paracorporeal design included the Thoratec pVAD (Abbott Labs, Chicago, IL) and Berlin Heart (Berlin Heart, Berlin, Germany).

Continuous Flow Ventricular Assist Devices

Numerous CF LVADs have been or are currently in clinical use, including such devices as the HeartMate II LVAS (Abbott Laboratories, Chicago, IL), Medtronic HVAD (Medtronic, Inc., Minneapolis, MN), EVAHEART®2 (EVAHEART, Bellaire, TX), Jarvik 2000 (Jarvik, Inc., New York, NY), BrioVAD (BrioHealth Solutions, Burlington, MA), Corheart 6 VAS (Shenzhen Medical Technology, China), and others. The current CF device with the widest global application is the HeartMate 3 (Abbott Laboratories), which will be described in detail in this section.

HeartMate 3

The HeartMate 3 LVAD (Abbott) is a compact intrapericardial-positioned CF rotary pump with centrifugal design.^{29,51} The unique feature of the HeartMate 3 is the full magnetic levitation of the internal rotor. The design differs from previous devices due to active control of rotation and levitation of the rotor that permits impeller movement farther away from the outer housing creating larger gaps for blood flow within the device. The larger gaps may reduce blood component trauma and improve hemocompatibility of the pump. The HeartMate 3 completed clinical evaluation in the Multicenter Study of MagLev Technology in Patients Undergoing Mechanical Circulatory Support Therapy with HeartMate 3 (MOMENTUM 3) clinical study.²⁹ The results of the MOMENTUM 3 clinical study demonstrated superiority of the HeartMate 3 over the HeartMate II and demonstrated the best overall survival ever reported in a clinical study of durable MCS (Fig. 20.10).^{29,38}

Total Artificial Heart

The major advantage of the TAH over VADs is the provision for biventricular support with an implantable device.⁵² The major disadvantage of the TAH is the necessity of removing the native



• **Figure 20.10** The HeartMate 3 LVAD. (Reproduced with permission of Abbott. © 2025 Abbott. All rights reserved.)

heart, which removes the option of recovery. The second major disadvantage of current iterations is the space requirement for the device, such that only patients with a body surface area of about 1.7 m² or greater are suitable.⁵²

The only TAH device currently available for clinical use is the SynCardia TAH-t (SynCardia Systems Inc., Tucson, Ariz.). The SynCardia TAH-t replaces the native ventricles and all four valves in the orthotopic position.

SynCardia Total Artificial Heart. The SynCardia TAH-t is a pneumatic pulsatile pump in which a rigid spherical outer housing around each artificial ventricle supports a seamless blood-contacting segmented polyurethane diaphragm, two intermediate diaphragms, and an air diaphragm (Fig. 20.11).⁵² Two Medtronic 27-mm inflow valves and two Medtronic 25-mm outflow valves provide unidirectional flow. The full ejection volume of each ventricle is 70 mL per beat, and the TAH typically generates a cardiac output of 7 to 8 L/min. The atrial components are sewn to the native atrial cuffs, and the atrioventricular and ventricular outflow graft connections are via “quick snap” connectors. The external console consists of one primary and one secondary pneumatic driver, air tanks, transport batteries, an alarm, and a computer monitoring system. Current portable pneumatic drivers and consoles allow out-of-hospital living with full ambulation.

Device Considerations. The HeartMate 3 is the only durable LVAD (dVLAD) currently available for use in the United States. Due to higher rates of adverse events, the HeartMate II, available for commercial distribution in the United States since 2008, is seldom used. The HeartWare HVAD, a centrifugal pump originally approved for commercial use in the United States for bridge to transplant in 2012 and for destination therapy in 2017, was removed from commercial distribution in June 2021 following reports of a device malfunction and higher rates of stroke and worse survival compared to the HeartMate 3.³⁸

Techniques of Operation

Intracorporeal Continuous Flow Left Ventricular Assist Device

Conventional CPB is recommended for most implants. In the setting of prior cardiac surgery, percutaneous catheters should be



• **Figure 20.11** The SynCardia TAH-t. (From Copeland JG, Smith RG, Arabia FA, et al. Cardiac replacement with a total artificial heart as a bridge to transplantation. *N Engl J Med.* 2004;351(9):859-867.)

placed in the femoral artery and vein, which facilitates rapid initiation of CPB in the event of bleeding or acute cardiac decompensation during sternotomy (see Chapter 5).

A pulmonary artery catheter is routinely advised to assess the response of pulmonary artery pressure to pharmacologic interventions and device implantation. Continuous measurement of pulmonary artery and central venous (or right atrial) pressures provides critical information in determining the need for inhaled nitric oxide and/or biventricular support to treat right HF (see Chapter 4).

Intraoperative transesophageal echocardiography (TEE) is necessary during MCS implants for assessment of biventricular function, evaluation of aortic regurgitation and presence of a patent foramen ovale, identification of intracardiac and aortic air bubbles during implantation and subsequent de-airing maneuvers, and assessment of inflow cannula position within the left ventricle.

Prior to sternotomy, formal discussions occur with anesthesia and echocardiography colleagues regarding the pulmonary artery and central venous pressures (CVP) and interventions to optimize right ventricular (RV) function, the presence and severity of aortic insufficiency and any need to address the aortic valve surgically, the presence or absence of intraventricular thrombus, and the presence of a patent foramen ovale (which if present has to be surgically closed to prevent postimplant right to left shunting with arterial desaturation).

The most common approach to implantation involves a median sternotomy incision (Table 20.1). The pericardium can be incised to the left of the midline to create a pericardial “flap” to ensure adequate coverage of the outflow graft during closure. Once incised longitudinally to the diaphragm, a “T” is created by incising the pericardium toward the right and then the left. At the apex, the pericardium is scored laterally several centimeters above the left phrenic nerve so as not to constrict the device in the

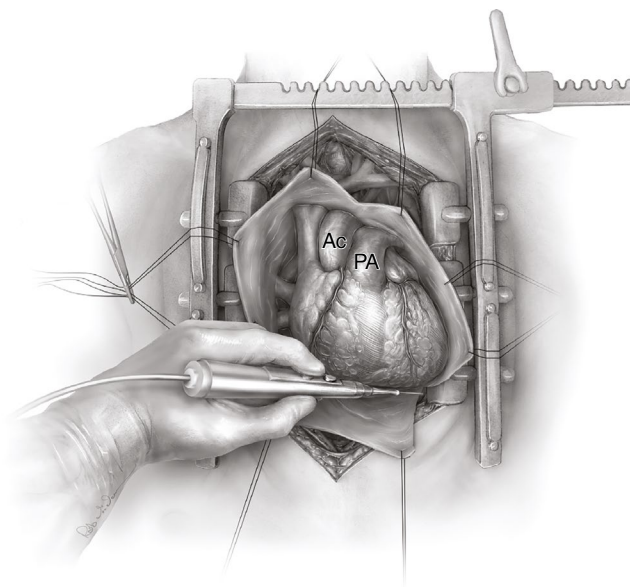
TABLE 20.1 Major Steps Involved with Placement of a Durable Implantable Left Ventricular Assist Device

- Identify and mark the exit site for the percutaneous lead followed by prepping and draping.
- Sternotomy (standard incision length).
- Incision of pericardium and creation of the pericardial cradle.
- Tunnel percutaneous lead.
- Administration of heparin and placement of arterial and venous cannulation stitches.
- Cannulation for CPB. Conformation of activated clotting time and initiation of CPB.
- Identify apex of the LV and sew apical cuff to apex of LV.
- Core muscle at LV apex and inspect LV cavity for thrombus or obstructing muscle bundles.
- Insertion of LVAD device inlet cannula into the LV cavity and secure device to the apical cuff.
- Size outflow graft and anastomose the outflow graft to the ascending aorta using a partial side-biting aortic clamp.
- De-airing of the outflow graft just prior to completion of the anastomosis. Placement of a clamp on the outflow graft and removal of the aortic side-biting clamp.
- Placement of an ascending aortic vent in ascending aorta distal to the outflow graft anastomosis.
- Resume ventilation and inotrope support, reduce CPB support to one-half total flow, and initiate de-airing procedures.
- The clamp on the outflow graft is removed and the LVAD is actuated at low RPM speeds and the de-airing process is completed.
- Wean from CPB and adjust LVAD RPM speeds with TEE guidance and use of hemodynamics.
- Administer protamine. Routine sternal closure and fixation of the percutaneous lead.

CPB, Cardiopulmonary bypass; LV, left ventricle; LVAD, left ventricular assist device; TEE, transesophageal echocardiogram

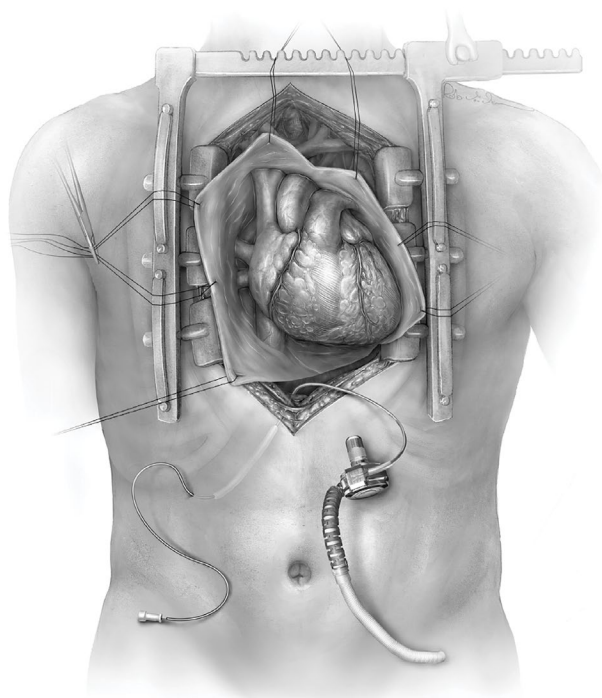
pericardium. It is important to ensure appropriate incision of the pericardium to obtain wide exposure of the apex for a device positioned within the pericardium or to guide creation of a small preperitoneal pocket if needed to prevent compression of the right ventricle that could impair right ventricular function (Fig. 20.12). Even for devices intended for intrapericardial placement, creation of a small preperitoneal space at the apex of the left ventricle can facilitate proper alignment of the inflow cannula. Insufficient leftward dissection will result in inappropriate positioning of the inflow cannula (particularly with a markedly enlarged heart and leftward displacement of the apex) such that the inflow portion points against that lateral wall rather than into the central portion of the LV cavity, producing chronic partial inflow obstruction.

Assembly of the pump and components are performed at this time specific to each device type. The tunneling of the percutaneous lead is performed before administration of heparin to prevent bleeding in the rectus sheath. This process is common to all devices designed with an external power source. The tunnel begins at the inferior aspect of the incision at the posterior rectus SHEATH 2 cm lateral to the midline and passes through the rectus muscle, exits the anterior rectus sheath, and continues in the subcutaneous tissue to exit the skin near the costal margin at the anterior axillary line (Fig. 20.13). It is important not to place the exit site posteriorly because this can interfere with the patient's ability to provide self-care with the exit site. The cloth-covered sections of the driveline are completely buried underneath the skin so that the interface



• **Figure 20.12** Patient and surgical preparation. Patients with advanced HF are preoperatively maintained with inotrope infusions to optimize organ function. To minimize infectious risks, a new central venous pulmonary artery (PA) catheter should be inserted no more than 24 hours preoperatively. The routine use of PA catheters for LVAD implantation may assist with perioperative therapies including pulmonary vasodilator therapy with nitric oxide as well as inotropic and vasoconstrictor support. The general preparation is similar to other cardiac surgical procedures with the patient prepared and draped in the standard sterile fashion. Routine broad-spectrum antibiotics are administered intravenously. In addition to the PA catheter, an arterial line and a Foley catheter are inserted. A transesophageal echocardiogram (TEE) is used to assess for a patent foramen ovale, valvular dysfunction, and intracardiac thrombi before implantation and is used to optimize pump speed while weaning from cardiopulmonary bypass following implantation. A standard median sternotomy is performed and the pericardium is incised; this incision is carried to the left, along the pericardial reflection to the apex. (Reprinted from Romano MA, Haft J, Pagani FD, HeartWare HVAD: principles and techniques for implantation, *Oper Tech Thorac Cardiovasc Surg.* 2013;18:230-8, with permission from Elsevier). Ao = aorta.

at the skin exit site is with the smooth surface of the driveline. This technique significantly reduces the risk of infection at the skin exit site of the percutaneous lead. A fixation point for the percutaneous lead can be created approximately 10 cm beyond the exit site with a 3-0 Prolene (Ethicon, Somerville, NJ) suture to distribute any tugging to this point rather than the exit site to prevent trauma to the exit site in the early postoperative period when the percutaneous lead is potentially vulnerable to dislodgement. A 5-0 subcuticular Prolene suture can be used to create a purse string around the exit site to encourage apposition of the skin around the exit site of the lead. While the traditional approach for percutaneous lead exit site placement during LVAD implantation is the right upper quadrant of the abdomen, alternative sites have been explored to mitigate complications such as infection and driveline trauma. Some centers utilize the left upper quadrant or subcostal areas, aiming to allow for a shorter, more direct route to the pump, potentially reducing tension and driveline wear. Additionally, a tailored approach, considering individual patient anatomy, body habitus, and risk factors for driveline complications, is increasingly advocated in contemporary practice.²

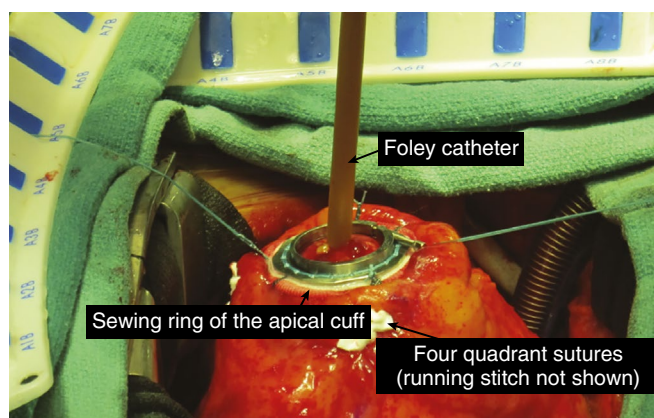


• **Figure 20.13** The percutaneous lead is tunneled through the right rectus sheath and exits just below the costal margin at the anterior axillary line. (Reprinted from Romano MA, Haft J, Pagani FD, *HeartWare HVAD: principles and techniques for implantation, Oper Tech Thorac Cardiovasc Surg.* 2013;18:230-8, with permission from Elsevier).

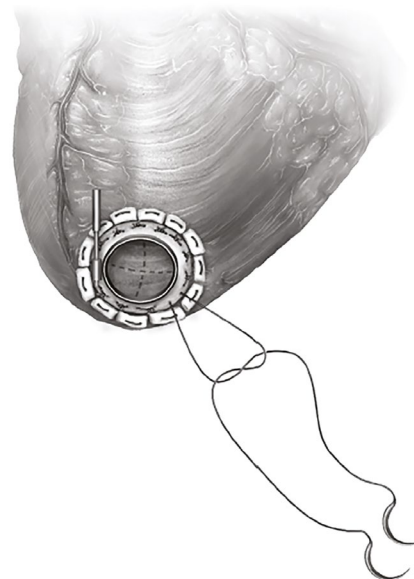
CPB is established with ascending aorta and right atrial cannulation (unless emergent femoral cannulation is required). Selective cannulation of the superior and inferior vena cavae with snares would be required for venous drainage if a concomitant procedure was planned on the right side (i.e., closure of patent foramen ovale or tricuspid valve repair). After initiation of CPB, the heart is maintained in a continuously beating state without cardioplegic arrest unless additional cardiac procedures are needed such as aortic valve repair or replacement. Suction is maintained on the aortic needle vent and continuous surveillance by TEE is important for detection of any air in the ascending aorta.

The apical cannula for current devices is designed to insert directly into the LV apex. The pump is connected to the left ventricle via an inflow collar or apical cuff. The apex of the heart is exposed by lifting the apex of the heart and placing several sponges underneath the diaphragmatic base. The apex of the LV is identified and marked. We generally identify the apical dimple by marking the course of the left anterior descending coronary artery, drawing a mark along the acute margin of the RV, and projecting this mark onto the obtuse margin. The apical dimple is typically located just anteriorly and laterally to the confluence of these two marks. A stab incision is placed into the apex, a 16F Foley catheter is inserted into the LV, the balloon is inflated, and gentle retraction is placed on the catheter to lift the apex. The LV is vented through the catheter to prevent LV distension and facilitate sewing of the apical cuff to the myocardium. The apical cuff is sewn to the LV apex before coring the LV apex (ie, “sew-then-core” method). The Foley catheter is passed through the apical connector before insertion in the LV (Fig. 20.14).

The apical cuff is first attached to the LV apex using four stitches of 2-0 Ethibond suture with pledgets placed in an interrupted,



• **Figure 20.14** The Foley catheter is inserted into the apex of the left ventricle. The apical cuff of the LVAD is sewn to the apex of the left ventricle using 4 sutures of 2-0 Ethibond (Ethicon, Somerville, NJ) with pledgets, placed in an interrupted, horizontal mattress fashion. A running suture of 4-0 or 3-0 Prolene around the cuff is then performed for hemostasis (not shown). (Reprinted from Stulak JM, Abou El Ela A, Pagani FD. *Implantation of a Durable Left Ventricular Assist Device: How I Teach It. Ann Thorac Surg.* 2017;103:1687-92, with permission from Elsevier.)



• **Figure 20.15** Alternatively, the apical cuff can be secured to the apex using 12, 2-0 Ethibond sutures with pledgets placed into the myocardium in an interrupted, horizontal mattress fashion. The suture is then passed through the sewing ring of the apical cuff and the apical cuff is secured to the apex of the LV. (Reprinted from Romano MA, Haft J, Pagani FD, *HeartWare HVAD: principles and techniques for implantation, Oper Tech Thorac Cardiovasc Surg.* 2013;18:230-8, with permission from Elsevier).

horizontal mattress fashion at four quadrants of the apical cuff. The suture is passed through the sewing ring of the apical cuff and the apical cuff is then secured to the apex of the LV. Next, a 4-0 or 3-0 Prolene suture is then continuously stitched in a running fashion around the apical cuff (Fig. 20.14) to ensure hemostasis. Alternatively, the apical cuff can be secured to the apex using 12, 2-0 Ethibond sutures with pledgets placed into the myocardium and circumferentially around the perimeter of the apical cuff in an interrupted, horizontal mattress fashion. The suture is then passed through the sewing ring of the apical cuff and the apical cuff is secured to the apex of the LV (Fig. 20.15).

Next, using a coring knife specifically designed for each specific LVAD brand, a core of LV muscle is removed through the center of the apical cuff (Fig. 20.16). The inside of the left ventricle is carefully inspected to ensure the absence of LV thrombosis or large trabeculae that may obstruct the orifice of the inlet cannula (Fig. 20.17). The heart is then gently filled to evacuate air in the left ventricle. The apical inflow cannula is inserted and secured during continuous TEE surveillance of the ascending aorta for air. With the heart and pump filled with blood, the operating table is rotated away from the surgeon, and the pump is de-aired to allow residual air in the pump to collect in the outflow graft and be eliminated. When no further residual intracardiac air is identified by TEE, the outflow graft is measured to an appropriate length, trimmed, and clamped proximally. An end-to-side anastomosis is then constructed to the exteriorized right lateral aspect of the ascending aorta, usually with continuous 4-0 or 5-0 polypropylene suture (Fig. 20.18)

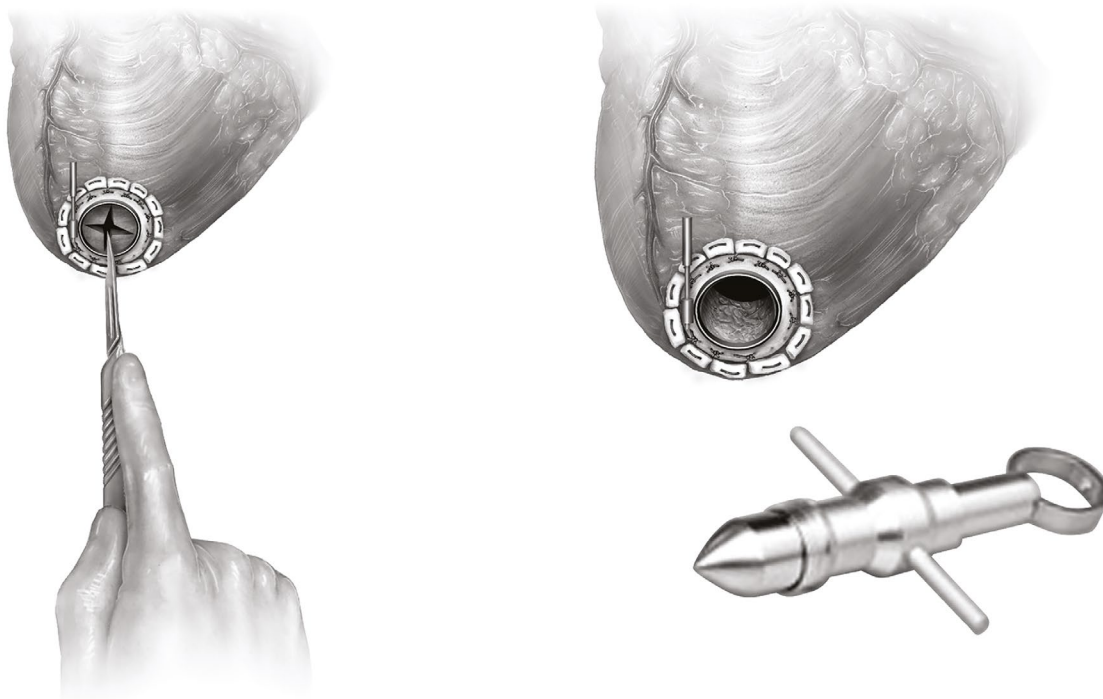
Appropriate inotropic support is initiated to optimize RV function, with or without the addition of inhaled pulmonary vasodilator such as nitric oxide or prostaglandin. The driveline and its connecting tubing are passed off the surgical field and connected to the pump console. Weaning from CPB is a critical phase of the operation. Data from multiple sources, CPB, LVAD device, TEE, hemodynamic monitor, and visual inspection of the heart are synthesized in real time to facilitate optimal weaning. CPB is slowly weaned to one-half total flow to facilitate native LV ejection and initial air removal steps. Air removal and filling are guided by TEE findings and hemodynamics as well as visual inspection of the right ventricle. After initial removal of air, the LVAD is actuated at its lowest rpm speed setting, and the clamp on the outflow graft is removed, permitting forward device flow. Air removal is

continued, CPB is weaned to one-quarter flow, and rpm speeds are increased slowly. After confirmation of air removal, the patient is weaned from CPB, and further adjustments to rpm speed are made with TEE guidance and hemodynamic assessments. Factors to consider include device flow, hemodynamics, LV septal wall position, atrioventricular opening, and degree of mitral regurgitation. It is important to prioritize septal position by TEE to ensure rightward deviation as the priority in assessment of optimal LVAD speeds while obtaining satisfactory device flows. After this, transient aortic valve opening at least every third beat can be attempted with changes in device speed to avoid stasis in the aortic root and ensure adequate LV filling, followed by further speed changes to reduce the degree of mitral regurgitation, if any.

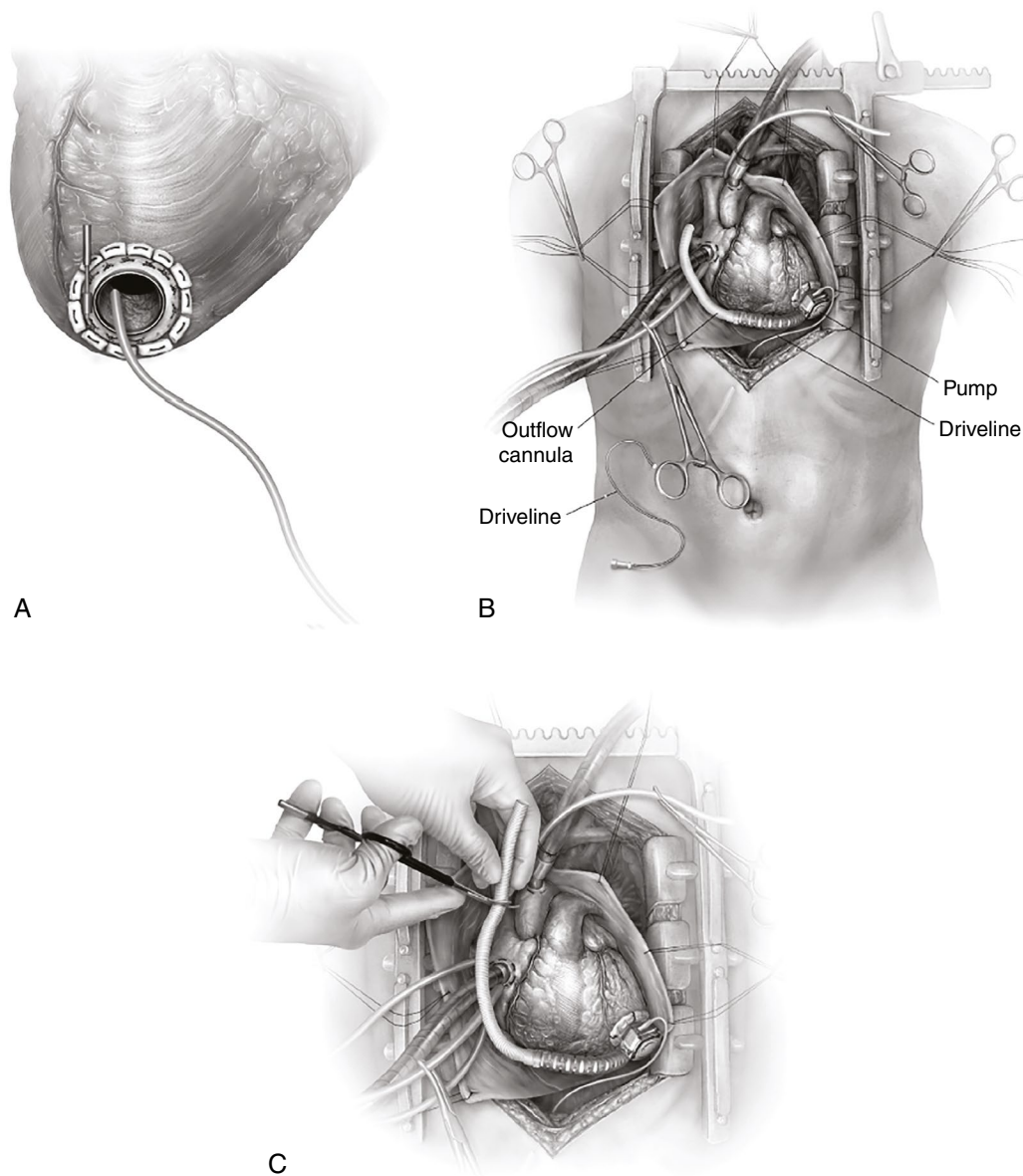
In the event of severe RV failure early post-bypass with a CF pump, left atrial or pulmonary artery diastolic pressure should be carefully observed for rapid fall, and pump speed should then be rapidly decreased while observing for aortic air by TEE. If air is seen in the ascending aorta, the outflow graft must be clamped and CPB rapidly reestablished when heparinization is adequate.

Prior to sternal closure, the outflow graft and apical cannulation site should be covered with native pericardium to facilitate safe sternal reentry and dissection at the time of cardiac transplantation.

The anterolateral thoracotomy approach (nonsternotomy) is an alternative approach to median sternotomy for durable LVAD implantation with reported lower blood transfusions, less incidence of severe RV dysfunction, and shorter time to extubation.³ The most common surgical approach for exposure for placement of the inflow cannula is a lateral thoracotomy usually in the fifth or sixth intercostal space. Access for anastomosis of the outflow



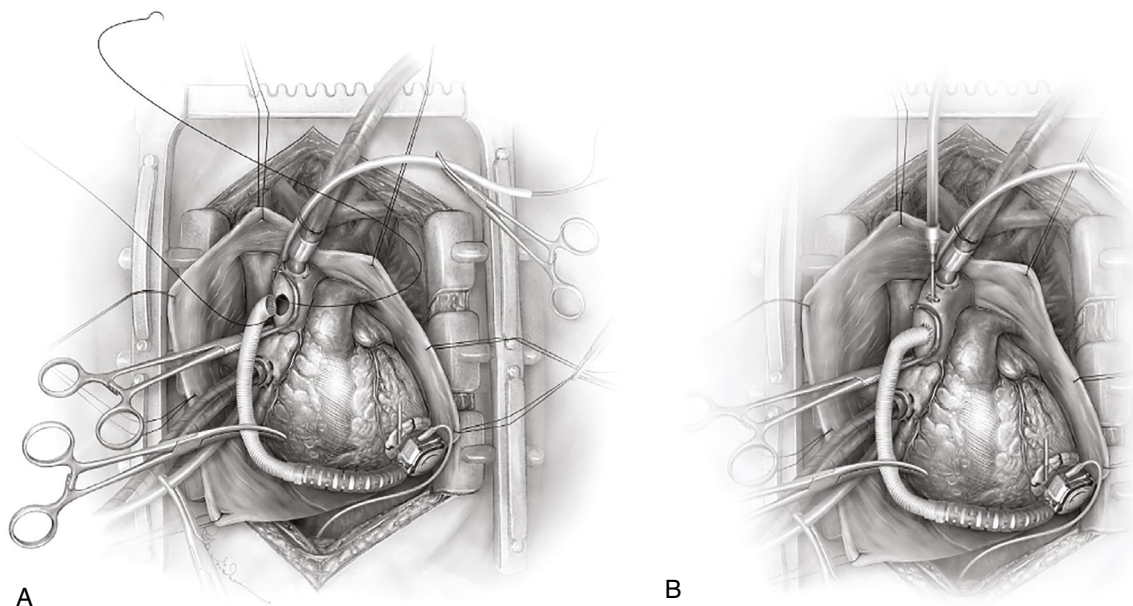
• **Figure 20.16** Next, using a coring knife specifically designed for each specific LVAD brand, a core of LV muscle is removed through the center of the apical cuff. (Reprinted from Romano MA, Haft J, Pagani FD, HeartWare HVAD: principles and techniques for implantation, *Oper Tech Thorac Cardiovasc Surg.* 2013;18:230-8, with permission from Elsevier).



• **Figure 20.17** (A) At this point, CO₂ can be infused into the operative field to facilitate deairing of the LV. (B) The inflow cannula is then inserted into the apical cuff and the device is positioned with the outflow graft and driveline parallel to the diaphragm. The outflow graft is positioned with a gentle curve around the perimeter of the acute margin of the right ventricle following the atrioventricular groove to the ascending aorta. Additional deairing is accomplished by passively filling the heart and pump and elevating the apex and gently shaking the ventricle. (C) The outflow graft is then distended, clamped, and trimmed to proper length. (Reprinted from Romano MA, Haft J, Pagani FD, HeartWare HVAD: principles and techniques for implantation, *Oper Tech Thorac Cardiovasc Surg.* 2013;18:230-8, with permission from Elsevier).

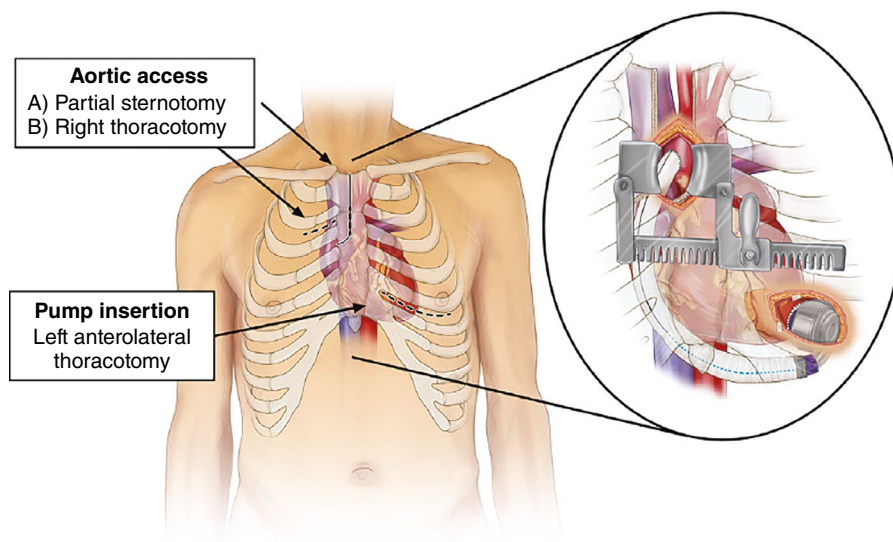
graft to the ascending aortic can be obtained with either an upper hemisternotomy approach or right anterior thoracotomy at the level of the second or third intercostal space (Fig. 20.19). Institution of CPB can be central or peripheral. One of the touted key benefits of the lateral thoracotomy approach is preservation of RV geometry by leaving most of the pericardium intact and not lifting the heart. The downside is the learning curve and potentially longer operative times. A group of patients who could benefit the most from this approach are those with previous cardiac surgery operations. The upper sternotomy or right anterior thoracotomy avoids the need for complete redo-sternotomy. The more important benefit is avoiding the need for complete

dissection around the heart and the need to lift the heart to core the LV apex. The Implantation of the HeartMate 3 in Subjects With Heart Failure Using Surgical Techniques Other Than Full Median Sternotomy (SWIFT) study is a prospective, postmarket, nonblinded, single-arm study of the HeartMate 3 that evaluated durable LV implant strategies other than median sternotomy.⁵³ The SWIFT study demonstrated no differences in length of hospital stay, blood product utilization, adverse events, functional status, or health-related quality of life between sternotomy and thoracotomy approaches. The study validated the noninferiority of the nonsternotomy surgical technique to the median sternotomy technique.



• **Figure 20.18** Outflow graft anastomosis. It is crucial to determine the appropriate length of the outflow graft as excess length may lead to kinking of the graft. Conversely, a short graft may cause added tension to the anastomosis and lead to bleeding or obstruction. The outflow graft should lie along the atrioventricular groove of the right side. The outflow graft should be trimmed at a 45 degree angle to prevent kinking at the aortic anastomosis and create laminar flow within the ascending aorta. (A) The anastomosis is performed along the proximal anterolateral ascending aorta. A partial-occlusion clamp is placed without dissecting the plane between the aorta and PA. The aorta is incised, and the anastomosis is sewn with a 4-0 polypropylene suture. The anastomosis is deaired. The partial-occlusion clamp is then removed to assess hemostasis. A vascular clamp should remain on the outflow graft while the device is off to avoid retrograde flow into the aorta. (B) A deairing cannula is placed in the ascending left ventricle between the outflow graft and aortic cannula. It is important to place the graft as proximal as possible to allow for future cannulation and aortic anastomosis at the time of heart transplantation. (Reprinted from Romano MA, Haft J, Pagani FD, HeartWare HVAD: principles and techniques for implantation, *Oper Tech Thorac Cardiovasc Surg*. 2013;18:230-8, with permission from Elsevier).

Thoracotomy-based implant technique



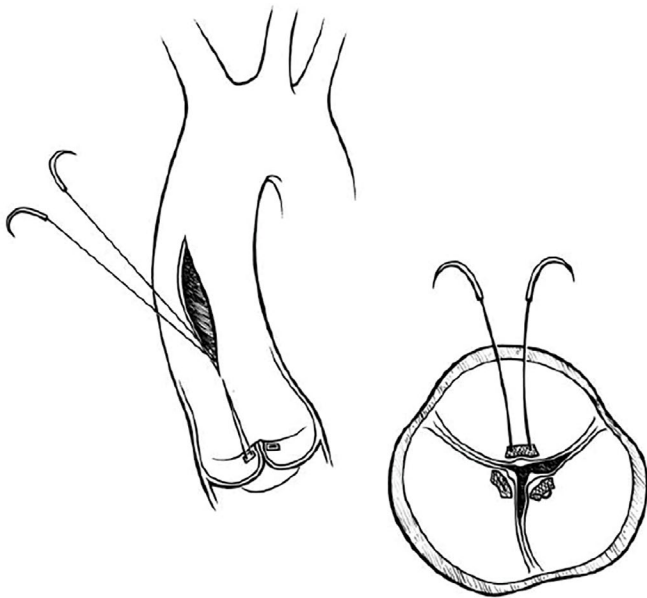
• **Figure 20.19** Pump insertion by left thoracotomy with aortic access by (A) partial sternotomy or (B) right thoracotomy. (Reprinted from Gosev I, Pham DT, Um JY, Anyanwu AC, Itoh A, Kotkar K, Takeda K, Naka Y, Peltz M, Silvestry SC, Couper G, Leacche M, Rao V, Sun B, Tedford RJ, Mokadam N, McNutt R, Crandall D, Mehra MR, Salerno CT. Ventricular assist device using a thoracotomy-based implant technique: Multi-Center Implantation of the HeartMate 3 in Subjects With Heart Failure Using Surgical Techniques Other Than Full Median Sternotomy (HM3 SWIFT). *J Thorac Cardiovasc Surg*. 2024;168(5):1474-1484, with permission from Elsevier.)

Concomitant Surgical Procedures

Tricuspid regurgitation is usually addressed at durable LVAD implant when severe due to the adverse effect on forward flow, RV overload, and hepatic congestion. Recent data suggest that although surgical correction of significant tricuspid regurgitation successfully reduces tricuspid regurgitation, correction of the tricuspid regurgitation does not reduce the incidence of right HF following durable LVAD implantation.⁵⁴

Aortic regurgitation should be addressed when a mild degree or more as aortic regurgitation will generally worsen over longer support times and aortic regurgitation adversely effects net forward cardiac output.⁵⁵ Aortic regurgitation is especially important to consider in patients who may be supported longer than a year. Options for addressing important aortic regurgitation include aortic valve repair (i.e., Park stitch) (Fig. 20.20)⁴, aortic valve replacement, or patch closure of the aortic valve anulus. Aortic valve repair is the more common procedure when feasible and has similar durability to AV replacement with optimal patient selection. Mechanical valves in the aortic valve position are usually replaced with a bioprosthesis to avoid thrombotic complications associated with incomplete valve opening following durable LVAD implantation. Mechanical valves in the mitral valve position are not routinely replaced as there are currently no data to support routine explant and exchange with a bioprosthesis.

Controversy surrounds the issue of whether important mitral valve regurgitation should be addressed at the time of durable LVAD implantation.⁵⁵ Recent data from the MOMENTUM 3 clinical study demonstrated that significant preoperative mitral regurgitation resolved from a preoperative incidence of 43.5% to a postoperative incidence of 6.2% with HeartMate 3 support and that mitral regurgitation present at baseline or follow-up did not affect late outcomes.



• **Figure 20.20** Park Stitch: Pledgeted 4-0 Prolene sutures are applied to approximate the fibrous nodules of Arantius to create a coaptation stitch. (Reprinted from Park SJ, Liao KK, Seguro R, Madhu KP, Miller LW. Management of aortic insufficiency in patients with left ventricular assist devices: a simple coaptation stitch method (Park's stitch). *J Thorac Cardiovasc Surg.* 2004;127(1):264-6, with permission from Elsevier.)

Closure of a patent foramen ovale or atrial septal defect is necessary at the time of durable LVAD implantation. LV unloading with MCS can reduce left atrial pressures below that of right atrial pressures, producing a significant right to left shunt resulting in significant systemic hypoxia and the risk of a paradoxical embolus. Shunting may not be immediately present as the patient is transitioned from CPB to LVAD support but may become evident later as right sided medical support (i.e., inotropes or inhaled pulmonary vasodilators) is weaned in the intensive care unit.

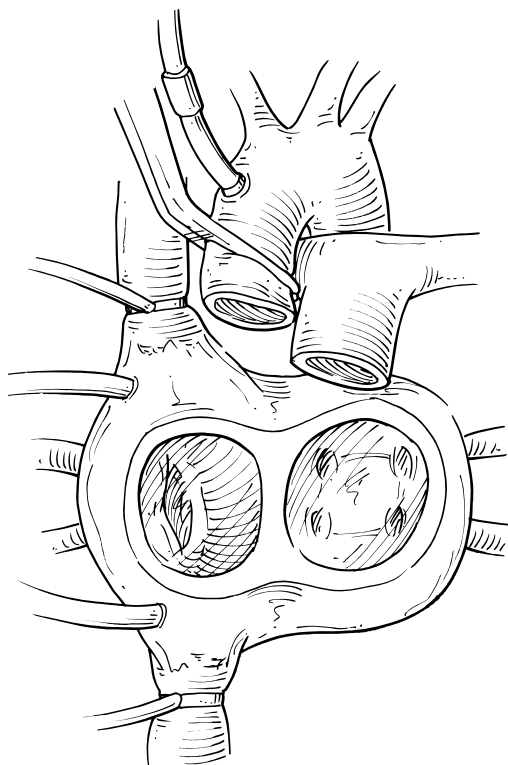
Management of Acute Right Heart Failure. After LVAD implantation, patients are gradually weaned from CPB to durable LVAD support. The key objective is to maintain LV filling and avoid leftward shift of the ventricular septum as pump speed is gradually increased. Leftward shift of the ventricular septum from excessive LV unloading distends the right ventricle and impairs the septal contribution to contractility of the right ventricle. Use of selective inhaled pulmonary vasodilators such as nitric oxide along with inotrope support of the right ventricle with epinephrine, dobutamine, milrinone, or isoproterenol can maintain adequate RV function and right-sided cardiac output. Pump speed is optimized with the goal of maintaining a cardiac index >2.2 L/min/m², which can represent contribution from the LVAD and native heart function. Medical management focused on optimizing RV preload with diuretics, RV contractility with inotropes, and RV afterload reduction remains the mainstay of therapy.

Despite optimal medical management, some patients are unable to maintain adequate cardiac output due to RV dysfunction. Early additional RV support with mechanical devices is the key to achieve good outcomes. Temporary right-sided MCS options include surgical implantation of the extracorporeal CentriMag pump (Abbott). Alternatively, a new percutaneous cannula has been used, the ProtekDuo (LivaNova, Pittsburgh, PA) along with an extracorporeal, centrifugal continuous-flow temporary right VAD (RVAD). The Protek Duo is a dual-lumen cannula that is inserted under fluoroscopy and/or TEE guidance, with the drainage holes positioned in the right atrium and the tip of the reinfusion cannula positioned in the main pulmonary artery. In a series of 27 patients, 85% survived to hospital discharge. Alternatively, the percutaneous Impella RP (Abiomed, Inc., Danvers, MA) placed through the right femoral vein can also be used to provide right-sided support.

Total Artificial Heart

The technique for implantation of the SynCardia TAH-t has been described in detail by Copeland and colleagues (see Fig. 20.11).⁵⁶ Before systemic heparinization, the arterial outflow conduits are appropriately preclotted, and drivelines are tunneled and brought out under the left costal margin. The left-sided ventricular driveline is positioned approximately 5 cm below the costal margin in the midclavicular line. The RV driveline is brought through the skin about 5 cm medial to the left driveline. Both ventricles are then covered with a surgical towel and placed on the left chest until implantation.

CPB is established and the aorta cross-clamped. The recipient heart is excised on the ventricular side of the atrioventricular groove, leaving a small rim of ventricular muscle. The great vessels are divided just above the aortic and pulmonary valves (see Fig. 22.21). The mitral and tricuspid valve leaflets are excised and the chordae trimmed. The coronary sinus ostium in the right atrium is oversewn to prevent backflow of blood to the AV groove.

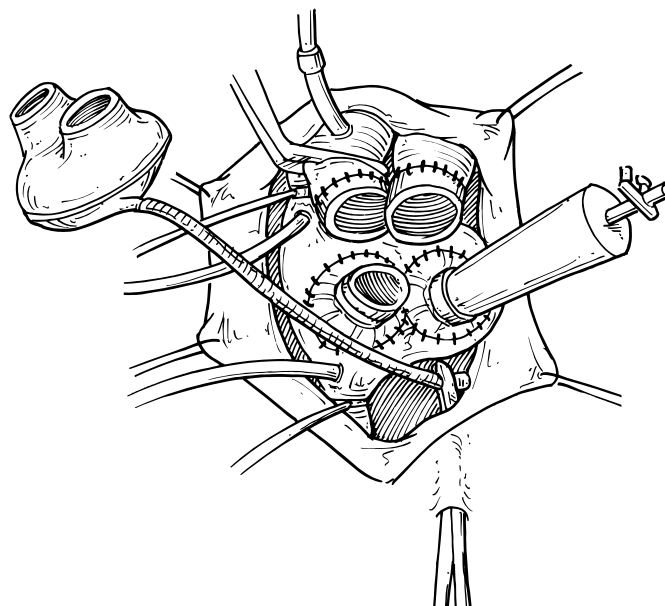


• **Figure 20.21** Ventricular rim with AV (atrioventricular) valves and chordae excised. Great vessels are transected just above sinutubular junction. (Redrawn from Copeland JG, Dowling RD, Tsau PH. Total artificial hearts. In: Frazier OH, Kirklin JK, eds. *ISHLT Monograph Series: Mechanical Circulatory Support*. Philadelphia: Elsevier; 2006:105.)

A “neopericardium” is constructed out of three 15- × 20-cm sheets of expanded polytetrafluoroethylene (ePTFE). The first sheet of ePTFE is sutured to the pericardial reflection as posteriorly as possible at the level of the superior vena cava, inferior vena cava, and pulmonary veins on the right side (without injuring the phrenic nerve); the second sheet is sutured anterior to the left pulmonary veins on the left side; and the third sheet is placed to cover the entire diaphragmatic reflection. The three ePTFE sheets are later folded over the artificial ventricles after completion of the implantation.

The outer rims of the left and right atrial cuffs, which consist of approximately 1 cm of ventricular muscle and fat in the AV groove, are buttressed circumferentially with 1-cm-wide strips of Teflon felt to reinforce the anastomoses between the atrial cuffs and the quick connects and provide maximal hemostasis. The atrial quick connects are trimmed to 6 mm from the connectors, turned inside out, and anastomosed to the atrial cuff (first left, then right), using 3-0 polypropylene. The suture line includes the felt strip—strengthened free ventricular wall and the interventricular septum, taking care to achieve a maximally hemostatic suture line. After completion of both anastomoses, the quick connects are everted back to normal configuration, and biological glue is applied to the suture lines (see Fig. 20.22).

The outflow conduits from each artificial ventricle are cut to an appropriate length, determined by briefly placing the artificial ventricles in the mediastinum. The entire length of the pulmonary artery is preserved and anastomosed to the conduit with 4-0 polypropylene. The aortic anastomosis is similarly constructed. All suture lines are tested for hemostasis using a plastic tester device with saline via a three-way stopcock (see Fig. 20.22).



• **Figure 20.22** Testing left atrial cuff for hemostasis after placement of atrial quick connector. (Redrawn from Copeland JG, Dowling RD, Tsau PH. Total artificial hearts. In: Frazier OH, Kirklin JK, eds. *ISHLT Monograph Series: Mechanical Circulatory Support*. Philadelphia: Elsevier; 2006:105.)

The ventricular connections to the quick connects are facilitated by grasping each side of the quick connect with a large clamp while pushing in the device. To avoid conduit angulation, the LV outflow should be as close as possible to the native aorta. Orientation of the ventricle and the outflow mount is fixed by the atrial connection. Prior to completing the aortic connection, the pump and conduit are filled with saline. The right atrial inflow quick connect and the right arterial conduit are attached to the right artificial ventricle similarly. The right side is filled by partially releasing the inferior vena cava tape, completing device implantation (see Fig. 20.23).

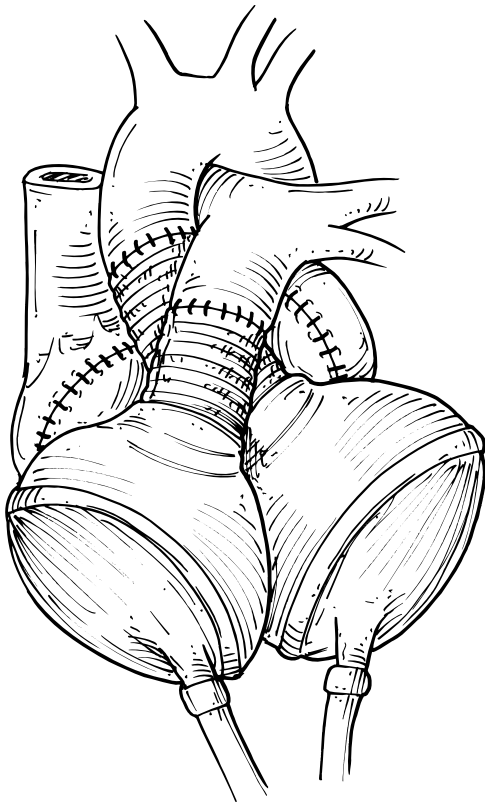
The patient is placed in steep Trendelenburg, and the aortic cross-clamp is removed with the ascending aorta vented. With the TAH pumping at 40 beats/min, vigorous de-airing is completed under TEE guidance. The TAH pumping rate is then increased and CPB discontinued while maintaining a CVP of 12 to 15 mmHg. Cardiac output from the TAH is usually about 7 to 8 L/min at this point. After obtaining meticulous hemostasis, the previously secured ePTFE “neopericardium” is closed over the TAH anteriorly. A rectangular piece of ePTFE is passed around the proximal aorta and secured. During chest closure, special attention is focused on device output and TEE monitoring of inferior vena caval and left pulmonary venous flows.

Special Features of Postoperative Care

Anticoagulation

The mainstay of pump anticoagulation therapy involves blockade of the intrinsic contact coagulation cascade and platelet aggregation. (See further discussion under “Biological Barriers”.)

Heparin. The primary anticoagulant effect of heparin results from heparin binding to antithrombin III, and that complex



• **Figure 20.23** Complete implantation of SynCardia TAH-t in orthotopic position. (Redrawn from Copeland JG, Dowling RD, Tsau PH. Total artificial hearts. In: Frazier OH, Kirklin JK, eds. *ISHLT Monograph Series: Mechanical Circulatory Support*. Philadelphia: Elsevier; 2006:105.)

inactivates thrombin (and thrombin-induced activation of factors V and VIII) and to some degree factors Xa, XIIa, XIa, and IXa.

Warfarin. Warfarin inhibits the vitamin K–dependent factors II, VII, IX, and X. During the 5 to 7 days generally required for effective inhibition of these factors, warfarin actually creates a relatively hypercoagulable state through inhibition of the natural anticoagulants protein C and protein S.

Antiplatelet Therapy. The mainstay of antiplatelet activity is aspirin (acetylsalicylic acid [ASA]). Aspirin irreversibly inhibits cyclooxygenase, an enzyme responsible for conversion of arachidonic acid to prostaglandin and eventually thromboxane A_2 , which promotes platelet aggregation. Effectiveness of aspirin therapy can be monitored by the *in vitro* platelet aggregation response to arachidonic acid. The antithromboembolic effect of aspirin does not increase with dosages greater than about 325 mg/day, and there is probably no advantage for doses greater than 160 mg/day. The ADP pathway for induction of platelet activation is not inhibited by aspirin, and ADP receptors play a central role in platelet activation secondary to shear stress.

Clopidogrel irreversibly inhibits platelet activation via ADP receptors on the platelet surface. Dipyridamole inhibits platelet aggregation through a different mechanism. Specifically, dipyridamole inhibits the uptake of adenosine into platelets and endothelial cells, effectively increasing local adenosine concentration. The excess adenosine stimulates platelet adenylate cyclase, which increases platelet cyclic adenosine monophosphate (cAMP) levels, inhibiting platelet aggregation response to ADP, collagen, and platelet-activation factor.

Anticoagulation for Continuous Flow Pumps. Current labeling for the antithrombotic regimen for the HeartMate 3 LVAD recommends anticoagulation with a vitamin K antagonist (warfarin) for an INR goal of 2 to 3 and antiplatelet therapy with aspirin 81 mg daily.

A recent prospective, multicenter, randomized, double blind clinical study of aspirin withdrawal from the antithrombotic regimen for the HeartMate 3 demonstrated noninferiority to an antithrombotic regimen including aspirin. Additionally, more patients were alive and free of hemocompatibility events at 12 months in the placebo group (74%) versus those taking aspirin (68%). Noninferiority of placebo was demonstrated (absolute between-group difference, 6.0% improvement in event-free survival with placebo [lower 1-sided 97.5% CI, -1.6%]; $P < .001$). Aspirin avoidance was associated with reduced non-surgical bleeding events (relative risk, 0.66 [95% CL, 0.51-0.85]; $P = .002$) with no increase in stroke or other thromboembolic events, a finding consistent among diverse subgroups of patient characteristics.

Anticoagulation for the Total Artificial Heart. Specific protocols have been developed for the SynCardia TAH-t.^{52,56} Warfarin and antiplatelet agents are routinely combined with additional monitoring by thromboelastography and platelet aggregation studies.

Blood Pressure Management with Continuous Flow Pumps

Although pulsatile flow can usually be detected by direct arterial line monitoring in the early postimplant period, ambulatory blood pressure may not be reliably measured with a standard blood pressure cuff or with automatic blood pressure devices. The normal Korotkoff sounds are typically absent with the minimal pulsation of CF pumps. The most reliable estimate of mean or systolic blood pressure (depending on degree of pulsatility) is use of a Doppler device over the radial artery while gradually deflating a standard blood pressure cuff. Monitoring outpatient blood pressure with CF pumps is important because of the occasional development of severe hypertension (possibly related to abnormal autonomic deregulation), which, if undetected and untreated, can result in a major intracerebral bleeding event.

Ventricular Suction

When pump speed in a rotary pump exceeds the ability of the left ventricle to supply a continuous inflow of blood into the inlet cannula, the resultant rapid decrease in size of the ventricular chamber can result in cavitory collapse around the inlet cannula, which impedes entrance of blood and creates a “suction event.” The result is a sudden decrease in pump flow. Ventricular suction is remedied by decreasing pump speed. The HeartMate 3 device detects sudden changes in pump flow by a change in *pulsatility index* (a measure of flow pulse through the pump described by the relationship: $\text{pulse index} = [\text{maximum flow} - \text{minimum flow}] / \text{mean flow}$). Sudden drops in pulsatility index and a decrease in pump flow indicate a likely suction event. The HeartMate 3 device incorporates a suction detection algorithm in which pump speed automatically reduces to a set rpm if a sudden change in pump flow pulsatility is detected, after which the pump speed slowly returns to the set speed. Observation of this phenomenon indicates the need to reduce the set pump speed.

Pump Thrombus in Continuous Flow Pumps

If thrombus develops on the pump rotor, an abnormal increase in power is required to maintain impeller speed. The increase in power consumption causes an inaccurately high estimate of pump flow and a decrease in pulsatility index caused by a reduction in cyclic power consumption (because continuously high-power consumption is caused by the rotor drag induced by thrombus). In contrast, inlet obstruction (secondary to inlet cannula misalignment or thrombus on the inlet cannula orifice), or outflow obstruction (such as kinking of the outflow graft or anastomotic narrowing) produces a decrease in power consumption along with decreased pump flow and pulsatility index. The diagnosis of inflow or outflow obstruction can be aided by echocardiographic imaging, which may reveal a dilated left ventricle, which is not improved by increasing pump speed.

Pump Stoppage of Continuous Flow Pumps

In contrast to pulsatile devices (in which a hand-pumping device is usually available), CF pumps carry no manual option to maintain forward pump flow in the event of pump stoppage. The physiologic sequelae of pump stoppage are discussed in “Continuous Flow Ventricular Assist Devices.”

When pump stoppage occurs, most commonly resulting from damage to the driveline, rapid decompensation and death are common secondary to a combination of severe reduction in cardiac output and a variable degree of “pump/aortic” insufficiency due to the absence of valves. If the patient is viable, emergency transport to an experienced MCS center is mandatory while cardiac output is maximized with inotropic support. Upon hospital

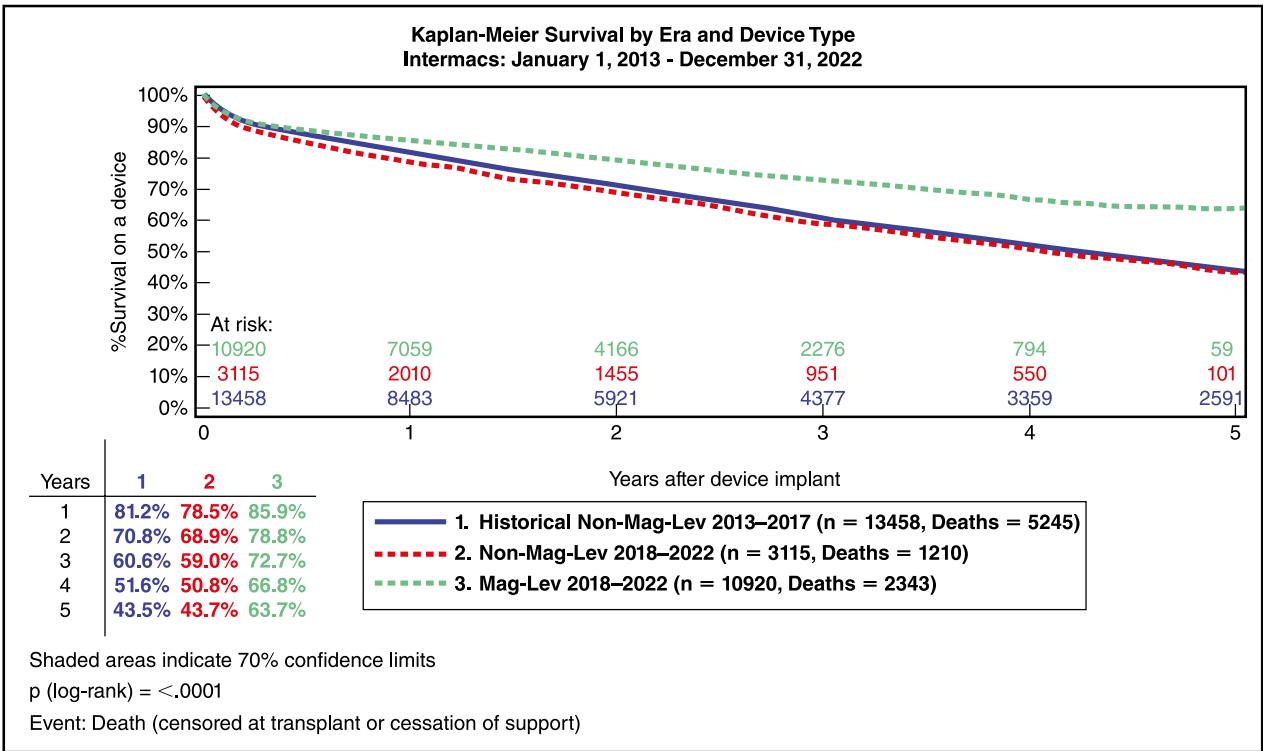
arrival, a surgical team should be ready for emergency device exchange. An echocardiogram is necessary to evaluate for possible intracavitary LV thrombus. If no thrombus is identified and soldering of damaged electrical wiring is an option, full heparinization followed by device restart (with the small possibility of thromboembolic stroke) may provide a more favorable risk/benefit ratio than emergency reoperation for pump exchange.

Results

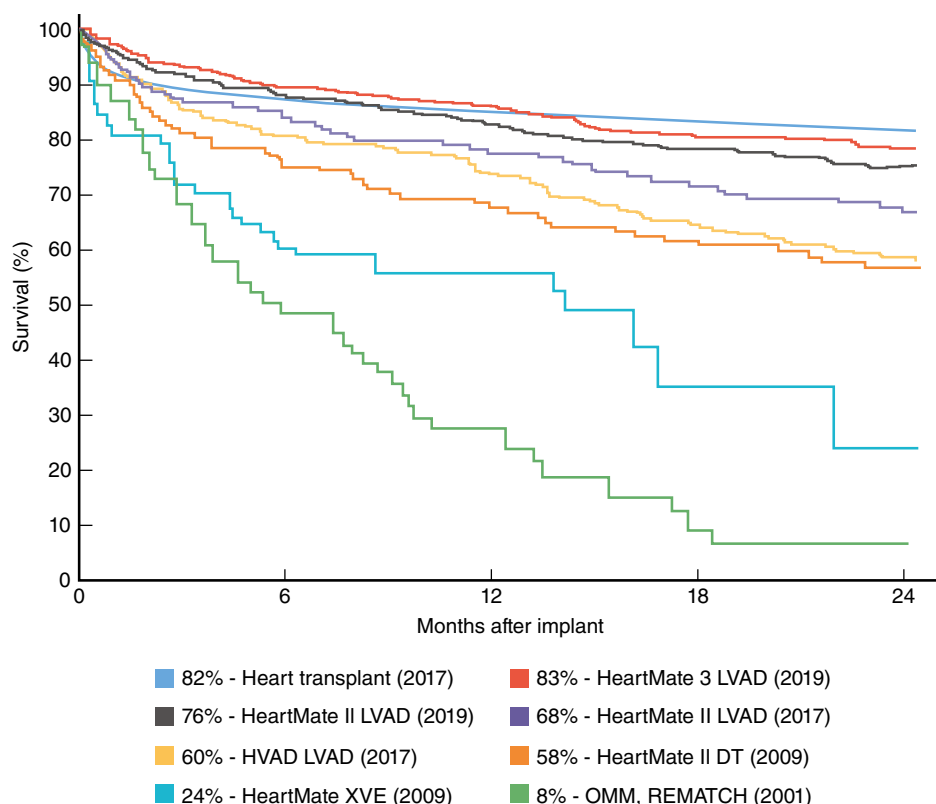
Survival

Survival rates for patients with advanced HF undergoing implantation of a durable mechanical circulatory device have significantly improved over the years. Current data indicate a 1-year survival rate of approximately 86% for patients supported by contemporary continuous-flow durable LVADs (Fig. 20.24).⁵⁷ Over the past 18 years, 1-year survival rates with durable LVADs have increased by approximately 35% (Fig. 20.25).^{22,29,58-60} In the 2001 seminal publication of the Randomized Evaluation of Mechanical Assistance for the Treatment of Congestive Heart Failure (REMATCH) trial, treatment with an early generation pulsatile durable LVAD (HeartMate XVE; Abbott) demonstrated a 1-year survival of 52% compared to 25% for medical management.²² Subsequent technologic advances improved 1-year survival to 68% with the next generation of axial rotary pumps (HeartMate II; Abbott)⁵⁸ and to 77% with centrifugal rotary pumps (HeartWare HVAD; Medtronic).⁵⁹

From 2018 to 2022, the Multi-center Study of MagLev Technology in Patients Undergoing MCS Therapy With HeartMate 3 (MOMENTUM 3) compared outcomes of patients receiving the HeartMate II to the contemporary HeartMate 3 device (Abbott). In



• **Figure 20.24** Kaplan-Meier survival estimates compared by patients with non–totally magnetically levitated devices (HeartMate II and HeartWare HVAD, Medtronic) in the previous era (2013–2017), non–totally magnetically levitated devices (HeartMate II, Abbott and HeartWare HVAD, Medtronic) in the contemporary era (2018–2022), compared to a totally magnetically levitated device (HeartMate 3, Abbott) in the contemporary era (2018–2022).



• **Figure 20.25** Historical survival of patients with advanced heart failure supported with durable LVADs from pivotal clinical trials.^{22,29,58-60} The green and blue curves depict the survival associated with medical therapy and early pulsatile LVAD technology (HeartMate XVE), respectively. Following the REMATCH trial, pulsatile pumps were replaced with smaller CF rotary pumps (HeartMate II, Abbott, Chicago, IL and HeartWare HVAD translating into improved survival (orange, yellow, purple, and black curves). The contemporary CF rotary pump with total magnetic levitation of the internal impeller (HeartMate 3, red curve) is associated with a 2-year survival of 83%, paralleling that of heart transplantation. (HeartMate II and HeartMate 3 are trademarks of Abbott Laboratories or its related companies.) (Reproduced with permission of Abbott. © 2025 Abbott. All rights reserved.)

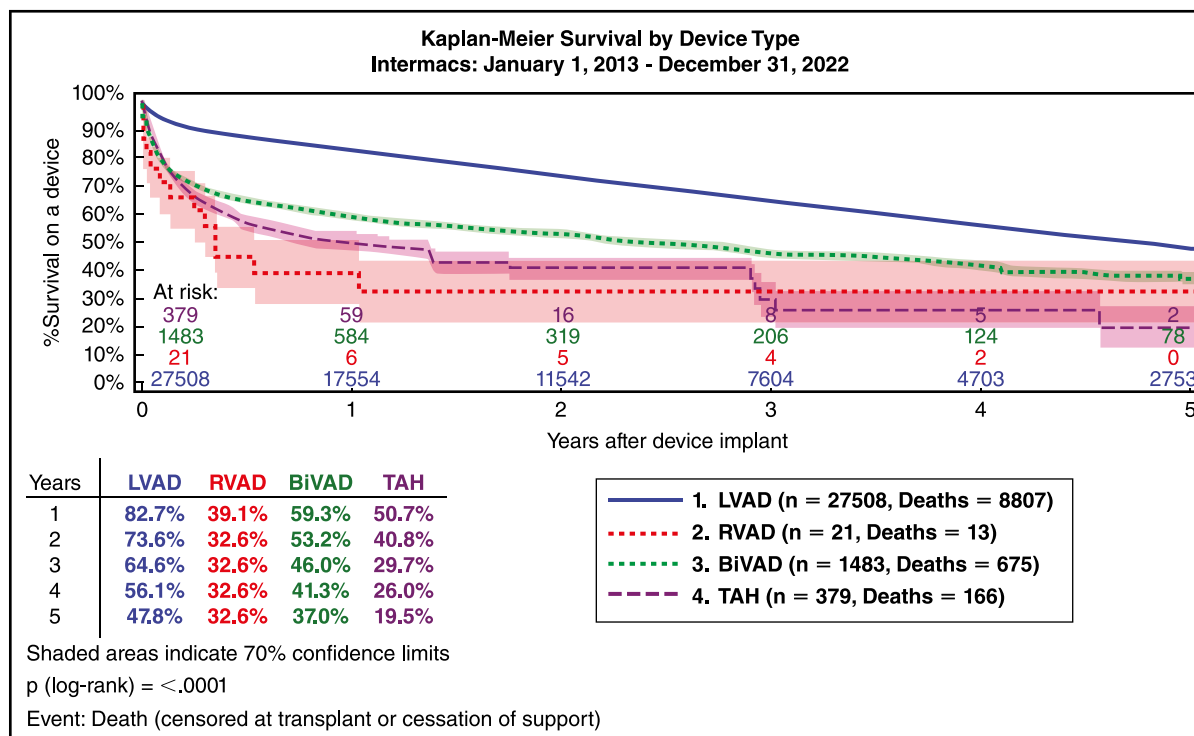
MOMENTUM 3, the 1- and 5-year survival was 84% and 52%, respectively, for patients receiving the HeartMate 3 compared to 82% and 44%, respectively, for patients receiving the older HeartMate II. The HeartMate 3 arm of the MOMENTUM 3 achieved the best survival in a durable LVAD clinical trial to date.^{29,61} “Real world” survival observed within The Society of Thoracic Surgeons Intermacs National Database (STS Intermacs) has validated survival achievements observed in MOMENTUM 3 (see Fig. 20.24).^{57,62-65} Five year survival from STS Intermacs is approximately 64% for patients receiving a primary LVAD implant. For patients 50 years of age or less, survival at 5 years is approximately 75%. For patients 70 years of age or older, survival at 5 years is 53%.

Survival on durable LVAD support, regardless of device intent, far exceeds the associated mortality from medical management of American Heart Association Stage D HF. Given that approximately 50% of patients receiving a durable LVAD demonstrate some degree of physiologic signs of cardiogenic shock or impaired tissue perfusion (i.e., STS Intermacs Patient Profiles 1 and 2) and with another 36% deemed stable but inotrope dependent (i.e., STS Intermacs Patient Profile 3), these outcomes exceed those of medical management of cardiogenic shock alone (survival of approximately 50% to 70% at 30 days). These high mortalities from cardiogenic shock and ambulatory American Heart Association Stage D HF support consideration for durable LVAD therapy to improve survival.⁶⁶⁻⁶⁸

A number of patient-related risk factors are associated with higher postimplant mortality. Important risk factors reported for mortality with rotary pumps include older patient age, female gender, higher body mass index, lower serum albumin as a marker of nutritional state, renal dysfunction requiring dialysis, higher serum bilirubin as a marker of right HF, and the need for concomitant cardiac operations.⁶⁹

The requirement for biventricular support has been consistently identified to confer a significantly worse prognosis. Survival at 1 year for patients requiring concomitant RVAD support at the time of durable LVAD support is approximately 59%.^{57,62-65} For patients with biventricular HF who require a TAH, survival is approximately 50% at 1 year (see Fig. 20.26). Among patients who ultimately need RV support, mortality appears to increase if the decision for RVAD placement is delayed until after the primary LVAD implant.⁷⁰

The causes of mortality in LVAD patients vary over time. Within the first 30 days, the leading causes of death include cardiac failure, multiorgan failure, and stroke (Table 20.2).⁷¹ Beyond the first month, infections become the most common cause. Current hazard functions for mortality exhibit a steep decline in the early postimplant phase, followed by a gradual increase after 6 months. Risk factors for long-term mortality include advanced age, severe right HF, cardiogenic shock at implant, and compromised organ function (Table 20.3).



• **Figure 20.26** Kaplan-Meier survival estimates for patients by mechanical circulatory support device type. BiVAD, biventricular assist device.

TABLE 20.2 Causes of Death after LVAD Implantation^a

Primary Cause of Death	EARLY (≤1 MO)		LATER (>1 MO)		TOTAL	
	n	% of 69	n	% of 122	n	% of 191
Cardiac failure ^b	21	30.4%	21	17.2%	42	22.0%
Infection	6	8.7%	25	20.5%	31	16.2%
CNS event	8	11.6%	19	15.6%	27	14.1%
Multiorgan failure	11	15.9%	9	7.4%	20	10.5%
Respiratory failure	4	5.8%	6	4.9%	10	5.2%
GI bleeding	0	0.0%	2	1.6%	2	1.0%
Surgical bleeding	5	7.2%	1	0.8%	6	3.1%
Bleeding, other	1	1.4%	4	3.3%	5	2.6%
Device failure	0	0.0%	9	7.4%	9	4.7%
Renal failure	3	4.3%	3	2.5%	6	3.1%
Hepatic failure	2	2.9%	1	0.8%	3	1.6%
Malignancy	0	0.0%	2	1.6%	2	1.0%
Arterial embolism	0	0.0%	1	0.8%	1	0.5%
Cardiac tamponade	1	1.4%	0	0.0%	1	0.5%
Post-explant failure to recover	0	0.0%	1	0.8%	1	0.5%
Other	7	10.1%	18	14.8%	25	13.1%
TOTAL	69	100.0%	122	100.0%	191	100.0%

^aInteragency Registry for Mechanically Assisted Circulatory Support (INTERMACS): June 2006-March 2009. Primary LVAD n = 1092.

^bCardiac failure includes RV failure and ventricular tachyarrhythmia (VT/VF).

CNS, Central nervous system; GI, gastrointestinal.

From Kirklin JK, Naftel DC, Stevenson LW, et al. Second INTERMACS annual report: more than 1,000 primary left ventricular assist device implants. *J Heart Lung Transplant*. 2010;29:1.

TABLE 20.3 Risk Factors for Death after LVAD Implantation

Variable	MULTIPHASE HAZARD MULTIVARIABLE MODEL OVERALL COHORT (<i>n</i> = 4448)			
	EARLY HAZARD		CONSTANT HAZARD	
	HR	<i>P</i> Value	HR	<i>P</i> Value
Preimplantation risk factors for death				
CF-HL	...	...	3.18	<.0001
Probability of receiving a CF-HL	...	...	...	...
Demographics				
Era 4: January-June 2019	0.73	.014	...	...
Age at implantation	...	...	1.02	.0006
Age ² (60 y vs. 50 y)	1.41	<.0001	...	...
Male	...	...	1.49	.0213
Body mass index	1.02	.0363	...	...
Clinical status				
Device strategy: bridge to decision	...	...	1.50	.0055
Device strategy: destination therapy	1.31	.0371	...	...
Intermacs patient profile 1	2.15	<.0001	...	...
Intermacs patient profile 2	1.47	.0095	...	...
Phosphodiesterase-5 inhibitor preimplantation	1.62	.0059	...	...
Cardiac factors				
Right ventricular ejection fraction: severe	1.69	.0004	...	...
Tricuspid regurgitation: moderate/severe	...	...	1.37	.0178
Mitral regurgitation: moderate/severe	0.56	<.0001	...	...
Aortic regurgitation: moderate/severe	1.75	.0209	...	...
LVEF <20% (severe)	0.70	.0041	...	...
Pulmonary artery diastolic pressure	...	...	...	...
LVEDD	...	...	0.85	.0144
Noncardiac systems				
Intervention within 48 h-ventilation	1.84	.0043	...	...
Intervention within 48 h-dialysis	2.62	.0010	...	...
Blood urea nitrogen	1.02	<.0001	1.01	.0044
Sodium	0.96	.0005	...	...
International normalized ratio	...	...	1.23	.0003
Albumin	...	...	0.71	.0009
Aspartate aminotransferase	...	...	...	...
Severe diabetes	...	...	...	...
Ascites preimplantation	...	...	...	...
Surgical complexities				
Cardiopulmonary bypass time	...	...	...	...
Previous cardiac operation	1.47	.001	...	...
Modifier: temporary circulatory support	1.62	.0012	...	...

CF-HL, continuous flow - hybrid levitation; LVEF, left ventricular ejection fraction; LVEDD, left ventricular end diastolic dimension.

From Pagani FD, Cantor R, Cowger J, et al. Concordance of treatment effect: an analysis of the Society of Thoracic Surgeons Intermacs database. *Ann Thorac Surg.* 2022;113(4):1172-1182.

Major Adverse Events

The burden of adverse events, including associated readmissions, remains substantial for patients with durable LVADs. Within 1 year of LVAD implantation, only 30% of patients avoid hospital readmission.⁷² Major adverse events significantly influence subsequent survival, as conditional survival analyses indicate that these events are the primary drivers of mortality risk.⁶⁹

Certain adverse events associated with durable LVADs are predictable due to factors such as (1) the requirement for anticoagulation, (2) the presence of a mechanical device and percutaneous power lead, and (3) the progression of underlying cardiomyopathy. However, additional complications arise specifically from the physiologic and hematologic changes caused by continuous-flow LVAD support. Among these, mucocutaneous bleeding, most commonly gastrointestinal bleeding or epistaxis, affects 25% to 30% of patients within the first year of implantation^{73,74} and is the result of changes induced at the cellular level within the hematologic cascade (arteriovenous malformation and deficiencies in von Willebrand high-molecular-weight multimers). The risk of mucocutaneous bleeding has decreased with the current generation of durable LVAD technology, and studies have investigated modified antithrombotic protocols including withdrawal of antiplatelet therapy to further reduce bleeding risk (i.e., Antiplatelet Removal and Hemocompatibility Events with the HeartMate 3 Pump [ARIES HM3]).⁷⁵ Moderate or greater aortic insufficiency affects approximately 20% of patients within the first year of durable LVAD support. This condition is linked to an increased risk of recurrent HF hospitalizations and the progression of right HF.⁷⁵ Aortic insufficiency is believed to result from continuous LV offloading leading to a persistently closed aortic valve that is subjected to high shear stress on the arterial side of the leaflets, leading to subsequent leaflet mal-coaptation, microthrombus, inflammation, and calcific degeneration.

Table 20.4 outlines the incidence and rates of key adverse events during both the early phase (<90 days) and late phase (≥90 days) of durable LVAD support. While stroke and infection pose the highest mortality risks, even less severe complications, such as gastrointestinal bleeding and worsening chronic right HF, significantly affect survival and health-related quality of life. Notably, advances in contemporary LVAD technology have greatly reduced the risk of pump malfunction to approximately 0.01 events per patient-year and the risk of stroke to about 0.04 events per patient-year after the first 6 months of support.^{63,76}

A contemporary analysis of HeartMate 3 devices from the STS Intermacs registry examined the effect of specific adverse events during the first 6 months of LVAD support on subsequent mortality.⁷⁷ By multivariable analysis, dialysis (Hazard Ratio [HR] 31.4), stroke (HR 13.1), prolonged respiratory failure (HR 5.7), right HF (HR 5.0), and reoperation (HR 3.4) contributed to mortality at 180 days (all $P < .05$).

Neurologic Events. Severe neurologic injury, often resulting from embolization of particulate matter or air, is one of the most catastrophic *early* complications following LVAD implantation. The risk of air embolization is notably higher after device implantation compared to other types of cardiac surgery. This increased risk is due to potential air trapping in the left atrium or ventricle, as well as the pump itself being a significant source of air. Even with proper de-airing of the pump, negative pressures generated by the blood pump can draw in air through porous conduits or around cannulation sites if the left ventricle is severely underfilled. Key strategies to mitigate this risk include:

1. Continuous TEE monitoring of the left atrium, left ventricle, and aortic root during the terminal phases of CPB and for 10 to 15 minutes after discontinuation of bypass
2. Thorough de-airing of the pump
3. Use of carbon dioxide to flood the surgical field at the time of device implant and weaning

TABLE 20.4 Adverse Events Associated with Durable LVAD Support

Event	Period ^a	CONTINUOUS-FLOW LEFT VENTRICULAR ASSIST DEVICE			
		EVENT COUNT	ADVERSE	PATIENTS	
		(No.)	Event Rate ^b	No.	%
Rehospitalization (all-cause)	Early	6737	2.09	4669	33.5
	Late	30,618	1.55	8371	60.0
Arterial non-CNS thromboembolism	Early	80	0.025	77	0.6
	Late	56	0.003	49	0.4
Major bleeding	Early	3928	1.22	2788	20.0
	Late	5007	0.25	2639	18.9
GI bleeding	Early	1800	0.558	1355	9.7
	Late	3458	0.18	1831	13.1
Non-GI reoperation for bleeding	Early	670	0.21	603	4.3
	Late	128	0.006	118	0.8
Cardiac arrhythmia	Early	2873	0.89	2266	16.2
	Late	2093	0.11	1384	9.9

TABLE 20.4 Adverse Events Associated with Durable LVAD Support—cont'd

Event	Period ^a	CONTINUOUS-FLOW LEFT VENTRICULAR ASSIST DEVICE			
		EVENT COUNT	ADVERSE	PATIENTS	
		(No.)	Event Rate ^b	No.	%
Device malfunction/pump thrombus	Early	463	0.14	424	3.0
	Late	1644	0.083	1253	9.0
Device malfunction	Early	184	0.057	173	1.2
	Late	789	0.04	652	4.7
Pump thrombus	Early	287	0.09	262	1.9
	Late	891	0.045	705	5.1
Major infection	Early	4181	1.30	3123	22.4
	Late	8420	0.43	4386	31.5
MCS infection	Early	502	0.16	465	3.3
	Late	3167	0.16	1974	14.2
Non-MCS infection	Early	3093	0.96	2382	17.1
	Late	4100	0.21	2640	18.9
Hepatic dysfunction	Early	273	0.085	267	1.9
	Late	248	0.013	229	1.6
Myocardial infarction	Early	26	0.008	26	0.2
	Late	36	0.002	35	0.3
Neurologic dysfunction	Early	1556	0.48	1393	10.0
	Late	2161	0.11	1621	11.6
Stroke	Early	946	0.29	870	6.2
	Late	1356	0.069	1121	8.0
Pericardial drainage	Early	371	0.12	347	2.5
	Late	18	0.001	17	0.1
Renal dysfunction	Early	1506	0.47	1404	10.1
	Late	806	0.041	657	4.7
Dialysis	Early	65	0.020	64	0.5
	Late	27	0.001	27	0.2
Respiratory failure	Early	2213	0.69	1830	13.1
	Late	773	0.039	646	4.6
Venous thromboembolism	Early	217	0.067	206	1.5
	Late	71	0.004	67	0.5
Wound dehiscence	Early	142	0.044	135	1.0
	Late	52	0.003	47	0.3
Other serious adverse events	Early	3180	0.99	2106	15.1
	Late	2806	0.14	1717	12.3

Adverse event rates in 13,945 patients supported with a CF durable LVAD from January 1, 2017, through December 31, 2021, from the Society of Thoracic Surgeons Intermacs National Database.

^aEarly rates are within 90 days after implantation. Late rates are 90 days and beyond after device implantation.

^bRates are reported per patient-year of device support.

CNS, central nervous system; GI, gastrointestinal; MCS, mechanical circulatory support.

From Yuzefpolskaya M, Schroeder SE, Houston BA, et al. The Society of Thoracic Surgeons Intermacs 2022 annual report: focus on the 2018 heart transplant allocation system. *Ann Thorac Surg.* 2023;115:311-327 and Tedford RJ, Leacche M, Lorts A, Drakos SG, Pagani FD, Cowger J. Durable mechanical circulatory support: JACC scientific statement. *J Am Coll Cardiol.* 2023;82:1464-1481.

4. Appropriate support and observation of RV function after bypass to avoid sudden drops in RV output to the LVAD
5. Minimizing the use of vacuum pressures until the chest is closed

The potential for thromboembolic events during and after CPB can be greatly reduced by careful examination of the LV cavity by TEE prior to implant, and direct inspection of the LV cavity during implantation of the inflow cannula. Additionally it is important to assess the left atrial appendage for thrombus. Identification of thrombus in the left atrial appendage should prompt excision of the left atrial appendage with removal of the left atrial thrombus.

Later neurologic events are generally categorized as either hemorrhagic or thromboembolic. Neutralizing this complication requires detailed analysis of thromboembolic events and a better understanding of the role of the device itself, the patient's underlying medical condition, and the selection and management of specific anticoagulation regimens.

A relationship between clinical infections and subsequent thromboembolic strokes following MCS/D implant has been observed in multiple centers,⁷⁸ but a causal relationship and potential mechanisms such as activation of inflammatory mediators with procoagulant effect remain unproven.

Alterations in the underlying coagulation system that are not identified by the INR, partial thromboplastin time (PTT), and platelet count (see "The Coagulation System" under Biological Barriers) may contribute to development of hemorrhage or thromboembolic strokes post-MCSD. The imperfection of standard coagulation monitoring has prompted greater application of more detailed analyses of the coagulation system using thromboelastography and platelet aggregation studies.

In a comprehensive analysis of stroke in the long-term cohort of the MOMENTUM 3 study (189 HeartMate 3 and 172 HeartMate II), Colombo and colleagues reported 65 strokes (40 ischemic strokes and 25 hemorrhagic strokes) in 52 patients at a median of 131 (range, 1–733) days.⁷⁹ There was no difference in rate of stroke between 0 and 180 days of follow-up between the HeartMate 3 and HeartMate II. However, stroke incidence in the long-term period (181–730 days after LVAD implant) was 3.3 times higher for the HeartMate II group as compared to the HeartMate 3 group (HM3: 0.04 vs. HMII: 0.13 events per patient-year; odds ratio, 0.23; 95% CI, 0.08–0.63; $P = .01$). Treatment with the HeartMate 3 pump was the only independent predictor of lower stroke events. There was no direct association of blood pressure or antithrombotic regimens with observed stroke rates. The occurrence of a stroke event significantly lowered 2-year postimplantation survival regardless of subtype or initial severity of neurologic impairment compared with patients without a stroke ($43\% \pm 12\%$ for hemorrhagic stroke, $57\% \pm 9\%$ for ischemic stroke, $51\% \pm 11\%$ for disabling, and $51\% \pm 11\%$ for nondisabling compared with $85\% \pm 2\%$ 2-year survival for patients without stroke).⁷⁹ In a multi-institutional analysis of the HeartMate 3 implants through 2022 in the STS Intermacs US registry, freedom from any stroke with was 93% at 1 year, 89% at 3 years, and 87% at 5 years. This represents a dramatic improvement over prior devices. It is noteworthy, however, that the improvement is demonstrable only in the constant phase, with no yet demonstrable improvement in stroke freedom early after implant.⁵⁷

Infection. Major pump-related infections involve either the pump pocket or the driveline at the exit site. Driveline infections continue to plague long-term circulatory support and will undoubtedly be an ongoing limitation until totally implantable systems are available. The most important maneuver to minimize

late development of exit site infections appears to be total immobilization of the driveline at the exit site, preferably using a double stabilization technique to minimize transmitted driveline tension or movement at the exit site.

In a comprehensive report from the MOMENTUM 3 clinical study, Patel and colleagues reported that the cumulative number of infections among 1020 patients implanted with either the HeartMate 3 ($n = 515$) or HeartMate II ($n = 505$) pump was 1213 (634 HeartMate 3 and 579 HeartMate II) at 2 years, and that major infection occurred in 58% of patients with the HeartMate 3 and 56% of patients with the HeartMate II ($P = .57$).⁸⁰ Infections of a local nature unrelated to pump components were most common ($n = 681/1213$; 56%), followed by driveline-associated infection ($n = 329/1213$; 27%), sepsis ($n = 194/1213$; 16%), and other events ($n = 9/1213$; 0.7%). Bacterial pathogens were implicated in 806 of 1213 events (66%). Significant predictors of infection included sex (women vs. men; HR: 1.38, $P = .003$), preimplant use of intra-aortic balloon pump (HR: 1.33, $P = .02$), preimplant history of cardiac surgery (HR: 1.28, $P = .01$), and body mass index ≥ 30 (HR: 1.40, $P < .0001$). Most deaths in those with infection occurred owing to noninfectious causes.⁸⁰ Despite the improved freedom from several other major adverse events with the HeartMate 3 LVAD, no demonstrable improvement has been observed in freedom from infection. This is likely driven by the persistent need for the percutaneous driveline associated with frequent driveline infections. Further reduction in infections will await totally implantable systems.

Gastrointestinal Bleeding. Nonsurgical bleeding is the most frequent adverse event in patients supported with a CF LVAD.^{73,74} Bleeding is a major source of morbidity and an obstacle to improving outcomes, reducing costs, and expanding the public health impact of LVAD therapy. Gastrointestinal bleeding is one of the most common complications in LVAD recipients, with a prevalence ranging from 20% to 35%.^{73,74} While the relationship between gastrointestinal bleeding and increased mortality in LVAD patients has not been firmly established, gastrointestinal bleeding in this patient population is strongly associated with increased rates of hospitalization, thromboembolic events, morbidity, and healthcare-related costs.^{73,74} Patients who are admitted for one gastrointestinal bleeding event have a 20% risk for experiencing another bleeding episode, resulting in additional hospitalization.^{73,74} Recurrent bleeding events may result in repeated endoscopic procedures, temporary or permanent discontinuation or reduction in the intensity of anticoagulant or antiplatelet medications (potentially exposing patients to a greater risk of thromboembolic events), administration of multiple packed red blood cell units (increasing the risk of allosensitization), and prolonged hospital stays.

Growing evidence highlights the critical role of acquired VWF deficiency in bleeding complications associated with CF-LVADs. VWF, a multimeric plasma clotting protein, is essential for initiating primary hemostasis. Its activation is tightly regulated by shear stress, with normal intravascular levels ranging from 2 to 8 pascals (Pa) and activation occurring above 10 Pa. During activation, shear stress unfolds inactive, globular VWF into an elongated, active form, exposing binding sites for collagen and platelets, which promote thrombus formation. High-molecular-weight VWF multimers are particularly effective in clotting; however, they are also highly susceptible to enzymatic cleavage by ADAMTS-13, a VWF-specific protease. This cleavage reduces high-molecular-weight

multimers into low-molecular-weight fragments that are ineffective in clotting.

CF-LVADs produce shear stress levels far exceeding physiologic values, reaching up to 1500 Pa due to the impeller spinning at thousands of revolutions per minute to propel blood forward. This excessive shear stress leads to pathologic VWF degradation through two key mechanisms:

1. **Mechanoenzymatic degradation (major mechanism):** LVAD shear stress activates VWF, exposing cleavage sites for ADAMTS-13, which enzymatically degrades high-molecular-weight multimers.
2. **Nonenzymatic degradation (minor mechanism):** Shear stress alone, independent of ADAMTS-13, mechanically breaks VWF multimers into smaller fragments.

Simultaneously, LVAD shear stress activates platelets, causing partial degranulation and depletion of ultra-high-molecular-weight VWF multimers, a critical reserve needed for primary hemostasis.

The absence of high-molecular-weight VWF multimers is directly linked to bleeding risk. This phenomenon mirrors other conditions such as type IIA and type IIB von Willebrand syndrome, where patients bleed due to the absence of functional VWF multimers, often presenting with gastrointestinal bleeding from angiodysplasia. Similarly, in Heyde syndrome, elevated shear stress from a stenotic aortic valve degrades high-molecular-weight VWF multimers, leading to bleeding. The mechanisms of VWF degradation in Heyde syndrome and LVAD patients are strikingly similar, underscoring the importance of shear stress in VWF-related bleeding complications.

Laboratory derangements and clinical presentation of LVAD patients are similar to patients with type II von Willebrand syndrome and Heyde syndrome. Patients exhibit absence of high-molecular-weight VWF multimers, reduced VWF collagen binding and Ristocetin-induced platelet aggregation, and prolonged bleeding times that are independent of INR and antiplatelet therapy. Not surprisingly, a dose-response relationship exists. LVAD patients with the least high-molecular-weight VWF multimers and lowest VWF function are most likely to bleed.

In a study of adult patients receiving the HeartMate 3 LVAD as a primary operation in the Society of Thoracic Surgeons' Inter-macs National Database from January 2017 to December 2020, Hammer and colleagues investigated the relationships between preimplantation characteristics and postimplant gastrointestinal bleeding.⁸¹ Of 6425 patients who received a HeartMate 3 LVAD, 1010 (15.7%) patients experienced a gastrointestinal bleeding event. Thirteen preimplantation factors were independent predictors of post-LVAD. A risk score was created from these factors and calculated for each patient. By 3 years postimplant, gastrointestinal bleeding occurred in 11%, 26%, and 43% of low-, medium- and high-risk patients, respectively. Experiencing one post-LVAD gastrointestinal bleeding event was associated with an increased risk for further gastrointestinal bleeding events, with 33.9% of patients experiencing at least one recurrence. While post-LVAD gastrointestinal bleeding event was associated with mortality, there was no relationship between number of gastrointestinal events and death.⁸¹

Right Heart Failure. In general, about 5% of LVAD implants are accompanied by severe enough right HF to require RVAD support at the time of or soon after LVAD implant. Among patients with isolated LVADs, about 20% of patients have mod or worse right HF (two or more components of CVP >14, edema, ascites, inotropes) at 1 month, and the later right HF develops,

the more likely it is to persist. Among those with new onset right HF at 6 months, about 25% will have persisting right HF at 1 year.⁸² The prevalence of right HF during follow-up (out to 3 years) is about 8% to 10%, and persisting symptomatic right failure is associated with a marked reduction in survival.

Patient Reported Outcomes

A primary goal of durable LVAD therapy is to help patients “live well” while on support and minimize hospital readmissions. All major clinical trials evaluating durable LVADs have demonstrated significant improvements in health-related quality of life (HRQOL) and functional capacity. In the MOMENTUM 3 trial, HRQOL measured by the Kansas City Cardiomyopathy Questionnaire (KCCQ) improved by 28 points within 6 months of LVAD implantation—a gain far greater than what is typically observed with medical therapies. Patients also reported marked improvements in mobility, self-care, and reductions in anxiety and depression, with these benefits largely sustained for up to 24 months.²⁹

Functional capacity showed similar long-term improvements. Before LVAD implantation in the MOMENTUM 3 trial, 95% of patients using the HeartMate 3 were classified as New York Heart Association (NYHA) functional class IV. After 6 months of support, 77% of patients improved to NYHA class I or II, with these outcomes remaining stable at 24 months. Six-minute walk distances increased by 140 to 230 meters postimplantation, although this measurement is partly influenced by the inability of severely ill patients to perform the test preimplant.

These enhancements in functional capacity and HRQOL were consistent across different devices and were observed even in patients who experienced adverse events, albeit to a lesser extent. This highlights the broad and sustained benefits of durable LVAD therapy for patients with advanced HF.

Despite the consistent improvement in quality of life metrics following LVAD support, self-reported metrics reveal an important negative impact of the number and types of adverse events, which can have a dramatic effect on such measures as the virtual analog scale, as noted by Kilic and colleagues.⁸³

The data consistently show that patients generally thrive while on durable LVAD support. It is believed that shifts in health-related quality of life expectations occur, leading patients to reprimoritize or reset their expectations, focusing more on survival and the unanticipated benefits that come with LVAD support.

Indications and Patient Selection for Durable Mechanical Circulatory Support

Durable MCS, usually with a dLVAD should be considered the mainstay therapy for patients with American Heart Association Stage D HF. The 2022 American College of Cardiology/American Heart Association/Heart Failure Society of America Heart Failure Guidelines recommend dLVAD therapy to improve functional status, health related quality of life, and survival in select patients with advanced heart failure with reduced ejection fraction (HFrEF) with NYHA functional class IV symptoms who are deemed to be dependent on continuous inotropes or temporary MCS (Class of Recommendation [COR]: I, Level of Evidence [LOE]: A). For patients with HFrEF with NYHA functional class IV symptoms despite guideline-directed medical therapy but who do not yet require continuous inotropes or temporary mechanical circulatory,

similar improvements in symptoms, functional class, and mortality reduction are seen (COR: 2A, LOE: B).⁷⁶

Identification of patients with advanced HF who may benefit from durable MCS is challenging. Patients are often referred too late to advanced HF centers, and many times are not referred at all. A number of findings are suggestive of advancing HF that can be used to identify patients for referral for durable MCS. Given the complexities and ever-changing thresholds for MCS candidacy, providers should have a low threshold for referral to advanced HF centers for evaluation.

Implantable durable LVADs are indicated for selected patients with advanced HF who have both poor HRQOL and medium-term prognosis. Reversible causes of HF and significant comorbidity or other life-limiting illnesses need to be excluded before considering durable LVAD therapy. However, carefully selected patients with comorbidity burden that would typically exclude consideration for clinical trial enrollment still experience important survival benefit with durable LVAD therapy.

Irreversible end-organ damage (hepatic, renal, and neurologic) is generally considered an absolute contraindication. Reports of patients on durable MCS requiring permanent dialysis describe poor prognosis compared with nondialysis counterparts.

Advanced Age

As with any HF therapy, advanced age is a predictor of adverse outcome and is associated with a higher mortality risk. Advanced age may be considered a relative contraindication to dLVAD therapy. However, a recent report from STS Intermacs reported that 4.8% of dLVAD recipients between 2010 and 2020 were over the age of 75 years. Five-year mortality rates were higher in the >75-year-old population compared with younger recipients. However, in the more contemporary implant period 2017 to 2020, 1- and 3-year survival in recipients >75-year-old were 71.2% and 55.7%, respectively, survival rates that are much higher than would be expected from guideline-directed medical therapy. Most importantly, functional capacity and HRQOL metrics improved similarly in both the older and younger recipients.⁸⁴

Obesity

Obesity is common in the HF population, and many patients who are not candidates for heart transplantation (HTx) due to obesity (BMI >35–40 kg/m²) is a contraindication to HTx at most centers) are considered for dLVADs. Center-specific obesity thresholds for dLVAD candidates are quite varied, and very significant obesity is considered a relative contraindication by some, but not all, centers.

Psychosocial Risk

Social and behavioral health evaluations remain an integral part of the durable MCS evaluation process. Limited social support, substance abuse, noncompliance, psychiatric disease, and cognitive impairment can be barriers to candidacy. The complexity of devices and adverse events associated with early generation durable MCS led many implanting centers to develop strict guidelines regarding adequate caregiver support and patient compliance to ensure excellent durable LVAD outcomes. Despite the improvements in technology that make devices more reliable, some centers have not adjusted policies for psychosocial requirements. Careful relaxation of these requirements (e.g., eliminating the

need for a 24/7 care partner, reducing the intensive oversight of a care partner to the first 6 to 8 weeks after LVAD implant, and resuming driving privileges earlier during recovery) may result in both increased offerings of durable MCS to appropriate patients in-need as well as improving racial and gender disparities.

Right Heart Failure

Most durable MCS devices, except for the TAH, are designed to support the left ventricle only. In many cases, unloading the left ventricle leads to reduced pulmonary congestion and RV afterload, along with improvements in systemic blood flow, which in turn can enhance RV function. The presence of isolated RV dysfunction should never disqualify a patient from referral or candidacy for durable MCS. However, patients with significant preimplant RV dysfunction, especially when it is disproportionate to LV dysfunction, are at increased risk of RV failure after MCS implantation. Predicting RV failure following durable MCS remains challenging, and while several risk scores have been developed, none has shown strong performance in independent validation cohorts. Some studies suggest that evaluating the preimplant RV response to LV unloading may help predict postimplant RV failure.⁸⁵ Although none of these prediction tools should, on their own, exclude patients from consideration for durable MCS, they can be valuable in shared decision-making with potential candidates. They may also help identify higher-risk patients for further study into methods or therapies aimed at preventing postimplant RV failure. It is important to recognize that certain factors during the MCS implantation process itself can worsen intrinsic RV function, which adds to the challenge of predicting postimplant RV failure.⁷⁶

Timing of Therapy

Appropriate timing of durable MCS implant relies on early referral patterns as well as risk prediction of worsening HF. Both remain imperfect. Recipients with the highest acuity prior to implant (lowest STS Intermacs Patient Profiles) have worse outcomes.^{62–65} Although not always possible, implantation of durable MCS ideally should occur prior to the development of end-organ damage related to HF.

Risk Scores

Predicting outcomes, including survival, with durable MCS remains challenging.⁸⁶ The HeartMate 3 Risk Score, developed from data in the MOMENTUM 3 trial, utilizes six simple preimplant metrics: (1) age, (2) history of prior valve procedures or coronary artery bypass grafting, (3) LV size, (4) serum sodium levels, (5) right atrial pressure to pulmonary artery wedge pressure ratio, and (6) serum blood urea nitrogen.⁸⁶ These factors help stratify potential recipients into categories of better-than-average, average, or lower-than-average expected survival. While this tool may assist in shared decision-making with patients, it still requires validation in real-world cohorts.

MCS Support Strategy: Short-Term and Long-Term Applications

Traditionally, the intent of durable LVAD use has been categorized into various strategies based on preimplant patient characteristics (bridge to transplantation [BTT], bridge to decision/candidacy [BTC], bridge to recovery [BTR], or destination

therapy [DT]). In the absence of absolute and irreversible contraindications to heart transplantation, the reality is that these clinical categories are arbitrary and often change following durable MCS implantation. As durable MCS extends life and improves HRQOL of patients in all classifications, some have called to abandon this categorization altogether.

Special Situations and Controversies

Adult Congenital Heart Disease, Including Fontan Circulatory Failure

Adults with congenital heart disease (ACHD) may require VAD therapy. Often, transplant eligibility is not known, because the reversibility of the end-organ dysfunction may not be known until cardiac output has improved. Overall, ACHD patients with a VAD have an increase in mortality at 1 year compared to non-ACHD patients (27.6% vs. 15.9%).⁸⁷ However, despite the complex anatomy/physiology, adverse event rates and improvement in functional class and HRQOL are similar in cohorts with and without ACHD. Specifically, VAD implantation for both pediatric and adult patients with Fontan circulation has become an acceptable practice with favorable outcomes of 79% at 1 year.⁸⁸

Myocardial Recovery

Durable LVADs significantly reduce volume and pressure load on the heart, enhance systemic perfusion, and help reverse stress-induced compensatory mechanisms that can harm patients with advanced HF. When combined with guideline-directed medical therapy, LVAD support offers additional benefits for cardiac structure and function. The process of reverse cardiac remodeling and improvement with this combined approach is not an all-or-nothing occurrence but rather progresses along a spectrum of responses, including full responders, partial responders, and nonresponders.⁷⁶

Predictors of myocardial improvement with durable LVAD intervention include: young age, duration of HF, nonischemic cardiomyopathy, and other clinical characteristics such as rotational mechanics and other echocardiographic criteria.⁸⁹ In patient populations with clinical predictors of response, the incidence of myocardial recovery sufficient to allow for weaning off the durable LVAD has been found to be around 50% or higher. The ideal goal of any implant that does not require cardiac replacement is to enable myocardial recovery, which may occur days, months, or even years after the MCSD implant. While the aim is to achieve sufficient myocardial recovery to allow for successful device explant, the successful explant of durable long-term devices has remained rare thus far.

Recovery outcomes are generally more favorable in cases of acute HF caused by recent onset nonischemic cardiomyopathy or acute myocarditis. Even patients with acute myocarditis presenting in cardiogenic shock and requiring emergent mechanical support often experience gradual improvement. Management of these patients should prioritize recovery and the likelihood of device explantation rather than moving prematurely toward cardiac transplantation.

SECTION II: PEDIATRIC MECHANICAL CIRCULATORY SUPPORT

Although adult pump configurations are suitable for the support of adolescents, small children and infants require appropriately

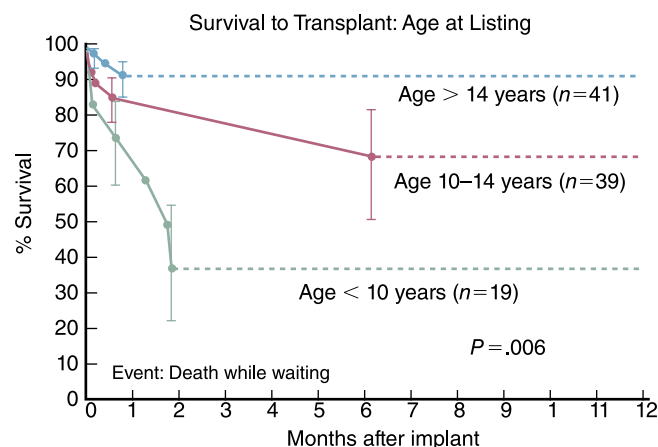
sized devices. This section will focus on those aspects of MCS which are unique to pediatric patients.

Historical Note

The application of durable MCS systems in pediatric patients was initially restricted to adult devices implanted in adolescents as bridging support for heart transplantation. When the marked inferiority of extracorporeal membrane oxygenator (ECMO) support compared to VADs became evident in the adult population, interest in appropriate devices for children became a focus of the pediatric HF community. In 2006, Blume and colleagues reported the dismal survival of MCS support among patients <10 years of age, for whom adult devices were not suitable (Fig. 20.27).⁹⁰ The first durable pediatric VADs were developed in Germany, based on the concept of paracorporeal pneumatically driven pulsatile technology. The Berlin Heart EXCOR system was introduced in 1988 in Europe, initially as an adult paracorporeal pump. In 1990, the support of an 8-year-old boy to successful heart transplantation with an adult EXCOR pump⁹¹ stimulated the development of pediatric size devices. The Berlin Heart AG group created a series of pediatric pumps with small pump chambers, with appropriate size cannulae and polyurethane valves. Roland Hetzer implanted the first infant pediatric pump in 1992.⁹² The first Berlin Heart EXCOR pediatric device in the United States was implanted by Copeland in 2000. This device gained FDA approval for use in the United States in 2011.

As CF technology gained acceptance in the adult population, a few pediatric applications were reported. Miera and colleagues reported the first use of an implantable CF pump in a child in 1998, using the MicroMed DeBakey pump. This device was briefly applied in the pediatric population, but the pump was flawed with an unacceptable incidence of pump thrombosis. The HeartWare VAD became the mainstay of CF device application in older children for the next decade after Miera reported the first such implant in Europe in 2011,⁹³ and Kirklin reported the first HVAD pediatric implant in the United States in 2012, in which a 9-year-old girl was supported for 60 days to successful heart transplantation.⁹⁴

Successful support of patients with single ventricle (SV) has provided the greatest challenge in pediatric MCS. Kirklin first successfully bridged a baby (age 15 months) with functional SV to successful heart transplantation with a durable pump (Berlin Heart EXCOR) in 2007 (7 weeks of support).⁹⁵ Use of adult LVADs to support the systemic circulation post-Fontan



• **Figure 20.27** Survival on mechanical circulatory support stratified by age.

was established early, but isolated support of the pulmonary circulation with a durable VAD to successful heart transplant was first accomplished with a Berlin EXCOR in 2008 by Pretre in a 27-year-old patient post-Fontan (11 months of support).⁹⁶

Attempts to develop a size-appropriate implantable CF-LVAD for small children and infants were initiated in the United States with the (NHLBI) Pediatric Circulatory Support Program, initiated in 2004. Early funding targeted the development of five circulatory support systems: three CF pumps, one pulsatile pump, and one ECMO system.⁹⁷ Ultimately, only the Infant Jarvik 2015 (Jarvik Inc, New York) implantable CF device was selected for a clinical trial. The Pumps for Kids and Infants (PUMPKIN) NIH clinical trial featured the Infant Jarvik pump in a randomized trial with the Berlin EXCOR paracorporeal pump. In 2014, the initial Infant Jarvik 2000 was not approved by the FDA due to excessive hemolysis during in-vitro testing. The revised Infant Jarvik 2015 showed significantly less in-vitro hemolysis, and Adorisio and colleagues reported 3 pediatric implants in 2021.⁹⁸ The mean weight was 11 kg, and the mean time on device was 338 days. All patients had dilated CM, and all pumps were successfully explanted for recovery. Although initiated in 2015, the trial never gained sufficient traction to complete enrollment.

The Interagency Registry for Mechanically Assisted Circulatory Support (INTERMACS), initiated and funded by the US national heart, lung, and blood institute, established a pediatric component (PEDIMACS) in 2012, in which data collection and analytics were maintained at the University of Alabama at Birmingham under the direction of James Kirklin. The first analysis reported on the first 200 device implants.⁹⁹ By 2023, over 1300 devices had been analyzed in the registry.¹⁰⁰

Clinical Features

The increased application of VAD support in the pediatric population during the decade following 2010 is evidenced by a report from the Society of Thoracic Surgeons PEDIMACS registry for pediatric VAD support in the United States between 2012 and 2020. Over that 9-year span, over 1200 devices in more than 1000 patients were reported to the registry. The most common cardiac diagnoses among patients supported as bridge-to-transplant were cardiomyopathy (58%), congenital heart disease (25%), and myocarditis (10%).^{100,101}

Cardiomyopathies

The vast majority of childhood cardiomyopathies (CM) are dilated CM. Less common causes include LV noncompaction, restrictive CM, and hypertrophic CM. Dilated CM with decompensation is well-supported with either CF or paracorporeal devices. Restrictive or hypertrophic CMs pose a greater challenge because of marked diastolic dysfunction and often small LV chamber size that can limit VAD filling.

Myocarditis

Acute/fulminant myocarditis can lead to rapidly progressive HF and cardiogenic shock, which can be fatal if not promptly reversed. MCS therapy plays an important role in the treatment strategy. Of importance, rapid resolution of advanced HF is common after a period of support, with favorable long-term prognosis. Acute myocarditis accounts for about 2% of HF admissions in

the United States, with a 30-fold increase in the risk of cardiac arrest compared to other forms of HF.¹⁰²

Congenital Heart Disease

Congenital heart disease (CHD) is the most common cause of HF in pediatric patients, particularly among infants, with CHD accounting for >50% of infant heart transplants. However, pediatric patients with CHD are less frequently supported with MCS prior to transplant than CM patients. Compared to the heart with CM, CHD poses additional challenges related to positional considerations, anomalies of systemic and pulmonary venous return, previous cardiac surgery, pulmonary vasculature abnormalities, the frequency of SV physiology, and multiple challenges with failing Fontan patients.

Neonates and Infants

Neonates are generally <5 kg and infants <10 kg, and these small sizes greatly limit options for MCS therapies. Currently the only durable device available for these age groups is the Berlin Heart EXCOR, although a number of centers have employed CF extracorporeal devices using the EXCOR cannulae to provide extended support.

Modes of Support

In the pediatric population, the major MCS support is through ECMO, temporary devices, and durable pumps. The application of ECMO will not be addressed in this chapter. Based on an analysis of the STS PEDIMACS US database for pediatric MCS, implantable CF devices account for 40% of implants (most commonly in older children and adolescents), paracorporeal pulsatile (PP) for 30%, paracorporeal continuous (PC) flow for 25%, and percutaneous devices for 5%.¹⁰³ The devices most commonly used in pediatric patients are summarized in [Table 20.5](#).¹⁰⁴

Temporary Ventricular Assist Device Support

Temporary VAD support can be generally defined as support intended for up to 30 days as a bridge to recovery, transplantation, transplant decision, or placement of a durable VAD. The most commonly used temporary devices in infants and small children are the Rota Flow (Maquet) centrifugal pump and the CentriMag/PediMag (Abbott) magnetically levitated devices. Temporary support is usually selected with or without prior ECMO support when neurologic or other organ system dysfunction is severe enough to limit the application of more durable devices or the cardiac dysfunction is deemed to be potentially readily reversible. Examples include acute myocarditis, early graft dysfunction following transplantation, and acute cardiac allograft dysfunction secondary to acute rejection. Temporary LVAD support with direct ventricular cannulation usually provides greater systemic ventricular decompression compared to ECMO alone. The few specific contraindications to temporary VAD support include ongoing poorly controlled bleeding, uncontrolled infection, contraindications to anticoagulation, and vascular limitations for cannulation or access.

Paracorporeal Pulsatile Support

Effective bridging of infants and children to transplantation requires availability of a variety of pump sizes suitable for the

TABLE 20.5 Mechanical Circulatory Devices Used in Children and Adolescents

Device	Manufacturer	Pediatric FDA Indication	Type	Output	Patient Size Industry Recommendation	APPROVAL	
						Support Time	Configuration
Short-term VAD							
RotaFlow™	Getinge		PC/CF	Up to 10 LPM	All	6 hr (US)/up to 30 d (Europe) ⁶	LVAD, RVAD, BIVAD
PediMag™	Abbott	X	PC/CF	Up to 1.5 LPM	<20 kg	LVAD: 6 hr (US), 30 d (Europe)/RVAD: 30 d	LVAD, RVAD, BIVAD
CentriMag™	Abbott		PC/CF	Up to 10 LPM	Not studied in patients <18 yrs	LVAD: 30 d (US), 30 d (Europe)/RVAD: 30 d	LVAD, RVAD, BIVAD
TandemHeart™	LivaNova		PC/CF	Up to 5 LPM	>1.3 m ²	6 hr	LVAD, RVAD, BIVAD
Tandem Life Protek Duo™	LivaNova		PC/CF	Up to 4.5 LPM	Requires 29F sheath	6 hr	RVAD
Impella 2.5™, CP™, 5.0™, 5.5™	Abiomed		IC/CF/IV	2.5: up to 2.5 LPM/CP: up to 4.3 LPM/5.0: up to 5.0 LPM/5.5: >5.5LPM	2.5, CP: advisory board >1.0 m ² 5.0, 5.5: advisory board >1.5 m ²	4 day Impella 2.5, CP: to 14 d (Impella 5.0 and 5.5)	LVAD
Impella RP™	Abiomed		IC/CF/IV	Up to 4.0 LPM	>1.5 m ²	14 d	RVAD
Long-term VAD/TAH							
Berlin Heart EXCOR®	Berlin Heart	X	PC/PF	0.6-8 LPM	>2.2 kg (smallest patient)	BTT (US), BTT, DT (Europe)	LVAD, RVAD, BIVAD
HeartWare HVAD System™	Medtronic		IC/CF	2-10 LPM	≥1.2 m ²	BTT, DT	LVAD
HeartMate 3 System™	Abbott		IC/CF	Up to 10 LPM	≥1.2 m ² (Momentum trial inclusion criteria)	BTT, DT or short and long term	LVAD
Jarvik 2015™	Jarvik		IC/CF	0.5-3 LPM	Study cohort 8-30kg	No approval	
SynCardia TAH 50cc™	SynCardia	X	C/PF	Up to 7.5 LPM	>1.2 - 1.85 m ² Needs room in the chest	BTT	BIVAD
SynCardia TAH 70cc™	SynCardia		C/PF	Up to 9.5 LPM	≥1.7 m ² Needs room in the chest §	BTT/DT	BIVAD
FDA, Food and Drug Administration; VAD, ventricular assist device; PC/CF, paracorporeal continuous flow; LPM, liters per minute; LVAD, left VAD; RVAD, right VAD; BIVAD, biventricular VAD; d, days; CP, cardiac power; IC/CF, intracorporeal continuous flow; IV, intraventricular; RP, right percutaneous; BTT, bridge to transplant; DT, destination therapy; C/PF, cardiac pulsatile flow. From Lorts A, Conway J, Schweiger M, et al. ISHLT consensus statement for the selection and management of pediatric and congenital heart disease patients on ventricular assist devices Endorsed by the American Heart Association. <i>J Heart Lung Transplant</i> . 2021;40(8):709-732.							

anticipated cardiac output. To address the challenge of providing mechanical support in small patients, the Berlin Heart EXCOR was developed to include multiple pump sizes. The Berlin Heart EXCOR is a PP pump with an electropneumatic drive system and is the only durable device approved for pediatric implantation in the United States by the FDA. Pediatric blood pumps are available in sizes of 10, 15, 25, and 30 mL as well as larger sizes (50, 60, and 80 mL) for adolescents and adults (Fig. 20.28). A variety of silicone inflow and outflow cannulae are available in sizes 5, 6, 9, and 12 for aortic, atrial, and ventricular cannulation to support

patients from neonates through adults (Fig. 20.29). The device can be employed for isolated LV support, RV support, or biventricular assistance.

The pump contains polyurethane trileaflet inflow and outflow valves. Of importance, the blood chambers and polyurethane ports are transparent, allowing direct visualization for detection of thrombus formation. When thrombus is detected (Fig. 22.30), pump exchange can be accomplished at the bedside. Paracorporeal pneumatically driven volume displacement pumps are particularly prone to thrombus forma-



• **Figure 20.28** Multiple available cannula and pump sizes available for implants with the Berlin Heart EXCOR device. (© Berlin Heart GmbH, used with permission.)

tion when either pump speed is low (≤ 50 beats per minute) or when there is consistent incomplete filling of the blood chamber.

Selection of pump size is of special importance; an undersized pump may incompletely decompress the ventricle and provide inadequate cardiac output, whereas an oversized pump is more likely to generate pump thrombus secondary to incomplete filling and low pump rates. In general, a 10- or 15-mL pump is selected for infants up to about 9 kg, a 25- or 30-mL pump between 10 and 25 kg, and a 50-mL pump for children and adolescents over about 30 kg. In selecting pump size for infants between about 8 and 12 kg, it is important to remember that the 10-mL pump will produce a maximum flow of 1 to 1.2 L/min. If calculated/desired cardiac output exceeds this value, the 10-mL pump may inadequately decompress the left ventricle, with

resultant important elevation of left atrial (and potentially pulmonary arterial) pressure.

Implantable Continuous Flow Devices

Available implantable CF devices that have been implanted in pediatric patients are adult devices, which are discussed in Section I under “Durable Circulatory Support Devices.”

Technique of Operation

Cannulation Strategies for Temporary Support

In older children and adolescents, percutaneous options may be available. The TandemHeart (Cardiac Assist, Inc., Pittsburgh, PA)



• **Figure 20.29** The EXCOR ventricular assist device. (Courtesy Berlin Heart EXCOR.)



• **Figure 20.30** Thrombus on outflow valve of Berlin Heart.

represents an option for support with peripheral arterial and venous cannulation and a transseptal catheter option to decompress the left atrium. LV temporary support with transaortic valve devices like the Impella (Abiomed, Inc., Danvers, MA) is also an option for older children and adolescents, with the 2.5 or 5.0 version. Unless implanted via central cannulation, these devices usually require special procedures to prevent peripheral arterial obstruction and severe distortion/damage to the vessel upon device removal. To avoid the potential need for complex arterial reconstruction, a side graft of Dacron or polytetrafluoroethylene (Gore-Tex) facilitates safe arterial insertion. The graft is sewn to the artery, and, upon device removal and extraction of any thrombus, the graft is clamped close to the arterial insertion, divided, and oversewn.

In small children and infants, central cannulation is required, usually with CPB. Standard CPB cannulae or Berlin Heart EXCOR cannulae are most often employed and connected to centrifugal flow pumps. Left-sided support usually includes left (systemic) ventricle or left atrial cannulation and an outflow cannula to the ascending aorta, placed near the sinutubular junction to leave space for subsequent cannulation at the time of CPB for heart transplantation. RV support usually includes right atrial and pulmonary artery trunk cannulation. Another option in the presence of right HF is to create a mixed circuit with a separate small jugular venous cannula that is connected to the circuit before the pump. In this situation, a membrane oxygenator may need to be added after the pump to avoid marked desaturation.

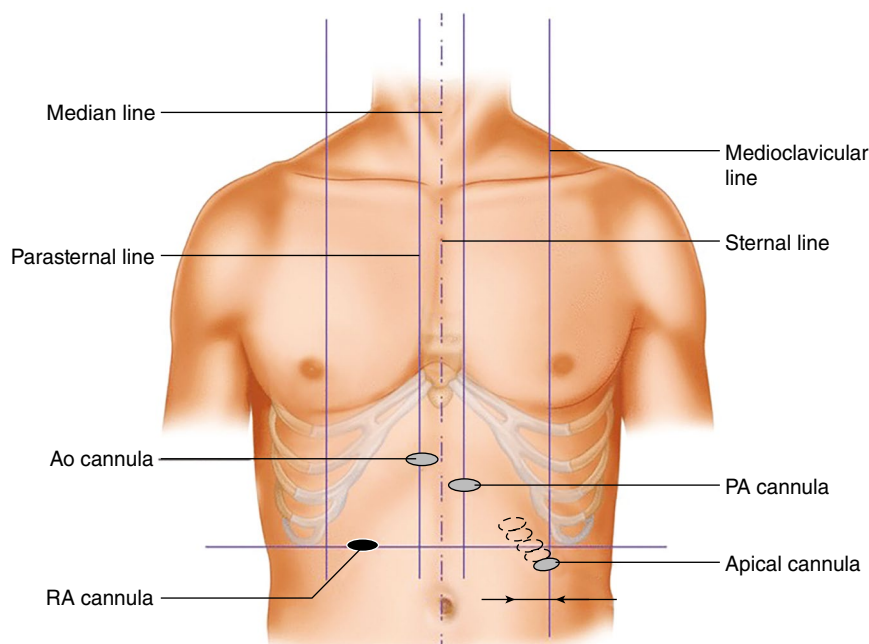
Paracorporeal Pulsatile Ventricular Assist Devices (Berlin Heart EXCOR)

The variety of pump and cannula sizes and configurations are discussed previously under “Modes of Support.” After selection of pump and cannula size and configurations, standard preparations are initiated for establishing CPB. Optimal exposure is obtained through a median sternotomy in most instances. Particularly in reoperative settings, the right neck and groins should be included in the surgical field. Axillary artery exposure or suprasternal exposure of the brachiocephalic artery may be advisable in high-risk situations in which the cardiac mass is likely to be adherent to the underside of the sternum. Especially in infants and small children, pen marks on the skin are useful to identify exit sites for the cannulae. Skin marking should always consider possible placement of a BiVAD configuration in case an RVAD is needed (Fig. 20.31). The LV apical cannula exits farthest to the left, at least several centimeters below the costal margin. The aortic and pulmonary outflow cannulae exit several centimeters below the xiphoid with the aortic to the right of the midline and the pulmonary artery cannula to the left. At least several centimeters of skin should be preserved between each cannula site to prevent skin erosion and breakdown.

Cannula positioning is crucial in order to achieve unobstructed pump filling, which is particularly important since the pump works best in fill-to-empty mode. When the failing ventricle is dilated, proper positioning can be readily achieved. However, in settings of ventricular hypertrophy or restrictive physiology, left atrial cannulation is preferable, especially since the left atrium is usually dilated in these conditions.

The aortic cannulation site for CPB should be as distal as possible to leave room for the EXCOR arterial outflow cannula on the proximal ascending aorta. Venous cannulation for CPB is usually with a single venous cannula in the right atrial appendage.

- Cannulae exit sites
- Best marked before starting surgery



• **Figure 20.31** Schematic drawing of optimal exit sites for Berlin Heart EXCOR cannulae in the biventricular configuration. Ao, aorta; PA, pulmonary artery; RA, right atrium; LV, left ventricle. (Reproduced with permission, Berlin Heart GmbH.)

If concomitant intracardiac procedures are planned, biatrial cannulation is utilized. All intracardiac holes should be repaired except for very small VSDs. Mitral valve competence is important since ventricular contraction into a regurgitant mitral valve during pump systole may result in severe mitral regurgitation with transient high left atrial pressure and possible pulmonary edema. Moderate or worse aortic regurgitation should be repaired at the time of implant. In the absence of concomitant procedures, the entire implant procedure can be performed on a beating heart with strong suction on an aortic needle vent. Alternatively, a brief period of cross-clamping with cardioplegic arrest can be employed to facilitate creation of the LV apex hole for the inflow cannula, after which the cross-clamp is removed with strong suction on the aortic needle vent while the LV apex is open without ejection. If concomitant intracardiac procedures are planned, aortic cross clamping with standard methods of myocardial protection are employed.

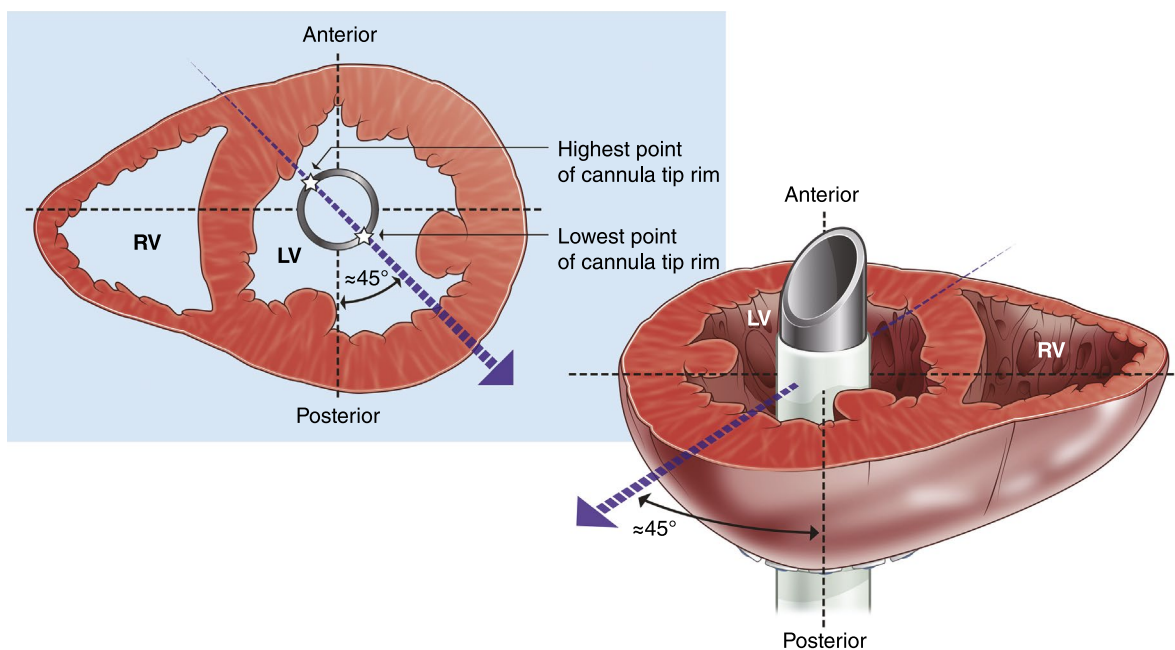
Apical LV cannulation is generally performed first in the cannulation sequence. Positioning of the apical cannula is the primary determinant of adequate pump filling. The heart is elevated on sponges to expose the LV apex, and the implant site is usually 1 to 2 cm to the left of the left anterior descending artery and just anterior to the LV apex. A full-thickness core of LV muscle is excised to accommodate the cannula, and a circular series of pledgetted sutures are placed around the cannulation site. The sutures are passed through the cannula sewing ring and secured. The orientation of the cannula must be such that the longer end of the tip is oriented toward the anterolateral LV wall with the oblique opening facing toward the mitral valve orifice (Fig. 20.32). Tissue sealant is placed around the cannulation site and a Gore-Tex membrane ring is placed over the apical site to facilitate device removal at the time of transplant, since the sewing ring becomes densely adherent to surrounding tissues.

The cannula is tunneled through the skin at the selected site to avoid distortion of the ventricle and any angulation of the cannula. This usually requires opening the pericardium laterally to the left, taking care to stay away from the phrenic nerve. The heart is transiently filled with blood to fill the cannula with no air retained in the cannula. The cannula is clamped externally, taking care to clamp only on the velour to avoid clamp damage to the remainder of the cannula.

If left atrial cannulation is selected, the skin exit site is carefully selected and the cannula is tunneled and brought out through the skin prior to inserting the cannula to avoid tension on the suture line when tunneling the cannula. Two purse-string sutures with multiple pledgets are placed to secure the cannula.

This aortic site for the outflow cannula should be placed as proximal as possible to facilitate cannulation and aortic reconstruction at the time of transplant. The silicone is beautifully designed to maintain its shape and avoid compression, but it is difficult to achieve a hemostatic anastomosis in a small patient. Therefore, it is recommended to routinely attach a short segment of slightly beveled Gore-Tex vascular graft to the end of the arterial cannula, secured with one or two silk ties (Fig. 20.33).¹⁰⁵ The cut end of the vascular graft is then anastomosed end-to-side to the right lateral aspect of the ascending aorta using one or two layers of continuous polypropylene suture. An accurate, hemostatic anastomosis is critically important, as this is a common site of postoperative bleeding. In small infants, more space on the ascending aorta may be made available by cannulating for CPB through a Gore-Tex graft anastomosed to the brachiocephalic artery as is common in neonatal operations involving the ascending aorta.

When biventricular support is needed, the LVAD is generally implanted first. The RV device can receive inflow from either the right ventricle or right atrium. The position of skin exit for RV and LV inflow and outflow cannulae on the anterior abdomen



• **Figure 20.32** Desirable orientation of the left ventricular Berlin Heart EXCOR apical cannula. The oblique opening of the cannula tip should face directly into the mitral valve to optimize drainage. RV, right ventricle; LV, left ventricle. (Reproduced with permission, Berlin Heart GmbH.)



• **Figure 20.33** Interposition of a vascular graft sutured on the distal end of the Berlin Heart EXCOR arterial cannula. (Reproduced with permission from Botha P, Hasan A, Perri G, Filippelli S, Griselli M. Modified technique for the implantation of Berlin Heart EXCOR ventricular assist device in children. *World J Pediatr Congenit Heart Surg.* 2012;3(3):373-377.)

must be carefully planned to allow appropriate spacing of the cannulae. The outflow cannula, usually with a Gore-Tex extension, is placed somewhat laterally on the pulmonary trunk to avoid sternal compression. The right atrial cannula should be positioned low in the atrial wall opposite the tricuspid valve. Placing additional holes in the cannula tip may improve drainage.

After cannulation is completed, the inflow and outflow cannulae are passed through subcutaneous tunnels and then connected to the pump itself. When cannulating the right or left atrium, it is most convenient to tunnel the cannula through the skin before placing the cannula into the atrium, which decreases the chances of inducing bleeding at the atrial site with cannula manipulation. The Dacron cuff of each cannula should extend at least 1 cm beyond the skin site to allow for tissue ingrowth at the exit site. Once all cannulae have been passed through the skin and clamped on the velour, the prefilled pump is brought onto the surgical field. All possible length of cannula should be retained for

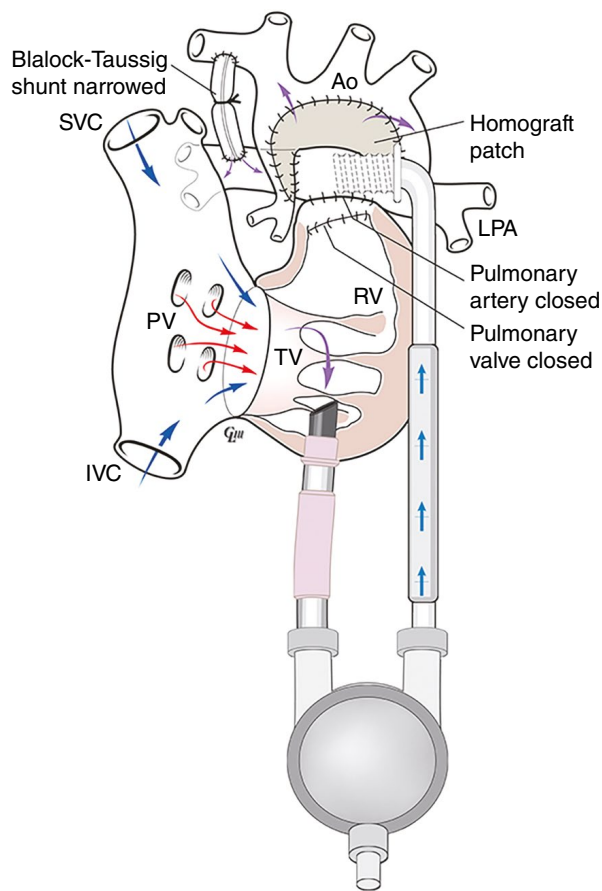
future possible pump exchanges at the bedside. The cannula lengths can be made even by adding extenders, if necessary, to avoid kinking of a cannula. In a biventricular configuration, the outflow cannulae are positioned to cross each other on the external abdominal wall, with the LV pump in an upside-down position compared to the RV pump.

Prior to initiating pumping, the heart is allowed to fill and all air is removed from the cannulae, after which they are again clamped at the velour. Each pump is filled with saline, and manual steps are performed to evacuate all air from the pump using the supplied de-airing needle in the de-airing nipple. A needle vent is inserted on the ascending aorta and placed on strong suction when pumping is initiated along with echo surveillance for air. Following discontinuation of CPB, complete filling of the pump with full diaphragm excursion should be established before leaving the operating room. If filling is incomplete, the cause must be identified and corrected to ensure optimal postoperative cardiac output.

An additional option with the EXCOR cannulae is to connect them to a short-term CF pump such as the CentriMag (Abbott), with later conversion to the pulsatile EXCOR if desired. The CF circuit facilitates inclusion of an oxygenator, continuous renal replacement therapy, and ease of sternal closure.

Abnormal arrangements of the great arteries pose an important challenge in infants and small children. Arterial cannulation for CPB via a Gore-Tex graft anastomosed to the brachiocephalic artery creates additional space on a posterior ascending aorta following the arterial switch procedure. A vascular graft extension on the aortic cannula can be trimmed to an appropriate length to optimize the geometry of the path from outflow cannula to ascending aorta.

In SV with a systemic right ventricle, placement of the ventricular inflow cannula frequently requires selection of the



• **Figure 20.34** Drawing of univentricular extracorporeal cannula positioning in single ventricle anatomy after first stage palliation of hypoplastic left heart syndrome. The inflow cannula is placed in the RV apex of the single ventricle and the outflow cannula in the reconstructed aortic arch, leaving the shunt open for pulmonary blood flow. SVC, superior vena cava; IVC, inferior vena cava; LPA, left pulmonary artery; PV, pulmonary vein; TV, tricuspid valve; RV, right ventricle; Ao, aorta. (Reproduced with permission from De Rita F, Hasan A, Haynes S, et al. Mechanical cardiac support in children with congenital heart disease with intention to bridge to heart transplantation†. *Eur J Cardiothorac Surg*. 2014;46(4):656-662.)

appropriate location for cannula placement, resection of RV trabeculae, and orientation of the beveled opening away from the interventricular septum. Given the variability of internal anatomy, this process is facilitated with aortic cross clamping, cardioplegia administration, and working through a relatively bloodless field until the cannula is positioned. With functional SV, pulmonary blood flow is provided through individually or a combination of a tight pulmonary trunk band, a systemic to pulmonary artery shunt, or a bidirectional cavopulmonary connection (Fig. 20.34).¹⁰⁶ Higher pump flows are required to match the increased volume return of both systemic and pulmonary blood flow to a functional single atrium, which may require selection of a larger pump size than which biventricular physiology.

Special Features of Postoperative Care

Circulatory Management

Management of CF devices follows the same principles as for adult patients and is discussed in Section I under “Special Features

of Postoperative Care.” Pediatric management during initial phases of LVAD support includes optimizing RV function with moderate inotropic support, pulmonary vasodilators (such as nitric oxide or sildenafil), and avoidance of excessive volume administration and resultant pulmonary congestion. Biventricular support is rarely used in pediatric patients, so these management principles are particularly important. Specific indications for biventricular support and the addition of oxygenator or dialysis components are discussed in “Special Situations and Controversies” later in this chapter.

When assessing VAD flow, a cardiac index of approximately 2.3 L/min/m² is a reasonable goal. When supporting a functional univentricular heart, a higher VAD flow is required to achieve both adequate pulmonary and systemic blood flow in the parallel circulation.

The Berlin Heart EXCOR pump is designed to operate with complete filling of the pump and complete emptying with ejection (fill-to-empty). This is important for accurate calculation of the pump output and to minimize stasis and tendency for thrombus formation. Completeness of filling and emptying can be ascertained by direct visualization of the pump membrane. The membrane should be smooth and nonwrinkled at the end of filling (diastole) and emptying (systole). If the pump is incompletely filling, maneuvers to improve filling include decreasing the percent of time in systole, increasing diastolic vacuum, and decreasing pump rate. Maneuvers to improve emptying include increasing the percent time in systole and increasing the pump systolic pressure. The most common causes of new-onset incomplete filling include decrease in ventricular filling (hypovolemia), RV dysfunction resulting in less blood coming to the left ventricle, and cardiac tamponade. Important causes of incomplete emptying include increased afterload and tamponade. These possibilities should be investigated if adjustment of pump parameters does not correct the problem.

Anticoagulation. Initial and maintenance anticoagulation for implantable CF devices follows the same protocols as in adults (see Section I). Separate strategies are employed for paracorporeal devices in children. The choice of anticoagulation has evolved from unfractionated heparin to infusion of direct thrombin inhibitors (DTIs) for paracorporeal devices.¹⁰⁷⁻¹⁰⁹ Based on studies from the Advanced Cardiac Therapies Improving Outcomes Network [ACTION]), infusion of a DTI (primarily bivalirudin) was shown to be superior to unfractionated heparin (UFH) in stroke prevention for both PP (Berlin Heart EXCOR) and paracorporeal CF (Rotaflow [Maquet, Wayne, NJ] and CentriMag/PediMag Abbott) devices.¹¹⁰ Decreased stroke prevention with heparin in infants and children may relate in part to dependence on inhibition of thrombin by antithrombin, which has quantitative and qualitative changes from fetal to adult life. In addition, prolonged use of heparin is associated with heparin-induced thrombocytopenia.¹¹⁰

Following implantation of the Berlin Heart EXCOR, standard administration of clotting factors can be employed to achieve hemostasis, with the exception that platelet transfusion should be avoided when possible because of the associated risk of sensitization and subsequent development of anti-HLA (Human Leukocyte antigen) antibodies that could affect donor matching. When bleeding and chest tube drainage are minimal, anticoagulation is commenced. Dosing with DTI was initiated within 48 hours in 85% of patients in a North American analysis.¹¹⁰ The median starting dose of bivalirudin was 0.3 mg/kg/h (range 0.1–1.4 mg/kg/h) with no bolus, assessing anticoagulant effect using activated partial thromboplastin time (aPTT). Patients are generally monitored with

aPTT every 4 hours until stable dosing is achieved, and then daily. Therapeutic target of aPTT ranges from 50 to 70 seconds early, with later targets of 80 to 100 seconds. Most regimens include one or more antiplatelet agents: aspirin with or without clopidogrel or dipyridamole. Specific testing of platelet function is advisable to determine adequacy of blockade of the arachidonic acid pathway (by aspirin) and the ADP pathway (by clopidogrel).

Results

Survival

General Results and Risk Factors. The vast majority of MCS devices in children are used as bridge-to-transplant support. Overall, the outcomes of MCS to bridge pediatric patients is generally favorable, with greater than 80% survival or successful transplant at 6 months. With implantable CF devices, successful bridge to transplant increases to 90%.¹¹¹ This superior outcome is partially device related and partly related to the requirement for larger (and older) children for support with CF devices. Risk factors for death prior to transplantation include biventricular support, diagnosis of CHD, infant age group, the need for ventilator support prior to implantation, Intermacs profile (cardiogenic shock), and renal impairment.¹⁰³ In this analysis, the 6-month survival on device (or successful transplant) was intracorporeal CF 92%, PP 84%, and paracorporeal CF 69%. For patients <10 kg, parametric modeling showed an increased early risk of death with CHD, intubation, and renal impairment. Among patients with CHD, risk factors for early mortality included biventricular support, renal impairment, and use of paracorporeal CF devices.¹¹² The use of ECMO support prior to VAD implant, if limited to brief periods to reverse organ dysfunction related to low cardiac output, is generally not an independent risk factor for poor outcomes during LVAD support.¹¹²

Implantable Continuous Flow Devices

Since about 2012, CF-LVADs have replaced paracorporeal pumps for longer term pediatric patients who are large enough to accommodate adult devices. Until 2021, the primary implantable CF device used in the pediatric population was the HVAD (Medtronic) centrifugal flow pump (see Section I). During US clinical trials for approval by the FDA, the HVAD was first used to support a pediatric patient in the United States in 2011.⁹⁴ The HVAD was applicable for pediatric patients down to about 10 kg, and this device accounted for >90% of implantable CF pediatric implants between 2015 and 2021.¹¹³ The smallest reported recipient was 3 years of age with a BSA of 0.6 m².¹¹⁴ Over half the patients were discharged from hospital, and 65% were successfully transplanted by 1 year, with 10% mortality on device.¹¹⁵ The HVAD device was removed from the market in June, 2021, after the FDA initiated a series of recalls related to pump failures. This left only the adult HeartMate 3 (a larger device than the HVAD) as the only implantable CF pump option for pediatric patients, leaving a care gap with no implantable device available for patients less than about 20 kg.

The pediatric experience with the HeartMate 3 implantable CF pump was reported by O'Connor and colleagues from the ACTION network.¹¹⁶ Between December 2017 and May 2022, 170 patients were supported with the HeartMate 3 device, with median age of 15.3 years and BSA of 1.68 m². The smallest patient was 17.7 kg and 0.73 m². Nearly all were BTT strategy, and 72% had dilated CM. The overall mortality during support was 8%, similar to the adult HeartMate 3 experience (see Section I).

The overall adverse event profile mirrored the adult experience, except that patients with a BSA <1.4 m² experienced a higher incidence of infection, renal dysfunction, and ischemic stroke.

Paracorporeal Devices

In a multicenter cohort of 204 pediatric patients between 2007 and 2010 supported by the Berlin EXCOR, 75% of patients survived to transplant or recovery.¹¹⁷ Risk factors for mortality included smaller patient weight, biventricular support, and renal or hepatic dysfunction. A postmarket approval surveillance study of Berlin Heart EXCOR implants from 2018 to 2020 reported successful bridge to transplant, recovery, or alive on device at 6 months in 86% of patients, up from 76% in the era from 2007 to 2014.

Small children and infants represent the most challenging groups, for whom the only devices available for use in the United States are paracorporeal pumps and ECMO. Multiple studies have demonstrated the dismal outcomes with isolated ECMO support unless a heart transplant can be performed within days to weeks.^{113,118} The outcomes of infants and small children were analyzed in a multi-institutional analysis of infants <10 kg, which included a cohort of 2049 patients under 10 kg listed for heart transplantation between 2010 and 2018, of which 28% were supported with MCS.¹¹⁹ ECMO support at the time of listing occurred in slightly less than 10% of patients with either CM or CHD, and about 5% after listing. In contrast, LVAD support was much more common among CM patients compared to CHD, both at listing (8% vs. 2.5%, $P < .001$) and after (23% vs. 5%, $P < .001$). Patients with CHD with VADs were generally smaller, younger, and more often had prior cardiac surgery compared to CM. The ongoing challenges of LVAD support in infants is most evident among infants with CHD, for whom mortality without transplant was about 30% by 3 months. Among those infants <5 kg, the mortality without transplant approached 60% by 3 months. Risk factors for mortality among infants during LVAD support include a diagnosis of CHD, elevated serum bilirubin as a marker of right HF, the need for biventricular support, renal impairment, and respiratory failure requiring intubation before implant.^{103,113} Despite the challenges in providing LVAD support in infants, these outcomes for CHD represent distinct improvements over the previous era between about 2007 and 2012.^{117,118,120} The outcomes have been consistently better for infants with CM, in which the likelihood of survival to transplant with Berlin Heart EXCOR support exceeds 80% in the absence of prior need for ECMO and without marked right HF.^{113,118}

Single Ventricle Before or After Stage I Palliation

The worst outcomes are among infants with functional SV before or after stage I palliation. The presence of collateral pulmonary circulation, variable pulmonary vascular resistance, and anatomic challenges adds to the difficulties of balancing systemic and pulmonary circulation with the combination of VAD plus pulmonary trunk banding if pulmonary blood flow is excessive or a systemic-pulmonary artery shunt. Kirklin and colleagues first successfully bridged a small patient with SV to successful heart transplantation with a durable pump in 2007.⁹⁵ A 15-month-old with SV and prior pulmonary trunk banding presented with cardiogenic shock and was supported with

ECMO for 3 days, after which a Berlin Heart EXCOR was implanted with tightening of the pulmonary artery band and a systemic to pulmonary artery shunt. The patient was supported for 7 weeks and underwent successful heart transplantation. Weinstein and colleagues reported survival to transplant in only one of nine infants with SV using the EXCOR PP pump.¹²¹ In 2014, Lal and colleagues reported the first successful bridge to transplant with initially a paracorporeal CF device and a modified Blalock-Taussig shunt, followed by a brief period of paracorporeal pulsatile support, and eventual successful heart transplantation after 65 days of support.¹²²

Since about 2015, outcomes have progressively improved. A US multicenter analysis identified 30 implants before or after stage 1 palliation between 2018 and October 2020. Age at implant ranged from 0.1 to 25 months with weight range 2.4 to 17 kg.¹²³ Device types included PP (50%), paracorporeal CF (37%), and mixed (13%). Successful support to transplant was achieved in 63%. However, another recent multicenter analysis reported mortality during support for SV patients < 10 kg exceeding 60% at 3 months.¹⁰³ Recently, Bleiweis and colleagues reported improved outcomes with bridge-to-transplant support with the Berlin EXCOR in a small group of infants < 5 kg, in which > 50% were successfully bridged to transplant.¹²⁴ These findings support the need for more effective small, durable devices for neonates and infants, particularly among patients with CHD.

Single Ventricle After Stage II Palliation

VAD support after the bidirectional Glenn stage of palliation requires clear understanding of the physiologic reasons for Glenn failure. VADs are most effective in the presence of primary ventricular failure rather than when elevated pulmonary vascular resistance (PVR) is an important component. If PVR is importantly elevated, cyanosis may actually worsen with VAD support, since the increase in cardiac output from the VAD may disproportionately increase venous return through the inferior vena cava and increase desaturated mixing within the common atrium. In addition, increase in venous return may increase flow through upper body veno-veno collaterals, with worsening cyanosis. In the Investigational Device Exemption study of the Berlin Heart EXCOR, PP support with the EXCOR was successful in bridging 7 of 12 patients after stage 2 palliation to successful transplantation, with a median support of 52 days.¹²¹

Adachi and colleagues first reported successful bridging to transplant after stage II with an implantable CF pump (HVAD).¹²⁵ Currently, however, few patients after stage II palliation would be of sufficient size for the adult HeartMate 3 device.

Failing Fontan Circuit

The etiologic factors in Fontan failure, either in isolation or more commonly in combination, are ventricular systolic dysfunction, ventricular diastolic dysfunction, increased pulmonary vascular resistance, and lymphatic abnormalities. Durable MCS therapy is most successful when the predominant factor is ventricular systolic dysfunction, and less so with primary diastolic dysfunction, for which MCS therapy is not well established. Patients with predominantly elevated PVR as the cause of failure are the least suitable for MCS therapy.

Early reports documented the effectiveness of VAD support in the setting of primary ventricular systolic dysfunction, initially in

adults^{126,127} and subsequently in children.^{128,129} Isolated successful support of the Fontan circulation was first reported by Pietre in 2008.⁹⁶ As first reported by Niebler and colleagues,¹³⁰ implantable CF pumps are currently the preferred option for support of the failing Fontan in adolescents and adults. Rarely, cardiac replacement with a TAH, as first reported by Rossano and colleagues,¹³¹ may be a high risk option for the failing Fontan with mixed causes.

An analysis of the Society of Thoracic Surgeons Pedimacs and Intermacs pediatric and adult MCS experience in the United States between 2012 and 2019 identified 55 Fontan patients who underwent systemic ventricular support (median age 10 years and weight 27 kg).¹³² Improved support of this patient cohort over the past decade is evidenced by the 75% survival on device at 6 months and a favorable outcome (alive on device, transplanted, or recovered) in over 80% of patients at 6 months. The median length of support was 3.8 months.

A multicenter analysis from the ACTION network by Cedars and colleagues reported an overall 70% successful support to transplant by 1 year with 21% mortality on device among 45 VAD supported Fontan patients.⁸⁸ The vast majority (83%) had moderate to severe ventricular dysfunction, with moderate or worse atrioventricular valve regurgitation in 65%. Seventy percent had a morphologic right ventricle. The most common device was the Medtronic HeartWare HVAD CF implantable VAD. This evolving experience among numerous centers indicates that selected patients with a failing Fontan, particularly with primary ventricular dysfunction, can be successfully supported in BTT strategy.¹²¹

Stroke

Stroke represents the most devastating complication during VAD support, and is particularly vexing with paracorporeal devices. Intracerebral hemorrhage and thromboembolic episodes are the major causes of morbidity and mortality during pump support.¹¹⁸ Specific time-related risk of these events is not precisely documented, but the incidence of serious neurologic events during support is estimated at 7% to 10%.

A multi-institutional analysis from the Society of Thoracic Surgeons Pedimacs registry from 2012 to 2021 (including eras of heparin-based as well as DTI anticoagulation) reported an overall stroke incidence of 10.7% during support on a variety of devices.¹³³ Comparing paracorporeal support methods, 17% of patients on pulsatile devices had at least one stroke compared to 10.4% with CF, but the stroke rate was lower with pulsatile devices (6.4 vs. 11.1 events/100 patient months). Among all patients, risk factors for stroke included presence of ascites and more seriously ill at implant ($P < .05$).

The transition from heparin-based anticoagulation to direct thrombin inhibitors (bivalirudin) has importantly reduced stroke incidence with paracorporeal pumps.^{107,110} The Berlin Heart EXCOR experience between 2018 and 2021 ($n = 72$), in which bivalirudin was the primary anticoagulant in >90% of patients, reported a 44% reduction in stroke incidence compared to the 2007 to 2014 ($n = 302$) experience with largely heparin-based anticoagulation. In the era with routine UFH use, nearly 90% of strokes were ischemic and 50% of strokes occurred in the first 14 days. The use of DTI compared to UFH was associated with a reduction in percent of patients with stroke from 29% to 12% with Berlin Heart EXCOR and from 24% to 10% with paracorporeal CF pumps.

Infection

Among all pediatric patients supported with MCS devices, about 20% experience infections, which are more common in younger patients with longer hospitalizations.¹³⁴ Patients with and without major infections during support have similar freedom from post-transplant infection, rejection, and graft loss (all $P > .1$). Device-related infections are uncommon but are associated with greater incidence of post-transplant rejection ($P < .03$) and graft loss ($P < .01$).

Renal Dysfunction

HF-associated renal dysfunction is well recognized in patients who require MCS support. As in the adult population, implantation of MCS therapy generally partially or completely reverses renal dysfunction. A linkage study between the STS Pedimacs registry and the pediatric heart transplant society (PHTS) registry documented the impact of MCS in improving renal dysfunction among pediatric patients with advanced HF. Mean estimated glomerular filtration rate (eGFR) increased from 85 ± 47 mL/min/1.73 m² preimplant to 131 ± 62 1 month postimplant.¹³⁵ Lower preimplant eGFR and failure to normalize renal function after initiation of support predicted the development of chronic kidney disease after heart transplantation.

Liver Dysfunction

Advanced HF in pediatric patients, as in adults, is known to cause progressive hepatic dysfunction, related to the combined effects of low cardiac output and hepatic venous congestion secondary to right HF. Elevated serum bilirubin is a known risk factor for mortality. A study of pediatric patients receiving PP pump support with the EXCOR had a 65% mortality within the first month with serum bilirubin >1.2 mg/dL.¹¹⁷ A more recent multicenter study from the PEDIMACS registry examined the combined impact of renal and hepatic dysfunction on post-LVAD mortality using the Model for End-stage Liver Disease eXcluding INR (MELD-XI) score.¹³⁶ Along with Pedimacs profile 1 (cardiogenic shock) (HR 1.6), CHD (HR 2.3), and being on a percutaneous VAD (HR 2.7), the MELD-XI score (HR 1.1 per 1 unit rise) was a better predictor of post-VAD mortality than bilirubin or creatinine alone. A high MELD-XI score cohort (score >13.6) predicted worse survival, increased bleeding, renal dysfunction, and respiratory failure postimplant compared to intermediate and low scores. The predictive value of this measure persisted after VAD implant, in that increasing or continued high MELD-XI scores early postimplant predicted worst survival.¹³⁶

Indications, Timing, and Decision-Making for Device Implantation

The standard indications for LVAD implantation in adults (see Section I) also apply to the pediatric population, particularly those with cardiomyopathy, in which outcomes are similar to those of adults. Current differences in decision-making relate in part to the nearly uniform application of devices in the pediatric population as bridge-to-transplant therapy, whereas most durable devices in adults are currently applied as longer term “destination” therapy. Unique indications in the pediatric population are discussed in this section.

Congenital Heart Disease in Two-Ventricle Situations

Pediatric patients with two-ventricle CHD are candidates for MCS when HF and low cardiac output, often with noncardiac organ dysfunction, are severe enough as to make survival to transplantation unlikely. In the event of rapid decompensation with multiorgan dysfunction, initial ECMO support (or in older children and adolescents, percutaneous device support) may be advisable prior to proceeding with more durable VAD support.

Pulmonary Hypertension

Pulmonary hypertension secondary to acute or chronic left atrial hypertension is an accepted indication for an LVAD in the setting of advanced HF. As in adults, when pulmonary hypertension is to a level that may preclude safe cardiac transplantation yet is still reactive, reduction of left atrial pressure and secondarily pulmonary artery pressure with LVAD support may convert a high risk transplant to a low risk operation. This logic is applicable to pediatric patients with cardiomyopathy after demonstration of reactive pulmonary hypertension. However, pulmonary hypertension related to CHD is often in part related to pulmonary vascular disease that may be difficult to examine for reversibility with multiple sources of pulmonary blood flow and, especially with functional SV, may have associated irreversible pulmonary vascular remodeling.¹³⁷ Thus, reversibility of pulmonary hypertension with LVAD support in CHD patients is less certain.

Infants with Two Ventricles Considered for Support as Bridge-to-Transplant

Infants with two ventricles considered for support with advanced HF as bridge-to-transplant therapy represent a higher risk group and require a more measured risk-benefit calculus given the lesser effectiveness of MCS in this age group. Based on information from multicenter studies, the following general expectations apply to infants considered for LVAD support¹³⁸:

- Infants weighing 5 to 10 kg with CM, no prior ECMO, and normal bilirubin (no right HF), are excellent candidates for BTT VAD support, with 90% likelihood of survival to transplant.
- Infants weighing 5 to 10 kg with CHD (but not SV), not on ECMO, and normal bilirubin have a 70% to 80% chance of survival to transplant on LVAD support.
- Infants weighing 5 to 10 kg with CM or CHD, normal bilirubin, but on ECMO, have about a 40% of survival to transplant if converted to LVAD support.
- Infants <5 kg not on ECMO, either CM or CHD (but not SV), and normal bilirubin have about a 60% chance of survival to transplant on LVAD support.
- Infants <5 kg with CM on ECMO have about 10% chance of survival to transplant on LVAD support. Risk-benefit ratio should be carefully considered.
- Infants <5 kg with CHD on ECMO have extremely poor chance of survival to transplant with or without conversion to LVAD support.

Infants with Functionally Single Ventricle Considered for Support as Bridge-to-Transplant

Infants with SV are at high risk for mortality with VAD support as BTT therapy. SV infants (and particularly those <5 kg) who

need VAD support have a considerably higher chance of successful bridge to transplant in pediatric cardiac centers experienced with VAD therapy in this anatomic population. Expected survival to transplant is <40%, and decreases dramatically after about 3 months of support.

Ventricular Assist Device Therapy in Failing Fontan

MCS in the failing Fontan is indicated in patients as bridge-to-transplant support when the likelihood of survival to transplant with preserved organ function is low and the predominant factor is systemic ventricular dysfunction. MCS is less established as bridging therapy for primary diastolic dysfunction. Isolated systemic VAD support is not indicated for Fontan failure primarily related to elevated PVR. In such situations, additional (or isolated) VAD support of the Fontan/pulmonary circuit is a high-risk option.

In all instances of MCS for the failing Fontan, reversal of major end organ dysfunction, with inotropes or preliminary ECMO, provides the best chance of success. When PVR is elevated and reactive, inhaled, intravenous, or subcutaneous vasodilator therapy is an important component of maximizing pulmonary blood flow and VAD filling. Extensive veno-veno collaterals can lead to increased cyanosis postimplant, so they should be clarified with plans and thresholds for intervention. Extensive aortopulmonary collaterals reduce systemic blood flow and should be investigated with proactive planning regarding closure.

Special Situations and Controversies

Atrial Versus Ventricular Cannulation for VAD Inflow

In most conditions, ventricular cannulation is preferable due to more reliable device filling or inflow and less stasis of blood in the ventricle with a greater risk of clot formation and embolization. However, in restrictive physiology associated with restrictive cardiomyopathy or hypertrophic cardiomyopathy, the optimal

cannula position within the ventricle to achieve unobstructed VAD inflow may not be possible. In such instances, atrial cannulation may be preferable.

Particularly in functional SV, if atrial cannulation is desired related to dense adhesions or unfavorable ventricular anatomy, bridging support with a CF pump (paracorporeal or implantable) may provide more effective initial venous drainage/decompression than a pulsatile device.

Indications for Biventricular Support, Addition of Membrane Oxygenator, and Renal Replacement Therapy

Biventricular support is generally indicated when target LVAD flows cannot be achieved with a CVP of about 15 mmHg or less despite moderate inotropic support and pulmonary vasodilator therapy. Particularly in small patients, biventricular support is challenging given the limited space available for multiple cannulae to exit the skin with sufficient skin bridges to prevent breakdown.

With accompanying respiratory dysfunction, a membrane oxygenator can be added to the VAD circuit. With a biventricular configuration, the oxygenator position should be after the LVAD components to avoid abrupt oxygen-induced pulmonary vasodilation, sudden increase in pulmonary blood flow and resultant increased pulmonary congestion.

When renal dysfunction requires dialysis, a continuous renal replacement therapy (CRT) circuit can be inserted into the VAD circuit. With additional connections in the circuit the risk of air embolism increases. Therefore, the CRT circuit may be safer with an additional membrane oxygenator placed beyond the CRT circuit.

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Complete references in this chapter can be accessed in the online version.

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